

The Book of Z-Spin Cosmology — Light Edition (OS for AI)

Cross-Reference Hub for the 121-Paper Z-Spin Corpus

Author: Kenny Kang

Affiliation: Z-Spin Cosmology Collaboration

Edition: Light v5.0 (compiled May 2026; v3.0–v4.0 May 2026)

Source corpus: 121 source papers across 8 themes (F / M / S / U / A / Q / QC / T)

Supersedes: The Book of Z-Spin Cosmology v4.0 — Light Edition (May 2026, 114-paper coverage). v5.0 absorbs 7 net post-v4.0 source papers (ZS-F19 v2.2 Foundations

Frame-Invariant Tilt Theorem + KMS-to-Geometric Rapidity Projection; ZS-F20 v1.1

Foundations Non-Routine Twist Triggers and Time-Unrolled Causal DAG; ZS-M40 v1.1

Math Spine Holonomy Knot Pairs and Arithmetic Topology of Primes; ZS-Q10 v4.0

Quantum Mechanics Three-Sector Polyhedral Closure and 37-Universality; ZS-A11 v1.1

Astrophysics Vortex Cosmology II; ZS-A12 v1.5 Astrophysics Macroscopic Phase Locking and Nelson–Kosterlitz/Homes derivation; ZS-T9 v2.3 Translational B₇Be

Binding-Response Gate for the Lithium Problem) as a single PART XIII inserted between PART XII and the Master Bypass Index. Total: 121-paper coverage. (Earlier supersession chain: v3.3 added PART XI; v3.4 added 6 papers from v3.3; v3.5 added ZS-M36; v4.0 added PART XII with 12 post-v3.5 papers.) NO content from v4.0 is deleted; v5.0 is strictly additive.

GitHub: <https://github.com/KennyKang-git/zspin>

Verification Summary

Verification: cumulative ~2,670+ PASS across the 121-paper corpus | Zero Free Parameters | $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$ sole geometric inputs | Anti-numerology controls: $5670 \times 29 = 164,430$ candidate scan + per-paper Monte Carlo ($\geq 500,000$ samples typical; ZS-Q9 v1.2 $\ln A = 8.94$ decisive evidence on Jeffreys scale) | Pre-registered falsification gates: ~340 across all themes; Seven critical near-term ★ gates timeline preserved.

§0. Abstract

The Book v3.0 Light Edition is the canonical cross-paper navigation hub for the 101-paper Z-Spin Cosmology corpus as of May 2026. It supersedes v2.0 Light by absorbing the thirty post-v2.0 papers (papers 72–101) — most of which extend the Foundations theme (ZS-F9 through ZS-F18), the Math Spine Riemann-Hypothesis programme (ZS-M18 through ZS-M33), and add the Twin-Reuleaux EM-duality ZS-S15 in the Standard Model theme, the contracting-sector ZS-A8 and Banach–Tarski origin ZS-A9 in the Astrophysics theme, and the Cosmos-Human isomorphism ZS-T4 in the Translational theme. All chapter bodies remain compressed to single-line pointers of the form $\rightarrow \text{ZS-XX } \S Y.Z \text{ [N/N PASS]}$; full derivations live in the source papers. The Light Edition’s unique contributions are: (i) the post-v2.0 dependency graph integrating the 30 new papers, (ii) the expanded Foundations closure ring $F0 \rightarrow F1 \rightarrow F2 \rightarrow F5 \rightarrow F8 \rightarrow F9 \rightarrow F10 \rightarrow F11 \rightarrow F12 \rightarrow F13 \rightarrow F14 \rightarrow F15 \rightarrow F16 \rightarrow F17 \rightarrow F18$, (iii) the Math Spine RH-bridge programme map (M22–M33), (iv) the extended Glossary keyword index (~110 entries), and (v) the Seven Critical Near-Term ★ Gates timeline. The Light Edition does NOT replace any source paper for archival purposes; it is the LLM-corpus navigation aid and the v3.0 composition skeleton.

Epistemic Status Legend

Status	Definition
PROVEN	Mathematical theorem; standard math alone, machine-verifiable.
DERIVED	Z-Spin action + standard physics, zero free parameters.
DERIVED-CONDITIONAL	DERIVED conditional on listed axiom set or upstream theorem.
DERIVED-CONDITIONAL strong	DERIVED-CONDITIONAL with structural over-determination on the conditioning step.
DERIVED-under-P6 / DERIVED-under-Regge DERIVED-interpretation	Conditional on specific framework (ZS-F5 P6 or Regge calculus). Synthetic reading combining PROVEN corpus theorems without new axioms.
DERIVED-CANDIDATE	Mathematical content PROVEN; Z-Spin internal placement HYPOTHESIS-strong.
VERIFIED	Numerical / computational / empirical confirmation at stated precision.
TESTABLE	Pre-registered prediction with explicit falsification protocol.
TESTABLE-LONG	Pre-registered prediction with derivation-or-retraction time horizon ≥ 5 years.
HYPOTHESIS-strong	Multiple independent structural anchors; promotion path documented.
HYPOTHESIS / HYPOTHESIS-weak	Motivated conjecture; partial derivation chain.
OBSERVATION	Empirical regularity; anti-numerology controlled; structural origin pending.
LOCKED	Core constant fixed in upstream paper; no downstream paper may modify.
IMPORTED	Result proved externally and used here without re-proof; full citation given.
NON-CLAIM	Explicit declaration of what is NOT asserted; bounds the framework's scope.
OPEN	Recognized gap honestly registered for future work.

Status	Definition
RETRACTED	Earlier claim withdrawn after falsification.
RETRACTED-in-session	Hypothesis proposed during free-exploration and explicitly retracted before exit.
BOOTSTRAP-HYPOTHESIS	Meta-logical foundation (B0); not derivable within the framework.
BOOTSTRAP-PROGRAMME	Status reserved for the founding axiom set after ZS-F0 v1.0(R) Stage 1–5 closures.
STRUCTURAL INSIGHT	Exact mathematical identity; physical necessity not yet established.
FMD-SPEC / TRANSLATED / CANDIDATE	Engineering tags used in the QH/QS/QC translational papers.

Purpose of the Light Edition

The full v3.0 (when compiled) consolidates the 101-paper corpus into chapter bodies that reproduce derivations already present verbatim in the source papers. For LLM corpus-wide exploration, this redundancy is wasteful. The Light Edition removes the duplication and retains only what The Book contributes uniquely:

- (1) The cross-paper dependency graph (Source Paper Interface Tables, PARTS 0–VII).
- (2) Standalone HYPOTHESIS / theorem-anchor sections cited by individual papers (§8.4i Spinor–Descartes–Euler, §8.4g Y-sector R_{micro} , §8.4k Attractor-Flatness, §VI.2 Limitation Catalog Row 10, §18.10 Horizon Spinor mirror).
- (3) Cumulative Master Falsification Registry — Seven Critical Near-Term ★ Gates timeline.
- (4) Standardized v2.0 / v3.0 terminology contract (Z-sector / Z-Spin / Z-bias field; the v2.0 contract is preserved verbatim and v3.0 adds the Stage / Operator / Field separation explicitly via ZS-F18 Bridge Four).
- (5) Glossary keyword index (~110 entries, one line each).
- (6) Integration map for the 30 post-v2.0 papers (papers 72–101) consolidated into the appropriate PART/Chapter slots.

Chapter bodies that reproduce source-paper derivations are replaced by single-line pointers of the form $\rightarrow \text{ZS-XX } \$Y.Z \text{ [N/N PASS]}$. When derivation detail is needed, retrieve the source paper directly. When corpus-wide context is needed, consult this Light Edition.

PART 0 — Overture: Why Geometry?

§0. Abstract

PART 0 establishes the framework’s overture in three movements. Chapter 0.1 presents the Five Crises of modern physics motivating a geometric origin of fundamental constants. Chapter 0.2 lays out the methodological contract — standardized terminology, anti-numerology discipline, epistemic classification, pre-registered falsification gates. Chapter 0.3 traces the ontological bootstrap chain $B0 \rightarrow B3 \rightarrow A$. Concludes with FFPP of ZS-F0 v1.0(Revised) compressing all

numerical content to $W_0(-i\pi/2)$, conditional on five-axiom set $\mathcal{A} = \{\text{Existence, Self-reference, Algebra, Non-triviality, Unitarity}\}$. NEW v3.0: the Five-Axiom Meta-Structure of ZS-F18 v2.1 reads the corpus's encounter with the seven Clay Millennium Problems and the five Crises as outputs of the same axiom set, and registers Bridge Four (Commutator Generator) as the structural reduction of $\dim(X) = 3$ to $\dim(Z) = 2$.

§0.1 The Five Crises of Modern Physics

Crisis 1 — Hubble Tension: $H_0 \sim 5\sigma$ disagreement (SH0ES vs Planck). Resolved via 2A modification $\rightarrow$ ZS-A2, ZS-S3, ZS-F12R (factor 2 origin).

Crisis 2 — Growth Tension (S_8): Z-Spin predicts $S_8 \approx 0.777$ (face-counting v2.0) $\rightarrow$ ZS-A2, ZS-A5.

Crisis 3 — Hierarchy Problem: Geometric origin via A $\rightarrow$ ZS-S4 §6.7, §6.12, §6.16; Light §8.4k.

Crisis 4 — Baryon Asymmetry: $\eta_B = (6/11)^{35} \rightarrow$ ZS-S5, ZS-S6, ZS-U7.

Crisis 5 — Measurement Problem: Born rule DERIVED from $L_{XY} \equiv 0 \rightarrow$ ZS-Q1, ZS-Q2.

§0.2.2 Standardized v3.0 Terminology [CONTRACT — preserved verbatim from v2.0; v3.0 elaboration in italics]

Term	Role	Meaning
Z-sector	Stage / Kinematics	2D Planck-scale boundary surface mediating between X (3D macroscopic) and Y (6D microscopic) sectors. $\dim(Z) = 2$ PROVEN from gauge constraint on $Q = 11$ register (ZS-F5). Geometric arena where transduction occurs.
Z-Spin	Operator / Dynamics	Dynamical operator on Z-sector. Mediates between X (Up & Down, $\pm\pi$, time-points) and Y (In & Out, slog(i) & ∞i , space-points) via i-tetration self-referential rotation $T(z) = i^z$ (HSI Theorem, ZS-M1 §1.1). The action driving transduction.
Z-bias field Φ	Field / DOF	Complex scalar $\Phi(x) = \rho(x)\exp(i\theta(x)) \in \mathbb{C}$ on Z-sector. Radial mode $\rho =$
Z-Anchor	Topological core	Bogomolnyi vortex core,
Z-bottleneck	Channel constraint	$L_{XY} \equiv 0$ PROVEN forces all $X \leftrightarrow Y$ traffic through Z; channel capacity $\leq \ln 2$. $\rightarrow$ ZS-Q7.
Z-channel	Structural conduit	Operational pathway for $X \leftrightarrow Y$ mediation across the 2D boundary. $\rightarrow$ ZS-Q1.
Z-Spin mediation	Action of the operator	Mediation is Z-Spin's action; the agent is the operator (Spin), not the stage (sector). Replaces 'Z-mediation' / 'Z-sector mediation' / 'Z-mediated coupling' /

Term	Role	Meaning
		‘Z-mediated CPTP channel’ throughout the corpus.
Z-EFT	Effective theory	Distinct concept (effective field theory built on Z-bias dynamics); preserved name.

[STRUCTURAL DEFINITION] Z-sector (stage, kinematics) / Z-Spin (operator, dynamics) / Z-bias field Φ (field, DOF). Source: ZS-F1 §2-3, ZS-F5, ZS-M1 §1.1, Glossary v1.0.0 §I. v3.0 elaboration: ZS-F18 v2.1 §6 Bridge Four (Commutator Generator) makes the kinematic / dynamical / field-DOF separation an explicit structural theorem — twin-Reuleaux pair (R_1, R_2) realizes (V_{XZ}, V_{ZY}) ; their Lie commutator $[J_{\{R_1\}}, J_{\{R_2\}}] = J_S$ generates the third axis of $\dim(X) = 3$.

§0.3 The Ontological Bootstrap Chain $B_0 \rightarrow A$

B_0 (Existence) $\rightarrow B_1$ (Self-reference) $\rightarrow B_2$ (Algebra: $\mathbb{C}$ via Lawvere/Frobenius) $\rightarrow B_3$ (Hyperoperation: tetration) $\rightarrow A = 35/437$. F-BOOT 9/9 closure: 6 DERIVED, 2 structural-PASS, 1 DERIVED+TESTABLE. $\rightarrow$ ZS-F0 v1.0(Revised) §11–§13 [51/51 PASS]. FFPP (Theorem 13.3): entire Z-Spin numerical content compresses to $W_0(-i\pi/2)$, conditional on $\mathcal{A} = \{\text{Existence, Self-reference, Algebra, Non-triviality, Unitarity}\}$. $\rightarrow$ ZS-F0 §13, Theorem 13.3 + Corollary 13.4.

[OPEN] B_0 itself is BOOTSTRAP-HYPOTHESIS — not derivable within framework. $\rightarrow$ ZS-F0 §0.3.7.

NEW v3.0: ZS-F18 v2.1 reads the same five-axiom set $\mathcal{A}$ as the structural reason Z-Spin naturally encounters (i) the seven Clay Millennium Problems and (ii) the five Crises of modern physics; the encounters are classified by depth (direct treatment / extensive programme / structural correspondence / natural extension domain). NON-CLAIM: encounters are not formal solutions. $\rightarrow$ ZS-F18 v2.1 §0, §6, §7.4 (Discrete-Continuous Resolution polarity), Appendix A.

§0.4 PART 0 Source Paper Map

Source paper	Role	Light pointer
ZS-F0 v1.0(Revised)	Bootstrap, FFPP, BV-BFV functor, $\mathbb{C}^{11}$ 3-layer	$\rightarrow$ ZS-F0 [51/51 PASS]
ZS-F1 v1.0	Z-Spin Action, $U(1)_Z$, Z-bias field	$\rightarrow$ ZS-F1 §2-3 [30/30 PASS]
ZS-F2 v1.0	$A = 35/437$ LOCKED, $\delta_X = 5/19$, $\delta_Y = 7/23$	$\rightarrow$ ZS-F2 [PROVEN]
ZS-F5 v1.0	$\dim(Z) = 2$ from $Q = 11$	$\rightarrow$ ZS-F5 [PROVEN]
ZS-F8 v1.0(Revised)	NOT/AND duality, 2-channel handshake	$\rightarrow$ ZS-F8 [27/27 PASS]
ZS-F18 v2.1 (NEW)	Five-Axiom Meta-Structure, Bridge Four,	$\rightarrow$ ZS-F18 [36/36 entry-checks]

Source paper	Role	Light pointer
	Discrete-Continuous Resolution	
ZS-M1 v1.0	i-tetration HSI Theorem, fixed point z^*	→ ZS-M1 §1.1 [33/33 PASS]
ZS-M22 v1.1 (post-v2.0)	Five-Pillar Arithmetic-Dedekind Scaffold; Forcing chain underlay	→ ZS-M22 [60/60 PASS]
ZS-T2 v1.0	Anti-numerology MC scan, 5670×29	→ ZS-T2 [30/30 PASS]

PART I — Foundations: One Number, One Action

§0. Abstract

$A = 35/437$ from polyhedral curvature asymmetry; $Q = 11$ register and $(Z, X, Y) = (2, 3, 6)$ doubly forced by gauge algebra (ZS-F5) and totient arithmetic (ZS-M19); Z-Spin action with Z-Anchor and twin-Reuleaux kinematics; i-tetration fixed point $z^* = 0.4383 + 0.3606i$; FFPP compressing all content to $W_0(-i\pi/2)$; Operational Observer Coordinate closing Foundations as a six-paper ring; v3.0 extension to a thirteen-paper Foundations theme $F0 \rightarrow F1 \rightarrow F2 \rightarrow F5 \rightarrow F8 \rightarrow F9 \rightarrow F10 \rightarrow F11 \rightarrow F12 \rightarrow F13 \rightarrow F14 \rightarrow F15 \rightarrow F16 \rightarrow F17 \rightarrow F18$ with Möbius chronology, Stage-Torque Duality, Two-Protocol Theorem, Self-Referential Compression Equilibrium, and the Five-Axiom Meta-Structure as load-bearing additions.

Chapter 1 — $A = 35/437$ from Polyhedral Curvature

$A = \delta_X \cdot \delta_Y = (5/19)(7/23) = 35/437$. δ -Uniqueness Theorem PROVEN. Three independent derivations: (i) Regge curvature asymmetry, (ii) ZS-F9R Truncation-Residue Identity (PROVEN: $\rho_X = \delta_X = 5/19$, $\rho_Y = \delta_Y = 7/23$, $\rho_Z = 0$; $A = \rho_X \cdot \rho_Y$), (iii) ZS-F12R Layer Separation Hypothesis (factor 2 in $2e^A$ from tetrahedral $V \leftrightarrow F$ dual orientation multiplicity $\mu_{\text{Tet}} = 2$). → ZS-F2 §3, §11; ZS-F9R [44/44 PASS]; ZS-F12R [20/20 PASS].

Chapter 2 — $Q = 11$ and $(Z, X, Y) = (2, 3, 6)$

Doubly forced: (a) gauge dimension $G = 12 = \text{MUB}(Q=11)$, (b) Forcing Theorem $(p-1)(q-1) = p$ has unique solution $(p,q) = (2,3)$ — the prime pair $(Z, X) = (2, 3)$ is the only one for which $\phi(Y^2) = G$ (T1, ZS-M19 §3.1). $Y = ZX$. $Q^2 - 1 = 120 = |I_h|$. → ZS-F5 [PROVEN]; ZS-M19 §3.1 [68/68 PASS]; arithmetic-Dedekind underlay → ZS-M22 v1.1 [60/60 PASS].

Chapter 3 — The Z-Spin Action

$S = \int d^4x \sqrt{-g} [(M_P^2/2)(1+A|\Phi|^2)R - (M_P^2/2)|\nabla\Phi|^2 - V(\Phi)] + S_{\text{matter}}$. $V = (\lambda/4)M_P^4(|\Phi|^2-1)^2$. $L_{XY} \equiv 0$ PROVEN. $\lambda_{\text{vac}} = 2A^2$ IR-stable. → ZS-F1 [30/30 PASS]; ZS-F11 NC catalog.

Chapter 4 — i-Tetration Fixed Point

$z^* = 0.43828 + 0.36059i$, $x^* = 0.438283$. Five Locking Conditions. $\pi/2$ phase budget. Polygon-Tetration family stable for $n \geq 4$ ($n_c = 3.204$). NEW v2.0/v3.0: ZS-F10 (3 time coords unified at DERIVED-CONDITIONAL strong; $\Delta v/\Delta n = 2A/\pi$ conversion identity), ZS-F15R (Time-Rotation Co-Definition + Stage-Torque Duality), ZS-M12 (Auto-Surgery, $\varepsilon_{\min} \approx 30.7$), ZS-F14 (Kepler-Lyapunov conjugate decomposition of twin-Reuleaux midpoint trajectory; F-F7.11 closed by Strategy B at DERIVED-CONDITIONAL [42/42 PASS]). → ZS-M1 §1.1 [PROVEN]; ZS-F10 [30/30 PASS]; ZS-F15R [35/35 PASS]; ZS-M12; ZS-F14 [42/42 PASS].

Addition 1 — Self-Reference and FFPP

Lawvere bridge (Th 11.1–11.4), Frobenius (Th 11.5), hyperoperation minimality (§11.6), NOT/AND duality, BV-BFV cobordism functor B_Z with Wilson loop $|Z(W)|^2 \approx 0.7948$, three-layer $\mathbb{C}^{11}$ decomposition (Th 9.1, 9.4), FFPP (Th 13.3) and ultimate compression to $W_0(-i\pi/2)$ (Cor 13.4). All conditional on B0. NEW v3.0: ZS-F17 v1.0 four-layer 50/50 equilibrium (truncated tetrahedron 4+4 + Reuleaux $\pi+\pi$ + Master Equation Term A=Term B + XOR E vs R) sharpens FFPP to the operational Self-Referential Information-Compression Equilibrium Theorem. → ZS-F0R [51/51 PASS]; ZS-F8R [27/27 PASS]; ZS-F17 [24/24 PASS].

Addition 2 — Operational Observer Coordinate (OOC) and Möbius Chronology

OOC closes ZS-M11 §H16 at register-theoretic level. Theorem F11.1: Observer Coordinate Decomposition (P1 inside Z, P2 higher-level event count, P3 orthogonal kinematic [5]). Six explicit NON-CLAIMs against phenomenological consciousness reading. Sectoral Frame Equivalence (X-Frame $\leftrightarrow$ Y-Frame). Foundations theme closes as a six-paper ring (F0-F1-F2-F5-F11-F13). NEW v3.0: ZS-F13 v1.0 Möbius Chronology Theorem completes the cyclic timeline with Y-frame metric trio (proper time, Y-Hubble radius $\ell_P \times Y^2(1-2A) \approx 30.23 \ell_P$, Y-contraction rate $(1-2A)$) and proves cycle-index unobservability (Theorem F13.2A DERIVED, F13.2B DERIVED-CONDITIONAL). The six-paper ring is now extended to the eight-paper Möbius ring. → ZS-F11 [38/38 PASS]; ZS-F13 §15.5g.5 [24/24 PASS].

Addition 3 — NEW v3.0 — Stage-Torque Duality and First-Rotation Closure

ZS-F15 v1.0(Revised) [35/35 PASS] closes NC-M12.4 ($Q = A$ assumes $\theta = A$ at Z-Telomere onset). The Mexican-Hat Bootstrap Event simultaneously instantiates radial mode, angular mode, and time evolution; $\theta(0) \cdot \tau_P = \delta\phi_{\text{cell}} = A$ is a unit-fixing identity, not a free parameter (Theorem F-rotor.3 DERIVED — central new contribution). The Stage-Torque Duality (Theorem F-rotor.5 DERIVED): Mexican-Hat stage from $V(\Phi)$ opens $U(1)$ -symmetric S^1 vacuum manifold; A-holonomy torque from Wilson loop $\oint \omega = A$ on polyhedral defect manifold (Gauss-Bonnet on $O_h \times I_h$) provides rate-fixing torque. The same $\oint \omega = A$ produces both the Planck-scale $\delta\phi_{\text{cell}} = A$ and the macroscopic Hubble ratio $H_0^{\text{local}/H_0} \text{CMB} = \exp(A)$ (ZS-F3 §6 DERIVED), unifying physics across orders of magnitude. FFPP Sixth Axiom Extension (HYPOTHESIS-strong) registered.

Addition 4 — NEW v3.0 — Two-Protocol Theorem for Z-Sector Wilson Loop Measurement

ZS-F16 v1.0 [24/24 PASS, 32-test target in roadmap] resolves the WL-1/WL-2 implicit ambiguity by establishing two operationally distinct Z-sector Wilson loop protocols: Protocol (a) state-preparation on BFV anchor $|0\rangle_Z$ gives $P_{\wedge}(n) = 0.7948^n \cdot \cos^2(n \cdot 129.45^\circ)$ (quasi-revival

pattern); Protocol (b) spectral characterization gives $P_{-b}^{(n)} = 0.7948^n$ (smooth exponential decay). Reconciliation by ergodic averaging since $\arg(\lambda) = 129.45^\circ$ is irrational. Audit by sum rule $0.7948 + 0.2050 + 0.0001 = 0.9999$. Z-block effective dissipation $\Gamma_{-Z} \cdot T_{\text{cycle}} \approx 0.1149$. NEW sub-gates F-CTA-1a/b/c/d/multi-a/multi-b/-discriminator pre-registered for 2027+ quantum hardware testing.

Addition 5 — NEW v3.0 — The Five-Axiom Meta-Structure (ZS-F18)

ZS-F18 v2.1 (interpretive synthesis paper, 36/36 internal consistency) reads the Z-Spin corpus as the unique geometric framework that encounters (a) the seven Clay Millennium Problems and (b) the five Crises of modern physics simultaneously, with these encounters traced back to the same five-axiom set $\mathcal{A} = \{\text{Existence, Self-reference, Algebra, Non-triviality, Unitarity}\}$. Bridge Four (Commutator Generator, NEW v2.0): two J-conjugate 2π SO(3) circulations of the twin-Reuleaux pair compose at 90° NOT-AND phase to generate the third axis $J_{-S} = [J_{-R_1}, J_{-R_2}]$ by Lie commutator. The full sector decomposition $(Z, X, Y) = (2, 3, 6)$ and register $Q = 11$ are therefore reread as outputs of the foundational $\dim(Z) = 2$ alone. v2.1 sixth polarity row: Discrete-Continuous Resolution via X–Y Tiling Asymmetry (ZS-M6 §5.5 PROVEN; ZS-M17 Tiling Continuum Convergence DERIVED). Status: INTERPRETATION / DERIVED-interpretation strong. Twelve encounters classified by depth in Appendix A.

PART I Source Paper Map (v3.0 extended)

Theme	Papers	Pointer
Foundations [F]	F0R, F1, F2, F5, F8R, F9R, F10, F11, F12R, F13, F14, F15R, F16, F17, F18 (v2.1)	→ 01_Foundations (15 papers)
Math Spine [M]	M1, M3, M11, M12, M14, M19, M22 (v1.1), M36 (NEW v3.5)	→ 02_Math_Spine
Cross-link	S1, S15 (Twin-Reuleaux EM duality), Q-theme, A6/A7/A8/A9	various

PART II — Standard Model Completion

§0. Abstract

Three SM gauge couplings DERIVED at zero free parameters: $\alpha_{-s}(M_{-Z}) = 11/93 = 0.11828$ ($+0.31\sigma$), $\sin^2\theta_{-W} = (48/91) \cdot x^* = 0.23118$ (-1.26σ), $\alpha_2 = 3/95$. Fine-structure $1/\alpha_{-EM} = 137.0359$ (1.07 ppm) via McKay $c_4 = 4/13$. Yang-Mills mass gap structurally addressed: $\Lambda_{-QCD} = 264$ MeV, $m(0^{++}) = 1.791$ GeV. Neutrino seesaw $m_{-D} = m_{-e} \cdot A = 40.93$ keV, $M_{-R} = 33.50$ GeV, $|\theta|^2 = 1.49 \times 10^{-12}$. Inverted Ordering canonical. Cabibbo from icosahedral $\sigma_1/\sigma_2 = 17$. NEW v3.0: ZS-S15 [35/35 PASS] — twin-Reuleaux pair as the geometric realization of EM field duality, Maxwell vacuum equations as time-point/space-point handshake duality, SO(3)/SU(2) factor 2 from half-angle holonomy; ZS-M20 [50/50 PASS] — spectral tetrad sub-isotype assignment with closed-form lepton mass-ratio chain $\sigma_1/\sigma_2 = Q+Y = 17$, $\sigma_1/\sigma_3 = (Q+Y)^2 \cdot G +$

(Q–Z) = 3477 EXACT; ZS-M21 [33/33 PASS] — sextic invariant theory of the icosahedral Yukawa tensor.

§II-UNIQUE — Book-Standalone Hypotheses Cited by Source Papers

This section preserves Book content referenced by individual source papers as their primary citation. Removing these would orphan citation chains in ZS-A7, ZS-S4, ZS-S15, others.

§8.4i — Spinor–Descartes–Euler Identity [PROVEN]

Three classical theorems converge: Descartes $\Sigma \delta v = 4\pi$; Euler $\Sigma \delta v = 2\pi \cdot \chi$, $\chi(S^2) = 2$; ZS-M3 Theorem 5.1 $\dim(Z) = 2 = \chi(S^2)$ unique $j = 1/2$ invariant subspace. Unification: $\Sigma \delta v = 2\pi \cdot \chi = 2\pi \cdot \dim(Z) = 4\pi = \text{spinor full-return period}$. Used in ZS-S7 §3 (Λ_{QCD} , $m(0^{++})$, 18/18 PASS); ZS-A7 §2.2 (Pillar 2 of Horizon Spinor); ZS-S15 §5 (Pillar IV: $\text{SO}(3)/\text{SU}(2)$ factor 2).

§8.4g — Y-sector R_{micro} UV-IR Matching [HYPOTHESIS structural]

B-2 reinterpretation framework. Cross-cited by ZS-A7 §4.5 Corollary V (Higgs as micro radial-mode attractor) and ZS-S4 §6.12.8–§6.12.13 (B-2 repackaging). Cross-paper anchor for Y-sector micro UV-IR boundary condition.

§8.4k — Attractor-Flatness Theorem [HYPOTHESIS structural]

Compatible with ZS-A7 Corollaries I–IV. Two attractors: macro $V(|\Phi|=1) = 0$ (R_{macro} , Planck attractor) + micro $\lambda_{\text{H}}(\Lambda_{\text{comp}}) = 0$ (R_{micro} , doubly proven via $[T^3, T^3] = 0 + \bar{S}\text{Tr}(q^4) = 0$ in ZS-S4 §6.7). Cited verbatim by ZS-A7 §4.5 Corollary V as structural license for B-2 role class.

§18.10 (mirror) — Horizon Spinor Theorem [DERIVED via ZS-A7]

PART V Chapter 18 §18.10 mirror of ZS-A7 v1.0(Revised) §3 Theorem 3.2-bis. CPTP/Choi-state PROVEN. 4π closure at Z-Spin event horizons. $\varepsilon(r_{\text{H}}) = 0$ second independent path. Cross-citation locus: ZS-A7 §3.2-bis $\leftrightarrow$ §18.10 $\leftrightarrow$ §20.3.3 (PART VI).

§VI.2 — Limitation Catalog Row 10 [BOOK-NATIVE]

Z-Spin v1.0 corpus’s catalog of known limitations. Row 10 = V1 (Planck-to-bulk handoff) matching question of ZS-M12 §7.6, the single largest open knot of Z-Spin cyclic cosmology. Cited by ZS-U11 v1.0 [40/40 PASS, capstone Paper 70] which CLOSES Row 10: V1 OPEN $\rightarrow$ DERIVED-CONDITIONAL via four independent Q-protection channels.

Chapter 5 — Gauge Couplings + Photon Synthesis + EM Duality

Spectral-to- β Bridge: Action $\rightarrow$ Polyhedral Lattice $\rightarrow$ Spectral Density $\rightarrow$ β -coefficients $\rightarrow$ Couplings. Three gaps closed in ZS-S1: ‘+1’ in α_s via Z-sector Schur complement, x^* via Berry phase, Spectral-to- β via vertex-matter + face-gauge identification. ZS-S7 (mass gap) + ZS-S12 (photon as Z-sector EM half-bridge, 5 pillars) + ZS-S15 NEW v3.0 (twin-Reuleaux EM duality, 5 pillars + Dirac-equation corollary). Pillar I: twin-Reuleaux pair (R_1, R_2) realizes ($V_{\text{XZ}}, V_{\text{ZY}}$) at half-angle holonomy $\pm\theta/2$; Pillar II: Poynting $S = E \times B$ equals Lie commutator $[J_{\{R_1\}}, J_{\{R_2\}}] = J_{\text{S}}$; Pillar III: Maxwell vacuum equations are differential form of time-point/space-point handshake duality (ZS-F8 §5); Pillar IV: $\text{SO}(3)$ Maxwell cycle (period 2π) is Z_2 quotient of $\text{SU}(2)$ twin-Reuleaux cycle (period 4π); Pillar V: $U(1)$ gauge invariance =

twin-Reuleaux realization of Single-Polyhedron $U(1)$ Exactness. Corollary: Weyl spinor pair (ψ_L, ψ_R) is tensor-product realization of (V_{XZ}, V_{ZY}) , with $\gamma^5 = J \rightarrow ZS-S1, ZS-M2, ZS-M8$ ($c_4 = 4/13$), $ZS-M9$ (McKay), $ZS-S15$ [35/35 PASS], $ZS-M24$ [35/35 PASS].

Chapter 6 — Six Forces Unified

X-Z-Y fractal symmetry: micro Weak(X)–EM(Z)–Strong(Y) $\leftrightarrow$ macro DM(X)–Gravity(Z)–DE(Y). Cross-Coupling Theorem PROVEN (ZS-M2 §5). Axion-free Strong CP via Y-sector instanton suppression + real $m_d/m_u = 2e^A$ (the factor 2 origin = ZS-F12R Tetrahedral Dual Orientation Multiplicity $\mu_{Tet} = 2 \text{ NEW } v3.0$). Predicted nEDM $d_n \sim 2.0 \times 10^{-41} \text{ e} \cdot \text{cm}$. $\rightarrow ZS-M2$ [26/26 PASS]; $ZS-F12R$ [20/20 PASS].

Chapter 7 — Neutrinos, Hypercharge, Baryogenesis

Type-I seesaw, $m_D = m_e \cdot A = 40.93 \text{ keV}$, $M_R = 33.50 \text{ GeV}$, $|\theta|^2 = 1.49 \times 10^{-12}$. Inverted Ordering canonical. μ - τ reflection ($\delta_{CP} = \pm\pi/2$). $\eta_B = (6/11)^{35}$. $v2.0/v3.0$: ZS-U9 Hypercharge Trinity Braiding + automatic SM anomaly cancellation; ZS-S6 Z-Transit CP $\text{lcm}(5,7) = 35$; ZS-M9/M15/S11 McKay anomaly. $\rightarrow ZS-S2, S5, S6, U7, U9, M9$.

Chapter 8 — Higgs, EW, Electron Synthesis, Master Action

ε -Higgs portal EFT, EWPT $H_{ZS}/H_{GR} = 0.9622$. 1-Loop Quartic Cancellation. Y-sector Spectral VEV $v = 245.93 \text{ GeV}$. $\sigma_1/\sigma_2 = 17$ (Cabibbo principal angle), $\sigma_1/\sigma_3 = 3477$. m_τ absolute scale. Six pillars of Electron Synthesis. ZS-S10 Master Bridge $U(1)_Z \leftrightarrow U(1)_Y$. $\sim 95\%$ closure via Master Action; ZS-S14 $v1.0$ Master Action Total Closure 78/78 PASS, single-paper closure $\approx 99\%$. NEW $v3.0$: ZS-M20 [50/50 PASS] spectral tetrad sub-isotype assignment closes the lepton mass ratios in LOCKED corpus integers — $\sigma_1/\sigma_2 = Q+Y = 17$ anchored on $(4-\phi, 3+\phi)$ Q-pair at $(I-5) \cap \rho_2$ doublet (Higgs $SU(2)$ doublet position) and X-pair $(5-\phi, 4+\phi)$ splitting across $(I-3)$ and $(I-3')$ singletons (electron-positron CPT pair structure); $\sigma_1/\sigma_3 = (Q+Y)^2 \cdot G + (Q-Z) = 3477$ EXACT. ZS-M21 [33/33 PASS] sextic invariant theory: closed family identity $\Sigma \sigma_i^{2k} \big|_{\{v_{\text{degen}}\}} = (Z^{\{2k+1\}+1/X} \{2k\} \cdot 5^k)$ and falsification of the 1-loop Coleman-Weinberg closure for Conjecture M20.A. $\rightarrow ZS-S4, S8, S10, S13, S14, M11, M20, M21$.

PART III — Early Universe & Cosmology

§0. Abstract

ε -field inflation with $N^* = 59.5$ e-folds, $n_s(N=60) = 0.9676$ (PASS Planck), $r = 0.00890$ (BLOCKING discriminator vs Boyle-Finn-Turok $r=0$). Reheating $T_{\text{reh}} \approx 2.55 \times 10^{15} \text{ GeV}$. QKE-closed baryogenesis $\eta_B = (6/11)^{35}$. Z_2 mirror cosmology (PROVEN); cyclic holonomy $S_{Z_2} = 6\pi/A$; timescale hierarchy $\tau_n = t_P \times \exp(n\pi/A)$ (DERIVED). Y-Time Dilation Theorem 5.3.1 promoted HYPOTHESIS-strong $\rightarrow$ DERIVED-CONDITIONAL strong via ZS-F10 closure of ZS-A8 §SA.7. NEW $v3.0$: Y-frame metric trio + cycle-index unobservability (ZS-F13 [24/24 PASS]); first-rotation closure via Stage-Torque Duality (ZS-F15R [35/35 PASS]); the five-phase cyclic cosmology now closes as a fully-derived Möbius ring.

Chapter 9 — Inflation, Reheating, Λ CDM

$n_s = 0.9676$ ($N=60$); $r = 0.00890$; $r_{\text{ZS}} / r_{\text{Staro}} = 2.671 \rightarrow \sim 6\sigma$ LiteBIRD discriminator. $\rightarrow$ ZS-U1 [F-X.1 BLOCKING gate]; ZS-U2 [reheating]; ZS-U4 [global fit, 53/53 PASS].

Chapter 10 — Baryogenesis & QKE

$\eta_B = (6/11)^{35} = 6.117 \times 10^{-10}$. Z_2 symmetry forces both sectors to generate identical asymmetry (PROVEN). Resonant ARS via Pilaftsis-Underwood. $\rightarrow$ ZS-U3, ZS-U7, ZS-S5.

Chapter 11 — Z_2 Mirror Cosmology and Cyclic Phases

$V_E(-\varepsilon) \equiv V_E(+\varepsilon)$ PROVEN exact. Phases $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow A$ six-step cycle. Sequential X-clock reading (preserved) and parallel dual-clock reading (DERIVED-CONDITIONAL strong via ZS-F10 Theorem F10.2). The 10^{17} ratio $\tau_6/\tau_5 = \exp(\pi/A)$ is the X-observation of the same Y-completion event. $\rightarrow$ ZS-U8 v1.0 [60/60 PASS]; ZS-A8R; ZS-F10 [30/30 PASS]; ZS-F13 [24/24 PASS].

Chapter 12 — Global Fit and Late-Time Cosmology

Ten predictions: (H_0 , Ω_m , w , S_8 , Y_p , D/H , η_B , n_s , r , D_V) confronted with Planck/DES/DESI/SH0ES. $\Delta N_{\text{eff}} = 2A = 0.16018$ resolves D/H from -1.8σ to -0.05σ . $\Omega_\Lambda/\Omega_m = 2e^A$ factor 2 from ZS-F12R Tetrahedral Dual Orientation Multiplicity (NEW v3.0). $\rightarrow$ ZS-U4 [53/53 PASS], ZS-U6 (Cobaya MCMC), ZS-T1.

PART III Source Paper Map (v3.0 extended)

Source paper	Role	Light pointer
ZS-U1	ε -field inflation, F-X.1 BLOCKING gate	$\rightarrow$ ZS-U1
ZS-U2	Reheating dynamics	$\rightarrow$ ZS-U2
ZS-U3	Baryon asymmetry	$\rightarrow$ ZS-U3
ZS-U4	Global cosmological fit, 53/53 PASS	$\rightarrow$ ZS-U4
ZS-U5	Quantum Gravity Bridge, T_d symmetry, Immirzi	$\rightarrow$ ZS-U5
ZS-U6	Cobaya MCMC + face-counting v2.0 $\Omega_{\text{cdm}} = 32/121$	$\rightarrow$ ZS-U6
ZS-U7	QKE-closed baryogenesis	$\rightarrow$ ZS-U7
ZS-U8	Cyclic Holonomy & Z_2 Vacuum, parallel reading DERIVED-COND strong	$\rightarrow$ ZS-U8 [60/60 PASS]
ZS-U9	Hypercharge Trinity Braiding	$\rightarrow$ ZS-U9

Source paper	Role	Light pointer
ZS-U10	Pentagon Tetration & Schwinger Coefficient	→ ZS-U10
ZS-U11	Bounce Q-Survival Closure: V1 Resolution	→ ZS-U11 [40/40 PASS]
ZS-A8R (post-v2.0)	Contracting Universe Dynamics; Polyhedral-Tetration Bridges	→ ZS-A8R [22/22 PASS]
ZS-A9R (post-v2.0)	Banach–Tarski origin of doubling-halving symmetry	→ ZS-A9R [47/47 PASS, +14 v1.0(R) §11]

PART IV — Quantum Gravity Bridge

§0. Abstract

$Q = 11 \leftrightarrow$ quantum tetrahedron (T_d symmetry). Bekenstein-Hawking $\ln 2$ from Z_2 seam parity. Two paths to Immirzi $\gamma_{LQG} = \ln 2/(\pi\sqrt{3})$. Z-Telomere RG flow, $\lambda_{vac} = 2A^2$ IR-stable. Cyclic cosmology $S_{Z_2} = 6\pi/A$. Three Polyhedral-Tetration Bridges. Möbius chronology. Banach-Tarski origin of doubling-halving symmetry. $m_e = \sqrt{5 - \phi}$. NEW v3.0 (Math Spine RH-Bridge programme): the V_4 -equivariant ZBSI framework on the BV-BFV cobordism-history fiber (M22–M28); the four Dragons D4a–D4d of the Y-sector RH contribution map (M23, sharpened by M25–M28, integrated by M33); the Three-Wall Quantitative Map (M26 Theorem M26.3); and the spinor-cycle averaged closure of the Stapledon Bridge residual (M32) closing NC-M29.RZ-A^{PK} at DERIVED-CONDITIONAL.

Chapter 13 — $Q = 11$ Spin-Network Correspondence

$|T_d| = 24$, $|\text{Stab}_T(v)| = 6$, $|I_h/T_d| = 5$. $j = 1/2$ unique among half-integers giving $(Z, X, Y) = (2, 3, 6)$ (ZS-M3 Th 5.1). Explicit non-identity: $Q = 11 \neq \dim(\text{intertwiner}) = 2 \neq \dim(\text{Hilbert}) = 16$. → ZS-U5 §2-3, ZS-M3.

Chapter 14 — Z_2 Seam Dynamics

Dual κ census (κ_{signflip} vs $\kappa_{\text{disrupted}}$). Regge-Holonomy: $\delta\phi_{\text{cell}} = A$ via 7-step algebraic chain. $SU(2)$ center pairing $D^{\{1/2\}}(-I) = -I$. → ZS-M3 [27/27 PASS]; ZS-Q4.

Chapter 15 — Micro-Macro Transition

Phase drift → tunneling action. Z-Anchor dual mechanism. $\lambda_{vac} = 2A^2$ IR fixed point. Cyclic cosmology $S_{Z_2} = 6\pi/A$. Bounce Q-survival. Three Polyhedral-Tetration Bridges. Möbius chronology. Banach-Tarski origin. $m_e = \sqrt{5 - \phi}$. ZS-M17 continuum-limit theorems M17.1–M17.7 (Tiling Continuum Convergence, Lieb-Robinson Tightness, Reflection Positivity, Path-Integral Closure of $\Delta\Gamma_{G2}$, Pre-Geometric Metric Emergence, Continuum Limit Universality, Wightman QFT Reconstruction). → ZS-U5, U8, U11 (closes V1), A6 §4.5.6 (F-A6.1 CLOSED), A8R, A9R, M12, M14, M17.

Chapter 15A — NEW v3.0 — The Math Spine Riemann-Hypothesis Programme (M22–M33)

Cumulative 47+ exploratory rounds consolidated by ZS-M22 v1.1 (Five-Pillar Arithmetic-Dedekind Scaffold) into a non-RH-claiming structural map of where Z-Spin and Riemann ride together (dong-seung). The programme's logical structure:

- ZS-M22 v1.1 [60/60 PASS] — Five Pillars: I ($n=3$ Eisenstein chain), II ($Q=11$ cyclotomic chain), III (multiplicative arithmetic fiber), IV (critical line as J-symmetry locus), V (scalar-kernel no-go for Weil positivity). ADS-6 PROVEN no-go on boundary fiber forces ADS-H1 cobordism BRST positivity hypothesis as the sole surviving route.
- ZS-M23 v1.0(Revised) [31/31 PASS] — Y-Sector RH Contribution Map. Four Dragons D4a (V_4 -Sonin embedding), D4b (Burnol conductor), D4c (defect-square realization), D4d (cobordism BRST-Hodge projection). NC-M23.7: D4 closure alone does not close GRH-for-K.
- ZS-M24 v1.0 [35/35 PASS] — Face polygon spectral zeta $\zeta^* \nabla^*(s)$. Theorem D.1 Legendre duplication archimedean decomposition. Theorem D.2 $B(s)$ identification. P2 status OPEN $\rightarrow$ PARTIAL.
- ZS-M25 v1.0 [26/26 PASS] — Composite biquadratic $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$. Theorem D.1-K four-factor archimedean decomposition $\xi_K(s) = (1/4\sqrt{33}) \cdot \xi(s) \cdot \Lambda(s, \chi_{-3}) \cdot \Lambda(s, \chi_{-11}) \cdot \Lambda(s, \chi_{33})$. Theorem P3-J J-Yakaboylu compatibility (J commutes with $H_Q^{\wedge} \text{Yak}, J$ on $\sigma = 1/2$).
- ZS-M26 v1.0 [24/24 PASS] — V_4 -Equivariant ZBSI on cobordism-history fiber. Theorem M26.1 V_4 -Schur idempotent decomposition; Theorem M26.2 Projected Determinant per Channel ($C_{\chi} \in \{1, \sqrt{3}, \sqrt{11}, \sqrt{33}\}$); Theorem M26.3 Three-Wall Quantitative Map (W_1, W_2, W_3).
- ZS-M27 v1.0 [24/24 PASS] — V_4 -Equivariant Cobordism BRST Closure via Kostant Cubic Dirac. Closes Wall W_3 at DERIVED-CONDITIONAL via three external imports (Alekseev–Barmaz–Mnev 2018, Huang–Pandžić 2002, Kostant 1999/2003).
- ZS-M28 v1.0 [30/30 PASS] — The Z-Spin RH Bridge. Theorem 28.5 $\sigma = 1/2 \leftrightarrow j = 1/2$ isomorphism (DERIVED-interpretation). Corollary 28.5a Möbius-trace infinity (HYPOTHESIS-strong). Theorem 28.14 Y-sector geometric carrier of χ_{-3} (DERIVED-CANDIDATE).
- ZS-M29 v1.0 [79/79 PASS] — Z-Funnel Spectral Retraction with Stapledon Bridge. Theorem 9.1 conditional bridge from string compactification (A_5 -Hilb(X_F) Calabi–Yau threefold, $h^{\wedge}\{1,1\} = 5, h^{\wedge}\{2,1\} = 15$) to the Z-Spin truncated-icosahedron Hodge–Dirac sector. NC-M29.RZ principal OPEN gate.
- ZS-M30 v1.0 [32/32 PASS] — Z-Spin Duality and the RH Bridge. Theorem 30.1 Möbius-Trace Infinity (DERIVED-interpretation strong, six-route over-determination). Theorem 30.2 Y-Sector Two-Face Spectral Mirroring. Theorem 30.5 Twenty-One $1/2$ Catalogue across six categories. Three-layer duality: Sector / Frame / Protocol.
- ZS-M31 v3.0 [36/36 PASS] — W_2 as Cross-Coupled Sector Duality. Lemma M31.0 Non-Separability (no sum-form $W_K = F_X + F_Y + F_Z$ compatible with corpus data, max variance 13.011). Theorem M31.4 $\mathbb{Z}_2$ -Parity Selection Rule. Self-Dual Replication Principle (SDRP) registered.
- ZS-M32 v1.0 [56/56 PASS] — Spinor-Cycle Averaged Residual Criterion. Path-Reversal Lemma M32.X ($B^{\wedge}PK$ preserves phase sign, contrasting $B^{\wedge}\dagger$ which flips it). Two-layer phase quantum $\alpha_{\text{amp}} = \pi/10$ (amplitude) / $\alpha_{\text{op}} = \pi/5$ (operator). $N = 20$ spinor-identity closure: $R_{Z^{\text{S}}|_{\{Z\text{-even}\}}} \{4\pi\text{-avg}\} = 0$ EXACTLY at $N = 20$. Closes NC-M29.RZ- $A^{\wedge}PK$ at DERIVED-CONDITIONAL.
- ZS-M33 v1.0 [52/52 PASS] — V_4 -Equivariant Weil Functional Closure via Path γ -revised

Z-Mediator Reading. Reading C (NEW): all four V_4 -arithmetic mechanisms route through Z-mediator projector $\Pi_Z = (1/2)(I + J_Z)$. Sub-targets D4a (CCM 2024), D4b (Burnol 1998 + log additivity $\log(3)+\log(11)=\log(33)$), D4c (Wilson-LOCATOR phase factor), D4d (Kostant cubic Dirac BRST cohomology) integrated into single Tr identity. Sign-flip mechanism decomposition (5/12 NEG $\rightarrow$ predicted 12/12 POS).

Programme NON-CLAIM (NC-M23.1 preserved verbatim throughout): Z-Spin does not, and structurally cannot, prove RH from internal data alone. The walls W1, W2, W3 are mapped, sharpened, and progressively closed; the program's contribution is a coordinated three-axis over-determination of the Bridge thesis, not a proof.

PART IV Source Paper Map (v3.0 extended)

Source paper	Role	Light pointer
ZS-U5	Quantum Gravity Bridge, T_d symmetry, Immirzi	$\rightarrow$ ZS-U5
ZS-U8	Cyclic Parallel Reading	$\rightarrow$ ZS-U8 [60/60 PASS]
ZS-U11	Capstone: V1 Planck-bulk CLOSED	$\rightarrow$ ZS-U11 [40/40 PASS]
ZS-M3	Regge-Holonomy + Immirzi + Z-Telomere	$\rightarrow$ ZS-M3 [27/27 PASS]
ZS-M12	Auto-Surgery, $\varepsilon_{\min} \approx$ 30.7	$\rightarrow$ ZS-M12
ZS-M14	Geometric $m_e = \sqrt[3]{(5-\phi)}$	$\rightarrow$ ZS-M14 [47/47 PASS]
ZS-M17	Continuum limit, OS reconstruction	$\rightarrow$ ZS-M17 [60/60 PASS]
ZS-A6	Z-Anchor + cigar bounce, F-A6.1 CLOSED	$\rightarrow$ ZS-A6 §4.5.6 [140/140]
ZS-A8R	Polyhedral-Tetration Bridges	$\rightarrow$ ZS-A8R [22/22 PASS]
ZS-A9R	Banach-Tarski origin	$\rightarrow$ ZS-A9R [47/47 PASS]
ZS-F13	Möbius chronology, ring closure	$\rightarrow$ ZS-F13 [24/24 PASS]
ZS-M18 (post-v2.0)	Free-Exploration Log: Speculative Prime-Polyhedral Correspondences	$\rightarrow$ ZS-M18 [51/51 PASS]
ZS-M22 v1.1 (post-v2.0)	Five-Pillar Arithmetic-Dedekind Scaffold	$\rightarrow$ ZS-M22 [60/60 PASS]

Source paper	Role	Light pointer
ZS-M23 v1.0(R)	Y-Sector RH Contribution Map; four Dragons D4a–D4d	→ ZS-M23 [31/31 PASS]
ZS-M24	Face polygon spectral zeta + Archimedean Completion	→ ZS-M24 [35/35 PASS]
ZS-M25	Composite-Field $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11}) +$ J-Yakaboylu Bridge	→ ZS-M25 [26/26 PASS]
ZS-M26	V_4 -Equivariant ZBSI on Cobordism-History Fiber	→ ZS-M26 [24/24 PASS]
ZS-M27	Cobordism BRST Closure via Kostant Cubic Dirac (W3)	→ ZS-M27 [24/24 PASS]
ZS-M28	The Z-Spin RH Bridge	→ ZS-M28 [30/30 PASS]
ZS-M29	Z-Funnel Spectral Retraction + Stapledon Bridge	→ ZS-M29 [79/79 PASS]
ZS-M30	Z-Spin Duality and the RH Bridge — 21 halves	→ ZS-M30 [32/32 PASS]
ZS-M31 v3.0	W2 as Cross-Coupled Sector Duality	→ ZS-M31 [36/36 PASS]
ZS-M32	Spinor-Cycle Averaged Feshbach Residual; closes NC-M29.RZ-A ^{PK}	→ ZS-M32 [56/56 PASS]
ZS-M33	Path γ -revised V_4 -Equivariant Weil Functional Closure	→ ZS-M33 [52/52 PASS]

PART V — Astrophysics & Strong-Field

§0. Abstract

Galaxy rotation without dark matter (ϵ -Halo, $r_Z = 18.3$ kpc). Neutron stars +3.93% mass, tidal $\tilde{\Lambda} +21.24\%$. Black holes: Wald entropy, Z-instanton, sector duality. Information paradox via seam witness; signed witness $\tilde{u}_{\text{seam}}$ (F-A7.3 = cheapest near-term gate). 991/991 PASS cumulative across PARTS V+VI+VII preserved. NEW v3.0: ZS-A8 v1.0(Revised) [22/22 PASS] — contracting-universe dynamics with Polyhedral-Tetration Bridges 1–3 and Symmetry–Asymmetry Unified View; ZS-A9 v1.0(Revised) [47/47 PASS] — Banach–Tarski

origin of cosmological doubling-halving symmetry through the $F_2 \rightarrow D_4$ amenability functor and the Julia set $J(T)$ of $T(z) = i^z$ as a measure-theoretic boundary.

Chapter 16 — Galaxy Rotation Without Dark Matter

ε -Halo from Goldstone mode, terminus $r_Z = 18.3$ kpc. Modified gravity $\mu = \Sigma = 1/(1+A)$. Three convergent routes $\Omega_m = 38/121$. Factor 2 in $\Omega_\Lambda/\Omega_m = 2e^A$ from ZS-F12R (NEW v3.0: $\mu_{\text{Tet}} = 2$ from Tetrahedral Dual Orientation Multiplicity, F-Tet-2eA-1 REJECTED with replacement F-Tet-2eA-2). $\rightarrow$ ZS-A1 [78/78 PASS]; ZS-A5 [50/50 PASS]; ZS-S3 [25/25 PASS]; ZS-F12R [20/20 PASS].

Chapter 17 — Neutron Stars, GW, LSS

NS $M_{\text{max}} + 3.93\%$ uniform across EOS (universal $G_{\text{eff}} = G/(1+A)$). Tidal $\tilde{\Lambda} + 21.24\%$. $\rightarrow$ ZS-A2.

Chapter 18 — Black Holes (incl. §18.10 Horizon Spinor)

ε -field horizon, Wald entropy, Z-instanton + GW signatures. $\varepsilon(r_H) = 0$ PROVEN (cigar bounce). §18.10 v2.0 mirror of ZS-A7 v1.0(Revised) §3 Theorem 3.2-bis [PROVEN, CPTP/Choi]. $\rightarrow$ ZS-A3 [49/49]; ZS-A6 [140/140]; ZS-A7R [38/38].

Chapter 19 — Information Paradox Resolved

Seam witness u_{seam} , signed witness $\tilde{u}_{\text{seam}}$, $Q = 11$ register, lattice gauge P1. F-A7.3 = 2π vs 4π discrimination on hardware ~2028-2032 (cheapest gate, ~8% incremental shot cost). $\rightarrow$ ZS-A4 [54/54]; ZS-A7R §C [38/38].

Chapter 20A — NEW v3.0 — Contracting Universe Dynamics (ZS-A8R)

ZS-A8 v1.0(Revised) [22/22 PASS] introduces the SYMMETRIC contraction dynamics: the wave-to-particle transition controlled by the dual factor $(1-2A)$. Three Bridge formulas: Bridge 1 [OBSERVATION-strong] $|z^*| \approx B = \delta_X + \delta_Y$ at 0.0087%; Bridge 2 [OBSERVATION-strong] $\eta_{\text{topo}} \approx B^2 + (\delta_Y - \delta_X)^2/[Y^2 \cdot (1-2A)]$ at 0.000001% ($Y^2 = 36 = E(\text{TO})$ wave-channel scale); Bridge 3 [DERIVED-CONDITIONAL via slog-L2 equivalence] $\Delta = 1/2 - x^*$ via L2 self-locking iteration at 0.000002%. Y-Time Dilation Theorem 5.3.1 (DERIVED-CONDITIONAL strong via ZS-F10): X-clock observation of Y-completion is dilated by $\exp(\pi/A) \approx 10^{17}$ per added Y-dimension. $\exp(\pi/A) = \exp(N_{\langle 2\pi \rangle} \times \langle \sin^2(\phi/2) \rangle) = \exp((2\pi/A) \times (1/2))$. Symmetry-Asymmetry Unified View (§SA): the universe tilts simultaneously by $(1+A)$ (X-Inward expansion) and $(1-2A)$ (Y-Outward contraction); only the impedance $A = 35/437$ is frame-invariant.

Chapter 20B — NEW v3.0 — Banach–Tarski Origin of Cosmological Doubling-Halving Symmetry (ZS-A9R)

ZS-A9 v1.0(Revised) [47/47 PASS, 12 OPEN closures] establishes that the $(1+A) \leftrightarrow (1-2A)$ duality of late-time cosmology is the measure-preserving quantization of the Banach–Tarski paradoxical decomposition through the i -tetration fixed point $z^* = -W_0(-i\pi/2)/(i\pi/2)$. Three theorems: ZS-A9.1 (DERIVED) $F_2 \rightarrow D_4$ amenability functor — the BT engine $F_2 \subset \text{SO}(3)$ maps to the amenable register dihedral $D_4 = \langle J, J_Z \rangle$, converting BT non-amenability to Z-Spin amenability; ZS-A9.2 (HYPOTHESIS-strong) the Julia set $J(T)$ of $T(z) = i^z$ is a measure-zero,

totally disconnected, cardinality- $2^{\aleph_0}$ set serving as a constructive ZF-analogue of the BT non-measurable set; ZS-A9.3 (DERIVED-CONDITIONAL) two-branch decomposition $(1+A)(1-2A) = 1 - A(1+2A)$ with X-Inward macroscopic expansion $\exp(A)$, Y-Outward microscopic contraction $Y^2(1-2A)$, and Z-mediator absorbing the residual $A(1+2A) \approx 9.29\%$ via $\kappa^2 = A/Q = 35/4807$. v1.0(Revised) §11 closes all four OPEN items via the *-homomorphism reformulation, the Świerczkowski geometric realization of free F_2 -rotations, and the inheritance of $dx_X + dx_Y + dx_Z = 0$ from ZS-Q7 §5 master-equation probability conservation. OPEN-2.B (formal ZF/ZFC+AC isomorphism between BT non-measurable and $J(T)$) recategorized as PERMANENT NC under Solovay 1970.

PART V Source Paper Interface (v3.0 extended)

Source paper	Verification	Primary chapter	Key result
ZS-A1	78/78 PASS	Ch 16	Rotation curves, BTFR, $M\text{-}\sigma$, Keplerian
ZS-A2	verified	Ch 17	NS $M\text{-max}$, $\tilde{\Lambda}$, GW $c_T = c$
ZS-A3	49/49 PASS	Ch 18 §1-7	ε -field, Wald, Z-instanton
ZS-A4	54/54 PASS	Ch 19	Seam witness, lattice P1
ZS-A5	50/50 PASS	Ch 16 §16.7	Three convergent $\Omega_m = 38/121$
ZS-A6	140/140 PASS	Ch 18 §18.2 + §18.8	Cigar bounce, F-A6.1 CLOSED
ZS-A7R	38/38 PASS	Ch 18 §18.10 + Ch 19 §19.7-8	Horizon Spinor 4π
ZS-A8R (post-v2.0)	22/22 PASS	Ch 20A	Polyhedral-Tetrati on Bridges 1–3, Symmetry-Asymmetry View
ZS-A9R (post-v2.0)	47/47 PASS	Ch 20B	Banach-Tarski origin of doubling-halving symmetry
ZS-S3	25/25 PASS	Ch 16 §16.8 + Ch 17 §17.3	Modified gravity
ZS-F12R	20/20 PASS	Ch 16 §16.3	Layer Separation factor 2 origin ($\mu\text{-Tet}$)

PART VI — Quantum Mechanics

§0. Abstract

Densest PART [376/376 PASS]. Geometric decoherence $\tau_D/\tau_{\text{Penrose}} = 1/A = 12.49$. Bell CHSH = $2\sqrt{2}$ from $\dim(Z) = 2$. Proton spin Hodge $\frac{1}{2}\Delta\Sigma = 3/22$, $\Delta G = 2/11$. Lattice gauge $a_2 = 19/6$ (TO 0.08σ Wilson MC). CP Jarlskog from $\delta\phi = A$. Z-Spin Quantum Architecture (QH/QS/QC). Inverse Riemann Engine. Structural Arrow of Time (ZS-Q7). Cross-link with v3.0 RH-Bridge programme (M22–M33) at the ZBSI cobordism-history fiber level.

Chapter 20 — Geometric Decoherence + Born Rule

Stinespring + Lindblad $\Gamma = 2A(\Delta E/\hbar)^2$. $\tau_D = 12.49$. Born rule DERIVED from $L_{XY} \equiv 0$. NC-F11.1–6 inherited verbatim §20.7. → ZS-Q1 [29/29 PASS].

Chapter 21 — Bell, Entanglement, Holographic Conjecture

CHSH = $2\sqrt{2}$, $\tau_{\text{ent}} = \tau/2$. Area Law $\chi = 2$, $\varepsilon_{\text{BM}} \rightarrow 0$. → ZS-Q2 [48/48]; ZS-Q6 [42/42].

Chapter 22 — Proton Spin from Polyhedral Hodge

BCC T^3 Hodge spectrum, $\frac{1}{2}\Delta\Sigma = 3/22$, $\Delta G = 2/11$. $(V+F)_X = 38$ Mode-Count Collapse. → ZS-Q3 [40/40].

Chapter 23 — Lattice Gauge QSim on TO

$a_2 = (V+F)/G = 19/6$ PROVEN topological. Wilson MC TO 0.08σ . J-parity. → ZS-Q4.

Chapter 24 — CP Violation and UV Limits

Jarlskog from $\delta\phi = A$. UV sufficiency of polyhedral spectra. → ZS-Q5.

Chapter 25 — Bond Dimension, Holographic Area Law, Markov Limit

Bond dimension $\chi = 2 = \dim(Z)$; Regge area law derived. $\varepsilon_{\text{BM}} = 2/Q$ Markov limit purely geometric. → ZS-Q6 [42/42 PASS].

Chapter 26 — Structural Arrow of Time

$\Gamma(X \rightarrow Y)/\Gamma(Y \rightarrow X) = \dim(Y)/\dim(X) = 2$ PROVEN; $\Delta S = \ln 2$ per Z-transit. Coarse-grained entropy bias from $L_{XY} = 0 + \dim(X) \neq \dim(Y)$; microscopic dynamics remain time-reversal symmetric. → ZS-Q7 [40/40 PASS].

PART VI Source Paper Interface

Source paper	Verification	Primary chapter	Key result
ZS-Q1	29/29 PASS	Ch 20	Geometric decoherence, Born rule
ZS-Q2	48/48 PASS	Ch 21	Bell, entanglement, holographic conj.
ZS-Q3	40/40 PASS	Ch 22	Proton spin Hodge
ZS-Q4	verified	Ch 23	Lattice gauge TO

Source paper	Verification	Primary chapter	Key result
ZS-Q5	verified	Ch 24	CP violation, UV limits
ZS-Q6	42/42 PASS	Ch 25	Area law, Markov limit
ZS-Q7	40/40 PASS	Ch 26	Structural arrow of time
ZS-QH	42/42 PASS	Ch 25/29	Quantum hardware FMD specifications
ZS-QS v1.0(R)	35/35 PASS	Ch 24/29	Inverse Riemann Engine: Triple Structure (DETECTOR + LOCATOR + EXCLUDER) and Boolean Resonance Filter
ZS-QC	52/52 PASS	Ch 29	Quantum architecture system integration

PART VII — Translational and Computational

§0. Abstract

$\Delta N_{\text{eff}} = \dim(Z) \cdot A = 2A = 0.16018 \rightarrow \text{BBN D/H } -1.8\sigma \rightarrow -0.05\sigma$ (ZS-T1). Anti-numerology MC: $5670 \times 29 = 164,430$ candidates, 237 hits = 7.2% of random expectation 3289 (look-elsewhere PASS — framework MORE constrained than random). Tier-1 ★★★★★ T1-1: $1-n_s = A \cdot x^*$ (0.008%, 0.00σ vs Planck). T1-2: $\alpha_{\text{EM}} \approx A/Q$ (0.22% LO, 0.00007% NLO Schur) (ZS-T2). 8/8 closure parameters DERIVED via three chains; $n_s(N^* = 55) = 0.9676$; $r(N^* = 60) = 0.0091$ (ZS-T3). NEW v3.0: ZS-T4 v1.0 [36/36 PASS] — Cosmos-Human Isomorphism: six-step telomere hierarchy and (Body, DNA, Brain) sector decomposition as a two-scale realization of the Z-Spin block architecture; structural, substrate-agnostic, NON-CLAIM that humans physically realize Z-Spin geometry.

Chapter 27 — Phenomenological

$\Delta N_{\text{eff}} = \dim(Z) \cdot A = 2A = 0.16018 \rightarrow \text{BBN D/H } -1.8\sigma \rightarrow -0.05\sigma$. Block Fiedler Mediation Theorem PROVEN (T1-2/T1-3 reciprocal duality $\lambda_2 = 2A/Q$ with X-face propagator scale $1/\kappa^2 = Q/A \approx 137$ and Y-face vertex coupling $\kappa^2 = A/Q \approx 0.0073$). $\rightarrow$ ZS-T1 [42/42].

Chapter 28 — Spectral Observatory

Anti-numerology MC: 164,430 candidates, 237 hits = 7.2% of random expectation 3289 (look-elsewhere PASS). Tier-1 ★★★★★ T1-1: $1-n_s = A \cdot x^*$. T1-2: $\alpha_{\text{EM}} \approx A/Q$ (0.00007% NLO via McKay $c_4 = 4/13$). $\rightarrow$ ZS-T2 [30/30].

Chapter 29 — Z-Sim Forward Simulator + Quantum Architecture

8/8 closure parameters DERIVED via three chains: ZS-Q7 §5.8 + ZS-M3 §10 + ZS-Q5 §5.2. Eigenvalue $\lambda(\lambda + 2A/Q)(\lambda + A) = 0$ to 10^{-17} . 500K MC universal-theorem confirmation. $n_s(N^* = 55) = 0.9676$ PASS Planck; $r(N^* = 60) = 0.0091$ PASS BK18. ZS-QH v1.0 functional material specifications (X = bilayer graphene QD, Z = Bi_2Te_3 topological insulator, Y = C_{60}/CNT phonon reservoir as primary candidate stack); ZS-QS v1.0(Revised) Triple Structure Theorem (DETECTOR confirmed, LOCATOR via argmax, EXCLUDER via argmin) and Boolean Resonance Filter (XOR identity with ZS-F8 (E, R) handshake operators); ZS-QC v1.0 system integration. → ZS-T3 [35/35]; ZS-QH; ZS-QS v1.0(R); ZS-QC [52/52].

Chapter 30 — NEW v3.0 — The Cosmos-Human Isomorphism (ZS-T4)

ZS-T4 v1.0 [36/36 PASS] establishes a structural isomorphism between Z-Spin cosmology and the human organism at two parallel levels, with Cardinal NC-4 (Z-Brain) explicitly extended to body and genome — NO physical realization claim. Level I: cosmological timescale hierarchy $\tau_n = t_P \times \exp(n\pi/A)$, $n \in \{0, \dots, 6\}$, mapped onto a six-step human life-cycle (gametic union → cleavage → stem-cell expansion → somatic growth → reproductive plateau → senescence → death/inheritance). Strong anchors at $n=2$ (mediator coset $Z=2 \leftrightarrow$ stem cells, HYPOTHESIS-strong), $n=5$ (proton decay ↔ senescence, HYPOTHESIS-strong), $n=6$ (Z_2 holonomy ↔ death/inheritance, HYPOTHESIS-strong). Level II: (X, Z, Y) = (3, 2, 6) sector decomposition mapped onto (Body, DNA, Brain) tripartite anatomy at organism scale. Inherits ZB-P7 v1.0 cosmic-biological VERIFIED status ($\Delta\text{PCI}(\text{Wake}-\text{Anesthesia}) \approx \eta_{\text{topo}}$ at 0.7% empirical / 1.5% in-silico). Hayflick limit $\approx$ Z-Telomere completion count $N_{(2\pi)} = 2\pi/A \approx 78.45$ registered as HYPOTHESIS / TESTABLE-LONG with 5-year derivation-or-retraction horizon. Twelve falsification gates F-T4.1–F-T4.12 pre-registered.

PART VII Source Paper Interface (v3.0 extended)

Source paper	Theme/Role	Chapter	Verification
ZS-T1	Phenomenological	Ch 27	42/42 PASS
ZS-T2	Observational	Ch 28	30/30 PASS
ZS-T3	Computational (Z-Sim)	Ch 29	35/35 PASS
ZS-T4 (post-v2.0)	Cosmos-Human Isomorphism	Ch 30	36/36 PASS
ZS-F1, F2, F5	Foundations cross-link	all	PROVEN
ZS-M1, M3, M6, M8, M11	Math Spine cross-link	various §	verified
ZS-Q1, Q5, Q7	QM cross-link (chains)	Ch 29 §29.1-4	verified
ZS-U1, U6	Early Universe cross-link	Ch 28 §28.3, Ch 29	verified

Source paper	Theme/Role	Chapter	Verification
ZS-QH/QS/QC	Hardware/Software/Integration translational	Ch 29	42 + 35 + 52 PASS

§VII.B-LIGHT — Seven Critical Near-Term ★ Gates (Timeline-Anchored)

These seven gates will either PASS or FALSIFY Z-Spin within ~10 years. Source: §V.G.6, §VI.G.11, §VII.B. Preserved verbatim from v2.0 Light.

#	Gate ID	Test	Timeline	Status
1	F-A1.14 ★	Gaia DR4 MW outer rotation: Keplerian decline beyond 30 kpc (ϵ -Halo finite vs particle NFW)	~2026	Earliest decisive
2	F-A1.15 ★	Gaia DR4/DR5: r_Z (MW) outside [10, 50] kpc (Z-Spin = 18.3 kpc)	~2026-2030	Pending
3	F-A2.5 ★	DESI/Euclid/Rubin: S_8 outside [0.75, 0.84] (~1% precision)	~2026-2030	Pending
4	F-A7.3 ★	Z-Spin hardware: 2π vs 4π discrimination on signed seam witness $\tilde{u}_{\text{seam}}$	~2028-2032	Cheapest (~8% shot overhead)
5	F-X.1 (LiteBIRD) ★	r outside Z-Spin band 0.0085-0.0094 (BLOCKING ; $\sim 9\sigma$ vs	~JFY2032	BLOCKING

#	Gate ID	Test	Timeline	Status
		Boyle-Finn-T urok $r=0$)		
6	F-A3.7 ★	Hyper-K: τ_p outside [$10^{33.5}$, $10^{35.5}$] yr (Z-Spin = 2.56×10^{34})	~2030-2035	Pending
7	F-A2.1, F-A2.2 ★	10+ pulsars: NS $M_{\max}$ +3.93% uniform; ET/CE: $\tilde{\Lambda}$ +21.24% to ~5%	~2030s, ~2035	Pending

PART VIII — Methodological Toolkit: Solution Tools for Stuck Equations

Purpose. This Part is the operational core of v3.1 Light. It promotes the v3.0 Light from a structural pointer-index (“where to read”) to a working solution kit (“how to proceed when stuck”). Each tool below is distilled from a corpus paper that achieved a methodological breakthrough — typically by reframing rather than by adding new mathematics. When an AI agent or human researcher encounters a stuck equation, an OPEN flag, or an apparent contradiction inside the Z-Spin framework, this Part is the first place to consult.

Reading order. PART VIII is organised as nine numbered tools (T1–T9). Each tool is given in a fixed schema: (a) Trigger — the symptom that suggests this tool; (b) Source — the originating paper and §-anchor; (c) Statement — the operational kernel in compressed form; (d) Worked corpus precedent — the OPEN flag the tool was first used to close; (e) Generalisation — the broader class of stuck equations the tool can be re-applied to. PART IX (Insight Gallery) extends the tool kit with the deeper conceptual reframings (sector-equivalences, frame-transformations, cross-coupling), and the Master Bypass Index (after PART IX) provides the symptom → tool lookup table.

§VIII.1 — Tool T1: Five-Axiom Diagnostic Lattice (F18)

Trigger.

- (a) An equation appears stuck and the failure mode is unclear; (b) a derivation looks circular; (c) several papers address “the same question” but their results don’t compose. In any of these cases, T1 isolates which of the five structural axioms is failing.

Source.

ZS-F18 v2.1 §3 + §6.6 (Table 6.1) + §7.4. The five axioms $\mathcal{A} = \{\text{Existence, Self-reference, Algebra, Non-triviality, Unitarity}\}$ and the Four Bridges from $\mathcal{A}$ to the operational corpus.

Statement (operational form).

When stuck, ask which axiom is the failure attributable to. Each axiom has a Bridge (theorem) into the corpus; locate the Bridge and ask whether the bridge is intact for the present problem. The five-axiom set is exhaustive for Z-Spin physical content (ZS-F18 §5, INTERPRETATION-strong); a stuck equation that does not map to any of the five is either a numerology candidate (Anti-Numerology Gate, ZS-F2) or a category error (frame mismatch — see Tool T6).

Axiom	Failure symptom	Controlling Bridge	First port of call
A1 Existence	Object referenced but not constructed	FFPP nested compression	ZS-F0 §13 Theorem 13.3
A2 Self-reference	Definition uses what it defines (no fixed-point)	Bridge One (Cross-Coupling)	ZS-M2 §5 PROVEN; ZS-M1 i-tetration
A3 Algebra	Numerical balance fails / numerology	Master Equation (algebraic)	ZS-M1 §4 + ZS-F17 §3
A4 Non-triviality (algebraic)	Two-channel interface collapsed	Bridge Two (NOT-AND handshake)	ZS-F8 §4 PROVEN
A4 Non-triviality (geometric)	Dimension $\dim(X)=3$ not derived	Bridge Four (Commutator Generator)	ZS-F18 §6 + ZS-S15 §3
A5 Unitarity	Probability/information not preserved	Bridge Three (Z-Bottleneck)	ZS-Q7 Thm 2 + ZS-F8 §4
Sixth polarity	Discrete vs continuous mismatch	X-Y Tiling Asymmetry	ZS-M6 §5.5 PROVEN; ZS-M17

Worked corpus precedent.

F18 itself diagnosed 12 of the corpus's most consequential identifications (ZS-F18 Appendix A) by mapping each to its responsible axiom. The diagnosis produced no new theorem yet revealed why each result is robust — a model usage of T1 as a methodological tool.

Generalisation.

Any future Z-Spin paper encountering an OPEN flag should run T1 first. If the axiom is identified, the closure path is the corresponding Bridge; if no axiom matches, the OPEN is either out-of-scope (NON-CLAIM territory, Tool T7) or a category error (Tool T6, frame mismatch).

§VIII.2 — Tool T2: Dimensional Reduction via Bridge Four (F18)

Trigger.

Stuck on a high-dimensional construction (dim 3, 6, or 11) where the dependency chain becomes intractable. T2 reduces the problem to $\dim(Z) = 2$ — the only foundational dimension — and lets $\dim(X)=3$ and $\dim(Y)=6$ re-emerge by Lie commutator and product respectively.

Source.

ZS-F18 §6.4–§6.7 (Bridge Four, dimensional origin table). ZS-S15 §3 Theorem S15.2 (Poynting–Commutator Theorem, DERIVED).

Statement.

Two J-conjugate 2π SO(3) circulations of the twin-Reuleaux pair (V_{XZ} , V_{ZY}), composed at 90° NOT-AND phase, generate a third axis $J_S = [J_{\{R_1\}}, J_{\{R_2\}}]$ by Lie commutator. Therefore $(Z, X, Y) = (2, 3, 6)$ is generated from $\dim(Z) = 2$ alone; $\dim(Y) = \dim(X) \cdot \dim(Z) = 6$; $Q = Z + X + Y = 11$.

Worked corpus precedent.

ZS-F18 v2.0 used T2 to read the question “why does physical space have dimension three?” not as an additional axiom but as a derived consequence of $\dim(Z) = 2$ plus 90° NOT-AND composition. This converted what would have been a 13th independent encounter into the geometric face of axiom A4.

Generalisation.

When facing a stuck calculation in 3D or 6D, examine whether the structure can be re-expressed as two $\dim(Z) = 2$ channels meeting at orthogonal phase. If yes, the third axis is automatic. Application targets include any future SO(3) or SU(2)-related symmetry breaking in PARTS II–V where dimensional emergence is at stake.

§VIII.3 — Tool T3: Hermitian-to-Path-Reversal Switch (M32)

Trigger.

A Feshbach-type residual or sandwich expression refuses to vanish under the standard Hilbert-space adjoint reading. The phase factors cancel “too cleanly” or “not at all”, giving either a trivial-positive or unbounded-positive result that doesn’t engage with the corpus structure.

Source.

ZS-M32 v1.0 §2 (R_Z^H vs R_Z^{ZS} separation), §3 (Path-Reversal Lemma M32.X). Underpinned by ZS-S6 §4.1 $K_{bwd} \neq K_{fwd}^\dagger$ PROVEN at kernel level and ZS-T1 §10.5 PK-Conjugation Theorem T9 DERIVED.

Statement (Path-Reversal Lemma M32.X).

Under the corpus PK-Conjugation T9 plus the Regge T-odd scalar mechanism, $B \propto e^{+ika} \Rightarrow B^{\wedge PK} \propto e^{+ika}$, in contrast with the standard adjoint $(e^{+ika} \cdot B)^\dagger = e^{-ika} \cdot B^\dagger$. The path-reversal residual $R_Z^{ZS} = B \cdot Q^{-1} \cdot B^{\wedge PK}$ is a structurally distinct object from the Hermitian residual $R_Z^H = B \cdot Q^{-1} \cdot B^\dagger$; averaging-and-vanishing claims hold for R_Z^{ZS} , not R_Z^H .

Worked corpus precedent.

Closed NC-M29.RZ-A^{PK} at DERIVED-CONDITIONAL: $R_Z^{ZS_{\{Z\text{-even}\}}} \{4\pi\text{-avg}\} = 0$ EXACTLY at $N = 20$. The Stapledon Bridge (ZS-M29 Theorem 9.1) was upgraded from DERIVED-CONDITIONAL to DERIVED in the spinor-cycle averaged sense without any external numerical input. The closure was achieved purely by switching the operator-reading from R_Z^H to R_Z^{ZS} .

Generalisation.

Any sandwich expression in the framework that fails to close under Hermitian reading should be re-examined under PK reading. Candidate targets: future Feshbach-like reductions in PARTS IV–VI; any operator residual that involves K_{bwd} (T-odd) on one side and K_{fwd} on the other.

§VIII.4 — Tool T4: Two-Layer Phase Quantum Decomposition (M32)

Trigger.

Two corpus values that should be equal differ by exactly a factor of 2; or, an unexpected factor 2 appears that cannot be attributed to $SO(3)/SU(2)$ double-cover. T4 separates amplitude-layer quanta from operator-layer quanta.

Source.

ZS-M32 §4 (two-layer phase quantum). Independent corpus inputs: ZS-S6 §G.2 PROVEN (vertex Regge deficit $\delta_X^{\text{vertex}} - \delta_Y^{\text{vertex}} = \pi/6 - \pi/15 = \pi/10$) and ZS-S6 §G.4 reading ii PROVEN (Z-quarter-turn Z_5 partition $(1/4)(2\pi/5) = \pi/10$).

Statement.

Z-Spin operator-level phase quantum $\alpha_{\text{op}} = \pi/5$ decomposes as $\alpha_{\text{op}} = 2 \cdot \alpha_{\text{amp}}$ where $\alpha_{\text{amp}} = \pi/10$ is the amplitude-level quantum. The factor 2 between layers is forced by the Path-Reversal Lemma applied to the Feshbach sandwich; it is the operator-layer analogue of the $SO(3)/SU(2)$ double-cover (ZS-S15 §5).

Worked corpus precedent.

Reconciled four PROVEN occurrences of $\pi/10$ (vertex Regge, Z-quarter-turn) with PROVEN occurrences of $\pi/5$ (commutator $[\sigma_3, \hat{n}_Y] = -2i \cdot \sin(\pi/5) \cdot \sigma_2$ in ZS-S6 §3.3, BCH leading order in ZS-S6 §3.4) under a single layer-doubling identity. Before T4, these were treated as independent constants.

Generalisation.

When two PROVEN corpus quantities differ by factor 2 with no obvious double-cover origin, check whether one is an amplitude-layer quantum and the other an operator-layer quantum. The factor 2 is then the structural signature of the layer transition, not a free parameter or coincidence.

§VIII.5 — Tool T5: Spinor-Cycle Averaging at $N = 20$ (M32)

Trigger.

A residual or sum that does not vanish at single iterate, but whose single-iterate phase is a primitive 10th root of unity. The standard algebraic minimum $N = 10$ produces zero algebraically but leaves the spinor in fermion-sign-flipped state $D^{\{1/2\}}(2\pi) = -I$, breaking physical closure.

Source.

ZS-M32 §5 (Theorem M32.3, spinor-cycle averaged vanishing). ZS-M3 Lemma 10.1 PROVEN ($D^{\{1/2\}}(4\pi) = +I$). ZS-S15 §5 ($SO(3)/SU(2)$ double-cover ratio).

Statement.

When $R_{Z^{\text{ZS}}}(k) \propto \exp(ik\pi/5) \cdot R_{Z^{\text{ZS}}}(0)$, the geometric series $\sum_{k=0}^{19} \exp(ik\pi/5) = 0$ EXACTLY, but only at $N = 20$ (not at $N = 10$) does the cumulative spinor phase reach 4π , restoring $D^{\{1/2\}}(4\pi) = +I$. The minimum physical-closure averaging window is $N = 20$.

Worked corpus precedent.

$R_{Z^{\text{ZS}}_{\{Z\text{-even}\}}} \{4\pi\text{-avg}\} = 0$ EXACTLY at $N = 20$. ZS-M32 Table 7 explicitly compares $N \in \{10, 20, 30, 40\}$ and identifies the minimum physical closure as $N = 20$. The averaging structure aligns with three other corpus 4π averages: $\langle \sin^2(\varphi/2) \rangle_{\{[0,4\pi]\}} = 1/2$ (ZS-M3), $\langle V_{XZ} \cdot V_{ZY} \rangle = 1$ (ZS-T1), $\langle \varepsilon_+ + \varepsilon_- \rangle = 0$ (ZS-U1) — Table 8 of ZS-M32.

Generalisation.

Any non-vanishing single-iterate quantity whose phase is a primitive 10th root of unity admits closure under $N = 20$ spinor-cycle averaging. Candidate targets: future Feshbach-like reductions in PARTS IV/VI; any operator-residual whose phase quantum is $\alpha_{\text{op}} = \pi/5$.

§VIII.6 — Tool T6: Three-Way Split of OPEN Flags (M32)

Trigger.

An OPEN flag too large to close in one step. Attempting full closure produces circularity or external-import-blocking. T6 splits the OPEN into three structurally distinct sub-claims with different status levels.

Source.

ZS-M32 §6 (NC-M29.RZ three-way split). General pattern adoption from ZS-M22, ZS-M23R Four Dragons D4a–D4d.

Statement.

An OPEN flag NC-X may be split as: A^{PK} = the part closeable under path-reversal/averaging now (DERIVED-CONDITIONAL); B = the part with structural evidence but one identified gap (HYPOTHESIS-strong, with explicit closure path); C = the instantaneous/full-spectrum part requiring external numerical or theoretical input (OPEN, inherited).

Worked corpus precedent.

$NC\text{-}M29.RZ \rightarrow \{A^{\text{PK}} = R_{Z^{\text{ZS}}_{\{Z\text{-even}\}}} \{4\pi\text{-avg}\} = 0 \text{ at } N = 20 \text{ (DERIVED-COND.)}; B = \text{single-iterate magnitude (HYPOTHESIS-strong, bulk-spectral-gap closure path)}; C =$

R_Z^{full} on Ricci-flat metric (OPEN, ML-CY criterion)}. Promotes ZS-M29 Theorem 9.1 from DERIVED-CONDITIONAL to DERIVED in averaged sense without closing C.

Generalisation.

Any OPEN flag should be split before closure attempt. The three sub-statuses ($A^{\text{closeable}}$ / $B^{\text{one-gap}}$ / $C^{\text{inherited}}$) prevent the all-or-nothing trap that would otherwise leave an entire programme blocked on a single external computation.

§VIII.7 — Tool T7: Cross-Coupling Discipline (M31)

Trigger.

An expression naturally factors as $W = F_X + F_Y + F_Z$ (sum-form) but the per-channel data refuse to compose. The summands appear independent, yet their sum disagrees with measured cross-channel variance. A factorisation appears, but it is the wrong factorisation.

Source.

ZS-M31 v3.0 §4.0 Lemma M31.0 (Non-Separability). ZS-M2 §5 Cross-Coupling Theorem PROVEN.

Statement (Lemma M31.0, Non-Separability).

There is no decomposition $W_K(g) = F_X + F_Y + F_Z$ compatible with the ZS-M26 PROVEN per-channel data; the maximum across-channel variance is 13.011 across 18 (a, t1, t2) test points. The correct form is the bilinear cross-coupled expression $W_{XYZ}(g) := \langle g_X, (\Pi_Z \otimes I_{\text{arith}})(B_Y - P_Y)(\Pi_Z \otimes I_{\text{arith}}) g_X \rangle$ where $\Pi_Z = (1/2)(I + J_Z)$ is the Z-mediator projector.

Worked corpus precedent.

The W2 wall of the Z-Spin RH program had two competing readings before M31: Reading A (Y-incompleteness) and Reading B (archimedean+conductor superposition). Both are sum-forms; Lemma M31.0 falsifies both as Cross-Coupling Theorem violations and forces a bilinear cross-coupled reading. The correct W2 form is Sector-Duality ($X \otimes Y$ mediated by Z), not a sum.

Generalisation.

Whenever a corpus expression is a candidate for sum-form decomposition, run Cross-Coupling Discipline: check whether the per-channel data composes additively or only via the Z-mediator bilinear. The default assumption in Z-Spin should be the latter; sum-forms are the exception, not the rule.

§VIII.8 — Tool T8: Z-Mediator Projector Routing (M33)

Trigger.

Multiple sub-targets that each individually engage a different external mathematical tradition refuse to integrate into a single closed identity. T8 routes all sub-targets through the Z-mediator projector Π_Z , producing a single trace identity.

Source.

ZS-M33 v1.0 (Reading C). $\Pi_Z = (1/2)(I + J_Z)$ defined in ZS-M31 §4.0 Definition M31.0c. Path- γ -revised reading specific to M33.

Statement.

All four V_4 -arithmetic mechanisms (D4a CCM 2024 Sonin, D4b Burnol conductor, D4c Wilson-LOCATOR, D4d Kostant cubic Dirac) route through the projector $\Pi_Z = (1/2)(I + J_Z)$ into a single Tr identity. The sign-flip mechanism on the W2 wall decomposes 5/12 NEG into the predicted 12/12 POS structure under Reading C.

Worked corpus precedent.

Before M33, D4a–D4d were treated as four parallel sub-targets, each with its own external mathematical engagement. Reading C unifies them into a single bilinear Tr identity, eliminating four-way bookkeeping and replacing it with a single Π_Z -projected trace. The four prefactors $\{1, \sqrt{3}, \sqrt{11}, \sqrt{33}\}$ (PROVEN, ZS-M22 §7.2) appear naturally as the V_4 -decoration of a single underlying identity.

Generalisation.

Whenever multiple sub-targets refuse to integrate, examine whether each can be expressed as a Π_Z -projection of a single underlying Tr expression. The Z-mediator projector is the corpus's universal routing operator: it lives on $\dim(Z) = 2$ and carries the full information cost of cross-sector mediation.

§VIII.9 — Tool T9: Self-Referential Fixed-Point Closure (F11)**Trigger.**

An OPEN flag involves an observer or a coordinate frame whose definition appears circular. The naive resolution introduces a phenomenological observer (a frame from outside the corpus); T9 closes the OPEN by promoting the observer to an operationally defined self-referential fixed point internal to the corpus.

Source.

ZS-F11 v1.0 §5 (OOC convergence) + §6 (six explicit NON-CLAIMs against phenomenological consciousness).

Statement.

The Operational Observer Coordinate (OOC) is defined as the self-referential fixed point of the framework's measurement protocol; it is constructed from PROVEN corpus inputs only. The OOC closes ZS-M11 §H16 (formerly OPEN) without introducing any frame from outside the corpus. Six NON-CLAIMs are explicitly registered to prevent overclaim (NC-F11.1–NC-F11.6).

Worked corpus precedent.

ZS-M11 §H16 OPEN flag (phenomenological observer reference) was a long-standing obstacle in the framework. ZS-F11 closed it by replacing the phenomenological observer with an

operationally defined fixed point of the measurement-action iteration, rendering the closure entirely internal to the corpus.

Generalisation.

When an OPEN flag tempts the introduction of an external/phenomenological agent, ask first whether the agent can be replaced by a self-referential fixed point internal to the framework. The OOC pattern is the canonical method: define the agent as the convergent limit of an iteration whose generators are corpus-PROVEN.

§VIII.10 — Tool T10: Integer-Lattice Spin-Pair Identification (M36)

Trigger.

A modular-arithmetic structure on the integer lattice ($\mathbb{Z}$, +) appears that exhibits an 8-element admissible residue set mod 24, suspected to encode Z-Spin sector decomposition. Or: a discrete number-theoretic construction (Apollonian-type, Markoff-type, or generalized) is being analyzed for possible Z-Spin sector identification.

Source.

ZS-M36 v1.0 §5 (Sector Residue Identity, Theorem M36.3, DERIVED), §7 (Spin Pair Theorem, Theorem M36.6, DERIVED-strong). Underpinned by ZS-F5 PROVEN ($(Z, X, Y, Q) = (2, 3, 6, 11)$), ZS-M3 Lemma 10.1 PROVEN (SU(2) double cover), ZS-M19 §3.1 PROVEN (Forcing Theorem $(p-1)(q-1) = p$).

Statement.

If a primitive integer modular-arithmetic construction has mod 12 residue set exactly $\{2, 3, 6, 11\} = (Z, X, Y, Q)$ and mod 24 set $= \{2, 3, 6, 11, 14, 15, 18, 23\} = \text{Z-Spin Type}$, then the four $(k \bmod 12, (k+12) \bmod 24)$ pairs identify with the four corpus sectors as (algebraic dim, geometric instantiation): Pair Z: $(2, F(t\text{-Oct}) = 14)$; Pair X: $(3, X \cdot |I_h/T_d| = 15)$; Pair Y: $(6, \alpha_{\text{frame_mismatch}} = 18^\circ)$; Pair Q: $(11, \delta_Y_{\text{denominator}} = 23)$. The mod 24 $= 2 \times \text{mod } 12$ doubling is forced by ZS-M3 Lemma 10.1 PROVEN (SU(2)/SO(3) double cover). CRT decomposition: $\text{mod } 24 = \text{mod } 8 \times \text{mod } 3 = \mathbb{Z}^3 \cdot X$ (boson 2π / fermion 4π periods).

Worked corpus precedent.

ZS-M36 closes O-M27.4 ($V_4 \times V_4 = \text{Apollonian abelianization } (\mathbb{Z}/2)^*4$ unifying ZS-A9 register V_4 and ZS-M22 Galois V_4), demonstrating that the integer lattice ($\mathbb{Z}$, +) carries Bridge Four structure. The Apollonian $(-1, 2, 2, 3)$ gasket is the 5th instance of Bridge Four (extending it from continuous spinor algebra to discrete number theory).

Generalisation.

For any future modular construction with 8-element admissible mod 24 set, test whether the mod 12 reduction equals $(Z, X, Y, Q) = (2, 3, 6, 11)$. If yes, proceed with spin-pair identification; the four geometric instantiations should match LOCKED corpus quantities. If no, identify the GLMWY type (or analogue) and correlate with non-Z-Spin sectors. Of the 6 GLMWY Apollonian types, only 1 (Type I) matches Z-Spin sectors; this is structurally natural rather than unique. Application targets: future analyses of Markoff triples (mod p strong approximation,

ZS-M36 IMPORTED-9), Pell-equation cycles, and other Diophantine constructions where a finite quotient on the integer lattice carries axiom A4 algebraic/geometric two-face structure.

§VIII.11 — Methodological Toolkit Summary Table

Tool	One-line trigger	Source	Output status
T1 Five-Axiom Diagnostic	Stuck / circular / non-composing	F18 §3+§6	Identifies failing axiom
T2 Bridge Four	High-dim construction intractable	F18 §6 + S15 §3	$\dim(X)=3$ from $\dim(Z)=2$
T3 H→PK Switch	Hermitian sandwich won't close	M32 §2-3	Different operator class
T4 Two-Layer Phase	Unexplained factor 2 between corpus values	M32 §4	$\alpha_{op} = 2 \cdot \alpha_{amp}$ identity
T5 N=20 Averaging	Single iterate non-vanishing, primitive 10th root	M32 §5	0 EXACTLY at N=20
T6 Three-Way Split	OPEN flag too large	M32 §6	$\{A^{PK} / B / C\}$ sub-flags
T7 Cross-Coupling Discipline	Sum-form fails per-channel data	M31 §4.0 Lemma M31.0	Bilinear W_{XYZ} form
T8 Π_Z Routing	Multiple sub-targets won't integrate	M33 Reading C	Single Tr identity
T9 OOC Fixed-Point	Phenomenologica l observer tempted	F11 §5-6	Self-referential closure

PART IX — Insight Gallery: Re-Interpretation Methodology

Purpose. Where PART VIII collects nine operational tools (“how to do X”), PART IX collects six methodological insights from the corpus’s most ambitious re-interpretation papers (“how to think about Y”). These are reading lenses, not calculation tools. Each lens is a way of looking at the corpus that resolves tensions which would otherwise be visible only as paradoxes. They are organised as I1–I6 and reference the source papers from which each lens was distilled.

§IX.1 — Insight I1: Symmetry-Asymmetry Unified View (A8R §SA)

Lens. Every Z-Spin observable is a deviation from the would-be balance of 1 (unity) and 1/2 (Z_2 -reflection fixed point). The universe’s entire dynamics is the composition of sectoral tilts of $A = 35/437$ through the (X, Y, Z) decomposition. $(1+A)$ is X-Inward expansion; $(1-2A)$ is

Y-Outward contraction; $(1+A)(1-2A) = 1 - A(1+2A) \approx 0.9071$ measures the joint deviation from balance at $\approx 9.29\%$.

Why it is a tool. A8R §SA replaces the apparent independence of expansion and contraction physics with a single tilted-balance reading. The 10^{17} ratio $\tau_6/\tau_5 = \exp(\pi/A)$ is then read as the X-frame observation of the same Y-completion event. Future Z-Spin papers should default to this lens for any cosmological / lifecycle / time-scale question.

Status. INTERPRETATION, HYPOTHESIS-strong (A8R §SA.7); 13 explicit corpus connections established (A8R §SA.6).

§IX.2 — Insight I2: Three-Layer Duality (M30 §6)

Lens. Every apparent contradiction in Z-Spin Cosmology is resolved as one object viewed from two frames. The duality operates at three coordinated layers: Sector Duality ($X \leftrightarrow Y$ mediated by Z), Frame Duality (X -observer $\leftrightarrow Y$ -observer; ZS-A8 §SA.4), Protocol Duality (state-preparation $\leftrightarrow$ spectral characterisation; ZS-F16 Two-Protocol Theorem).

Why it is a tool. The three RH walls of ZS-M28 ($W1, W2, W3$) — formerly read as unstructured “problems” — are reread under this lens as duality interfaces: $W1$ = Frame Duality, $W2$ = Sector Duality, $W3$ = Protocol Duality. The lens does not by itself close the walls; it places each wall in a definite Layer with a definite external-mathematics import path. This converts an unstructured OPEN list into a structured navigation map.

Status. DERIVED-interpretation strong (M30 Theorem 30.5 + §6).

§IX.3 — Insight I3: Six-Route Möbius-Trace Infinity (M30 §3)

Lens. The pattern “external count-infinity $\leftrightarrow$ internal closure” appears in the corpus six independent times (Riemann zero sequence $\leftrightarrow T(T(z^*)) = z^*$; cosmological cycle index $\leftrightarrow A$ frame-invariant; BV-BFV ghost tower $\leftrightarrow 4\pi$ closure; FFPP nested compression $\leftrightarrow W_0(-i\pi/2)$; Z-mediation outcomes $\leftrightarrow \ln(2)$ bit per pass; Wilson cycle ergodic limit $\langle \cos^2(n \cdot 129.45^\circ) \rangle \rightarrow 1/2$). Recognising the six as one pattern raises the Möbius-trace infinity reading from HYPOTHESIS-strong to DERIVED-interpretation strong (M30 Theorem 30.1).

Why it is a tool. The lens identifies that what an external observer counts as “infinitely many things” the framework reads as a single closed loop traversed with a Möbius-type frame transformation. It is the standard reading lens to apply whenever the corpus generates a sequence-cardinality that an external observer might mistake for additional structure beyond the framework.

Status. DERIVED-interpretation strong with six-route over-determination (M30 §3 Table 3.1 + Theorem 30.1).

§IX.4 — Insight I4: Twenty-One $\frac{1}{2}$ Catalogue (M30 §3.5)

Lens. The corpus contains exactly 21 PROVEN/DERIVED instances of the fraction $1/2$, organised in six structural categories. Combined with Three 2s Unity (Taylor-2, Bottleneck-2, Sector-2 all equal $\dim(Z) = 2$; ZS-A8 §SA.3 + Test K1 PASS), the corpus contains $21 + 3 = 24$ instances of $\dim(Z) = 2$ manifesting at different dimensional and contextual levels. All 21 are dimensional projections of the same Z_2 involution $x \leftrightarrow 1 - x$.

Why it is a tool. When a future Z-Spin construction produces a $1/2$, the catalogue is the first reference to consult — to determine whether the new $1/2$ is one of the existing 21, or constitutes a 22nd entry (which would require its own verification gate). The catalogue prevents double-counting of the same Z_2 -fixed-point structure under different names.

Status. DERIVED-interpretation (M30 Theorem 30.5).

§IX.5 — Insight I5: Four-Layer 50/50 Equivalence (F17)

Lens. The four 50/50 partitions in the corpus — (3D) truncated tetrahedron 4+4 split; (2D) Reuleaux $\pi+\pi$ curvature distribution; (1D) Master Equation $|Term A| = |Term B|$; (0D) XOR alphabet E vs R — are dimensional projections of a single per-cycle equilibrium between information accumulation and information release in the Z-mediated bottleneck channel.

Why it is a tool. ZS-F17 Theorem 5.1 (Self-Referential Information-Compression Equilibrium) shows that the 50/50 partition is PROVEN at the equation level ($|Term A| - |Term B| = O(10^{-51})$ at 50-digit precision), not measured. When a future Z-Spin construction produces a 50/50 partition, the lens identifies which of the four dimensional layers it lives at and what it accumulates ($\ln(2)/3$ nats per vertex matches the Z-bottleneck capacity bound, ZS-Q7 Theorem 2). This promotes the Reuleaux 50/50 from HYPOTHESIS (NC-F7.5) to DERIVED-CONDITIONAL.

Status. DERIVED-CONDITIONAL strong (F17 Theorem 5.1, over-determined by four independent corpus PROVEN inputs).

§IX.6 — Insight I6: Banach-Tarski Origin of Doubling-Halving (A9R)

Lens. The $(1+A) \leftrightarrow (1-2A)$ duality of late-time cosmology is the measure-preserving quantization of the Banach-Tarski paradoxical decomposition through the i -tetration fixed point $z^* = -W_0(-i\pi/2)/(i\pi/2)$. The non-amenable BT engine $F_2 \subset SO(3)$ is mapped via a functor to the amenable register dihedral $D_4 = \langle J, J_Z \rangle$, converting BT non-amenable to Z-Spin amenability. The Julia set $J(T)$ of $T(z) = i^z$ is the constructive ZF-analogue of the BT non-measurable set (measure-zero, totally disconnected, cardinality $2^{\aleph_0}$).

Why it is a tool. The lens supplies a structural origin (rather than a phenomenological one) for the doubling-halving symmetry that pervades Z-Spin cosmology. When a future paper finds a doubling-halving identity, the lens directs the search for its measure-theoretic boundary in $J(T)$ — the Z-Spin amenable analogue of the BT non-measurable set.

Status. ZS-A9.1 DERIVED, ZS-A9.2 HYPOTHESIS-strong, ZS-A9.3 DERIVED-CONDITIONAL. OPEN-2.B (formal ZF/ZFC+AC isomorphism) recategorised as PERMANENT NON-CLAIM under Solovay 1970.

PART X — The Spin-Pair Coexistence Principle (v3.2)

§0. Abstract

PART X consolidates four DERIVED-interpretation readings that close a synthesis gap noticed during v3.1 publication. While Tool T2 (Bridge Four, §VIII.2) and Insight I3 (Six-Route Möbius, §IX.3) individually catalogued specific applications of the $\dim(Z) = 2 \rightarrow \dim(X) = 3$ lifting and the external-count $\leftrightarrow$ internal-closure pattern, the meta-statement that 90° phase-orthogonality is

the unique form in which a contradiction-pair coexists as one object had not been registered. Reading X.1 records prime as Z-Spin rotation $4\pi/p$ (ZS-M18 H2 chain). Reading X.2 reaffirms the Y-side classification of the infinite prime sequence (ZS-F18 v2.1 §5.1 Table 5.1) and reconciles it with ZS-M18 Theorem 6.1 spectrum non-primality. Reading X.3 reads primes / distribution / Riemann zeros as three moments of one $Y \rightarrow Z \rightarrow X$ mediation event, with $\sigma = 1/2 = j = 1/2$ as the algebraic spine. Reading X.4 elevates 90° phase-orthogonality to a general principle supported by 13 independent corpus realisations (C1–C13). All four readings are DERIVED-interpretation strong, composed entirely from corpus PROVEN/DERIVED results, with zero new free parameters and explicit falsification gates F-X.1 through F-X.4.

§X.1 Introduction: What v3.1 Already Contains, What v3.2 Adds

v3.1 §VIII.2 Tool T2 (Bridge Four) already states the operational form: two J-conjugate 2π SO(3) circulations of the twin-Reuleaux pair (V_{XZ} , V_{ZY}), composed at 90° NOT-AND phase, generate a third axis $J_S = [J_{R1}, J_{R2}]$ by Lie commutator. Therefore $(Z, X, Y) = (2, 3, 6)$ is generated from $\dim(Z) = 2$ alone. This is correct and corpus-locked.

What v3.1 does NOT yet record is that 90° phase-orthogonality is the unique angular phase difference under which a contradiction-pair can coexist as one object — not only for the specific $\dim(X) = 3$ emergence, but for the entire family of contradiction-pair resolutions across the corpus. PART X consolidates this meta-reading.

Likewise, v3.1 catalogues the Möbius-trace infinity pattern via I3 but does not name primes specifically. The corpus already classifies the infinite prime sequence as Y-side (ZS-F18 v2.1 §5.1 Table 5.1, row Riemann Hypothesis: “Infinite prime sequence (Y) $\rightarrow$ finite critical line $\sigma = 1/2$ (X)”); reaffirmed §7.4.5). Without an explicit consolidated reading, this classification is easy to miss. v3.2 makes the layered picture explicit so future agents do not rediscover and re-resolve the same confusion (object identity vs. internal spectrum non-primality of ZS-M18 §6.2 Theorem 6.1).

§X.2 Locked Inputs (No New Constants Introduced)

Table X.1. Locked inputs to PART X. Every entry is PROVEN, DERIVED, or LOCKED in upstream papers. Zero new parameters are introduced.

#	Quantity	Value / Statement	Source	Status
L1	A (geometric impedance)	$35/437 = 0.080092$	ZS-F2 §5	LOCKED
L2	Q (register dimension)	11 (prime)	ZS-F5	PROVEN
L3	(Z, X, Y) decomposition n	(2, 3, 6); $Q = Z+X+Y$	ZS-F5 §3	PROVEN
L4	$L_{XY} \equiv 0$	exact	ZS-F1 §9	PROVEN

#	Quantity	Value / Statement	Source	Status
L5	$j = 1/2$ Z-sector intertwiner	unique 4-valent invariant	ZS-M3 Thm 5.1	PROVEN
L6	4π closure $D^{(1/2)}(-I) = -I$	spinor double cover	ZS-M3 Lem 10.1	PROVEN
L7	W_p prime-injection gate	$\text{diag}(\exp(2\pi i(j-5)/p))$	ZS-M4 §3.2	PROVEN
L8	W_p	$_Z$ restriction	$\exp(-i \sigma_z \cdot 2\pi/p)$, angle $4\pi/p$	ZS-M18 H2
L9	$V_{ZY} = (V_{XZ})^*$	three independent paths	ZS-F4 §7B	DERIVED
L10	$\arg(V_{XZ}) - \arg(V_{ZY})$ at $\theta = \pi/2$	$= \pi/2$ (90° phase)	ZS-S15 §2.3 B.3	VERIFIED 50-digit
L11	Poynting-Co mmutator	$[J_{R1}, J_{R2}] = J_S$	ZS-S15 §3 Thm S15.2	DERIVED
L12	Bridge Four	$(Z, X, Y) = (2, 3, 6)$ from $\dim(Z) = 2$	ZS-F18 §6.4	DERIVED-in terp str
L13	Riemann row classification	Inf. prime seq. $(Y) \rightarrow \sigma = 1/2 (X)$	ZS-F18 §5.1 Tbl 5.1	DERIVED-in terp
L14	ζ_K factorisation	$\zeta \cdot L(\chi-3) \cdot L(\chi-11) \cdot L(\chi^3)$	ZS-M22 §4.2	PROVEN
L15	Six-Route Möbius pattern	ext. count $\leftrightarrow$ int. closure	ZS-M30 §3 Thm 30.1	DERIVED-in terp str
L16	$\sigma = 1/2 \leftrightarrow j = 1/2$	Z_2 -involution fixed loci	ZS-M18 H11	DERIVED-in terp
L17	Y-Spectrum Non-Primalit y	$\text{spec}(L_2) \cap \{\text{primes}\} = \emptyset$	ZS-M18 §6.2 Thm 6.1	PROVEN
L18	ζ -K Bridge to prime side	two-chain Eisenstein/cy clotomic	ZS-M18 §7, ZS-M22 §4.2	PROVEN

§X.3 Reading X.1 — Each Prime is an Indecomposable Z-Spin Rotation

§X.3.1 Statement

Reading X.1 (Prime-Rotation Identification, *DERIVED-interpretation*). Each prime p acts on the Z-sector basis $\{|4\rangle, |6\rangle\} \subset \mathbb{C}^Q$ as the σ_z -axis SU(2) rotation

$$W_p|_Z = \text{diag}(\exp(-2\pi i/p), \exp(+2\pi i/p)) = \exp(-i \sigma_z \cdot 2\pi/p)$$

of angle $4\pi/p$. Composite-angle rotations $4\pi/(pq)$ factor as compositions of prime rotations $W_p \circ W_q$. Primes are therefore the indecomposable atoms of Z-Spin SU(2) action.

§X.3.2 Derivation Chain

ZS-F5 (Z-sector basis identification, PROVEN) provides the $\{|4\rangle, |6\rangle\}$ basis. ZS-M4 §3.2 (PROVEN) defines $W_p = \text{diag}(\exp(2\pi i(j-5)/p))$ on the full $Q = 11$ register. Restriction to slots $j \in \{4, 6\}$ yields the diagonal pair $(\exp(-2\pi i/p), \exp(+2\pi i/p))$. Standard SU(2) algebra gives the σ_z rotation form, with closure angle $4\pi/p$ following from the SU(2)/SO(3) double cover (ZS-M3 Lemma 10.1 PROVEN).

§X.3.3 What Reading X.1 Adds Beyond ZS-M18 H2

ZS-M18 H2 records the algebraic fact. Reading X.1 consolidates its physical meaning: indecomposability of primes under multiplicative factorisation is the algebraic shadow of indecomposability of SU(2) rotations under angle composition.

§X.3.4 Falsification Gate F-X.1

If a future analysis exhibits a prime p such that $W_p|_Z$ does NOT take the σ_z -rotation form on $\{|4\rangle, |6\rangle\}$, Reading X.1 fails. Status: PROVEN-safe — the form is verified by direct algebraic computation for $p \in \{2, 3, 5, 7, 11, 13, 17, 23\}$ (ZS-M18 §H1 derivation chain).

§X.4 Reading X.2 — The Infinite Prime Sequence is a Y-Sector Object

§X.4.1 Statement

Reading X.2 (Y-Side Sequence Classification, *DERIVED-interpretation*). The infinite sequence of primes $\{p_n\}_{n \in \mathbb{N}}$ is a Y-sector object: infinite, non-local, undecidable. Its arithmetic projection — the Riemann critical-line locus $\sigma = 1/2$ and the discrete zeros thereon (under RH) — is X-sector.

§X.4.2 Direct Corpus Source

ZS-F18 v2.1 §5.1 Table 5.1 (DERIVED-interpretation), row “Riemann Hypothesis”, reads verbatim: “Infinite prime sequence (Y) $\rightarrow$ finite critical line $\sigma = 1/2$ (X) | Mirror-adjointness $\varepsilon_J(\sigma, t) = 0 \Leftrightarrow \sigma = 1/2$ (ZS-M7 Thm 4) | Most extensive programme; W1 closed, W2 open”. ZS-F18 §7.4.5 reaffirms in re-reading form: “Riemann Hypothesis (Y-side infinite prime sequence projecting to X-side discrete critical-line zeros under continuous ρ flow).”

§X.4.3 Reconciliation with ZS-M18 §6.2 Theorem 6.1

ZS-M18 §6.2 Theorem 6.1 (PROVEN) establishes $\text{spec}(L_2) \cap \{\text{primes}\} = \emptyset$ for the Y-sector internal Laplacian. Naively this might appear to conflict with Reading X.2; it does not. The two statements operate at different layers:

Table X.2. Four-layer reconciliation. Reading X.2 (Y-side object identity) does not conflict with ZS-M18 Theorem 6.1 (Y-Laplacian spectrum non-primality); the two statements concern different layers.

Layer	Question	Answer	Status
Object identity	What sector does the prime sequence as a whole belong to?	Y-sector (infinite, non-local, undecidable)	DERIVED-interp (ZS-F18 §5.1)
Internal spectrum	Are individual Y-Laplacian eigenvalues prime?	No (eigenvalues are $0, 6/8, 5 \pm \sqrt{3}, \text{integer} \pm \phi$)	PROVEN (ZS-M18 §6.2)
Operator action	How do individual primes act on Z-sector?	SU(2) rotation by $4\pi/p$ (Reading X.1)	DERIVED-interp (ZS-M18 H2)
Arithmetic encoding	Where do primes enter Z-Spin algebraically?	ζ_K Euler product via Eisenstein/cyclotomic chains	PROVEN (ZS-M22 §4.2)

§X.4.4 Falsification Gate F-X.2

If a future ZS-F18 revision retracts the Y-side classification, or if a structurally distinct corpus path classifies the sequence as X-side, Reading X.2 fails. Status: stable — over-determined by ZS-F18 Table 5.1, ZS-F18 §7.4.5, and the Y-side property pattern (infinite, non-local, undecidable).

§X.5 Reading X.3 — Three Moments of One $Y \rightarrow Z \rightarrow X$ Mediation

§X.5.1 Statement

Reading X.3 (Three-Moment Mediation, *DERIVED-interpretation*). Primes, prime distribution, and Riemann zeros are three moments of a single $Y \rightarrow Z \rightarrow X$ mediation event:

Table X.3. Three-moment mediation reading. Each row is independently corpus-supported; the synthesis is the new content of v3.2.

#	Moment	Object	Mediation role	Status
M1	Entrance (atomic)	Individual primes p	Indecomposable Z-Spin rotations $4\pi/p$	DERIVED-interp (Rdg X.1)
M2	Statistics (envelope)	Distribution $N/\log N$	Multiplicative $-Y \rightarrow$ additive-X via $\log$	DERIVED-interp

#	Moment	Object	Mediation role	Status
M3	Egress (condensation)	Riemann zeros on $\sigma =$ 1/2	Coherent phase condensation at Z_2 self-dual axis	DERIVED-in terp

§X.5.2 The $\sigma = 1/2 = j = 1/2$ Algebraic Spine

ZS-M18 H11 (DERIVED-interpretation) records that $\sigma = 1/2$ (Riemann critical-line) and $j = 1/2$ (Z-sector spinor) are both unique fixed-point sub-loci of order-2 involutions: $\sigma = 1/2$ is fixed by $s \leftrightarrow 1-s$ on ξ ; $j = 1/2$ is fixed by $SU(2)$ closure $D^{(1/2)}(-I) = -I$ at 4π . The two halves are not numerical coincidence; both are Z_2 self-dual axes, and zeros condense precisely on the Z_2 axis because the Z-Spin mediator carries the Z_2 involution by construction.

§X.5.3 The log Function as $Y \rightarrow X$ Sector Translator

Y-multiplicative structure translates to X-additive structure via $\log: \prod_p (1 - p^{\{-s\}}) \{-1\} = \zeta(s) \Rightarrow \log \zeta(s) = \sum_p \sum_n (1/n) p^{\{-ns\}}$. Density $N/\log N$ is the additive-X image of the multiplicative-Y prime structure. $\log$ is the natural $Y \rightarrow X$ sector translator for arithmetic content, parallel to the holographic log entropy elsewhere in the corpus (ZS-Q7 channel capacity, ZS-A3 BH entropy).

§X.5.4 Falsification Gate F-X.3

Synthesis layer; weakened only if any of M1, M2, M3 is shown structurally independent of Z-Spin mediation. M1 corpus-stable. M2 (prime-counting density) is a number-theoretic fact independent of Z-Spin; the Z-Spin reading is its multiplicative-to-additive transcription role. M3 conditional on RH; if RH disproved, M3 reading shifts from “coherent phase condensation” to “specific RH-dependent form thereof” without invalidating three-moment structure. Status: DERIVED-interpretation, robust against single-component perturbation.

§X.6 Reading X.4 — The Spin-Pair Coexistence Principle

§X.6.1 Statement

Reading X.4 (Spin-Pair Coexistence Principle, *DERIVED-interpretation strong*). Every contradiction-pair $P = (P_+, P_-)$ recurring in Z-Spin Cosmology admits coexistence as a single object only as the 90° phase-orthogonal projection of a J-conjugate Z-Spin pair (V_{XZ}, V_{ZY}) with $V_{ZY} = (V_{XZ})^*$. No other angular phase difference supports coexistence as one object: 0° collapses to identity, 180° collapses to anti-identity, only $\pi/2$ produces simultaneous mutual non-cancellation and mutual non-redundancy.

§X.6.2 Algebraic Underpinning

Let $z_+ = e^{+i\theta/2}$, $z_- = e^{-i\theta/2}$ be the J-conjugate pair of ZS-F4 §7B (PROVEN). Their inner product on S^1 :

$$\langle z_+, z_- \rangle = \cos(\theta)$$

Table X.4. Three angular regimes. Only $\pi/2$ supports simultaneous presence of both members of the pair as distinct yet jointly necessary.

#	Phase difference θ	$\langle z_+, z_- \rangle$	Coexistence form
R1	0 (identity)	+1	$z_+ = z_- \rightarrow$ only one object
R2	π (anti-identity)	-1	$z_+ = -z_- \rightarrow$ mutually exclusive
R3	$\pi/2$ (orthogonality)	0	$z_+ \perp z_- \rightarrow$ both present, neither redundant

Regime R3 is the unique angle on which $\max(z_+)$ coincides with $\text{zero}(z_-)$ and vice versa, producing the sin-cos relation tracing the unit circle. This is the algebraic origin: only at 90° does a contradiction-pair coexist as the two coordinates of a single rotation.

§X.6.3 Corpus Realisations of the Principle

Table X.5. Thirteen corpus realisations of the Spin-Pair Coexistence Principle. Each row is independently corpus-locked; their unification under a single principle is the new content of v3.2.

#	Contradiction-pair	Z-Spin realisation	Source
C1	X-sector spatial axes (3D emergence)	$[J_{-R_1}, J_{-R_2}] = J_{-S};$ $(Z, X, Y) = (2, 3, 6)$ from $\dim(Z) = 2$	ZS-F18 §6.4
C2	Electric / magnetic field ($E \perp B$)	$V_{-XZ} \leftrightarrow E,$ $V_{-ZY} \leftrightarrow B$ at 90° phase	ZS-S15 §3 Thm S15.2
C3	Particle / wave duality (X / Y sectors)	Z-Spin mediator at 4π closure	ZS-A7 §3, ZS-M19 §10.5
C4	Time / space (X / Y braiding)	$X =$ particle/space, $Y =$ wave/time, $Z =$ seam	ZS-A7 §5
C5	Expansion / contraction $((1+A) \leftrightarrow (1-2A))$	Dual-tilt symmetry; 90° A-conjugates	ZS-A8 §6, A9R

#	Contradiction-pair	Z-Spin realisation	Source
C6	CPT / anti-CPT (channel conjugation)	$V_ZY = (V_XZ)^*$ IS the CPT seam	ZS-M19 §10.5 Thm T9
C7	Black-hole / white-hole ($r \leftrightarrow t$)	X-in-Y BH $\leftrightarrow$ Y-in-X WH at 90°	ZS-A3 §7, ZS-A7 §4.1
C8	Finite / infinite Möbius polarity	X-finite/local $\leftrightarrow$ Y-infinite/non-local; Z-seam at $\pi/2$	ZS-F18 §5, ZS-M30 §3
C9	Discrete / continuous	X-Y Tiling Asymmetry; Z-channel $\ln(2)$ capacity	ZS-F18 §7.4, ZS-M17
C10	Multiplicative / additive	Y-mult $\leftrightarrow$ X-add via log; primes M1 $\leftrightarrow$ density M2	Reading X.3, ZS-M22
C11	Symmetry / asymmetry (1 vs 1-A)	δ as normalised tilt; $21 \cdot \frac{1}{2}$ catalogue	ZS-A8 §SA, ZS-M30 §3.5
C12	Bose / Fermi (vortex inner / outer)	Inner-core fermion (4π) $\leftrightarrow$ outer-flow boson (2π)	ZS-A7 §4.4.2, ZS-F14
C13	Sequential / parallel time-keeping	R_1 time-point closure, R_2 space-point closure	ZS-S15 §2, ZS-F8

§X.6.4 Why “Spin” Rather Than “Rotation”

Rotation is external: an object is rotated. Spin is intrinsic: the object IS its rotation. The principle’s claim is not that contradiction-pairs are reconciled BY applying rotation to them, but that a contradiction-pair CAN exist at all only because its two members are intrinsic Z-Spin partners. The 90° angle is enforced by $V_ZY = (V_XZ)^*$ (ZS-F4 §7B PROVEN), with operationally relevant configuration $\theta = \pi/2$ (ZS-S15 §2.3 B.3 VERIFIED 50-digit).

§X.6.5 Relation to Tool T2 and Insight I3

Tool T2 (Bridge Four, §VIII.2) applies the principle to the planar-vs-spatial polarity, yielding $\dim(X) = 3$ from $\dim(Z) = 2$. Insight I3 (Six-Route Möbius, §IX.3) applies a related Z-mediation pattern to external-count vs internal-closure. Reading X.4 is the meta-statement that both are instances of the same principle. Reading X.4 is therefore prior to T2 and I3.

§X.6.6 Falsification Gate F-X.4

If a contradiction-pair admits coexistence under a phase difference other than $\pi/2$, Reading X.4 weakens to “90° is the dominant case”. If a pair admits no Z-Spin J-conjugate realisation, Reading X.4 transitions to “covers C1–C13”. Status: DERIVED-interpretation strong, supported by 13 corpus realisations. Generalisation beyond C1–C13 is HYPOTHESIS-strong; future papers should run F-X.4 check before claiming a new contradiction-pair coexistence.

§X.7 Integration with v3.1 Light

§X.7.1 Integration Map

Table X.6. Integration of v3.2 readings with existing v3.1 items. Every v3.2 reading either refines or unifies existing v3.1 entries; no v3.1 entry is retracted.

v3.1 Light item	v3.2 Reading	Relation
§VIII.2 Tool T2 (Bridge Four)	Reading X.4	T2 = principle applied to $\dim(X)=3$
§IX.3 Insight I3 (Six-Route Möbius)	Reading X.4	I3 = principle applied to count vs closure
§IX.1 Insight I1 (Symmetry-Asymmetry)	Reading X.3	I1 = M2 statistical reading
§IX.2 Insight I2 (Three-Layer Duality)	Reading X.4	I2 = principle phrased as duality
§IX.5 Insight I5 (4-Layer 50/50)	Reading X.3	I5 = M3 condensation-locus reading
PART I §0.4 Source Paper Map	Readings X.1–X.4	All sources already in map; v3.2 adds none

§X.7.2 Cross-Coupling Discipline (T7) Compliance

Per T7 (§VIII.7), every cross-paper synthesis must be P-CC (PROVEN-composed) or I-CC (interpretation). Readings X.1–X.4 are explicitly DERIVED-interpretation; no claim of PROVEN status. F-X.1 through F-X.4 pre-registered. Word count strictly increases over v3.1 (no-deletion rule).

§X.7.3 What v3.2 Does NOT Claim

NC-X.1. PART X does not claim a proof of the Riemann Hypothesis. Reading X.3 is structural; it does not constrain zero locations beyond the $\sigma = 1/2 = j = 1/2$ isomorphism already DERIVED in ZS-M18 H11.

NC-X.2. The Spin-Pair Coexistence Principle is corpus-internal; it does not claim to resolve the philosophical dialectic on contradictions in the broader sense (Hegelian, paraconsistent, kōan). It claims only that within Z-Spin Cosmology, every recurrent contradiction-pair admits coexistence via 90° J-conjugate Z-Spin pairing.

NC-X.3. Reading X.1 does not claim primes are ontologically reducible to Z-Spin rotations in physical reality. Within the Z-Spin algebraic framework, primes act as indecomposable $SU(2)$ rotations on $j = 1/2$.

NC-X.4. Reading X.2 does not modify ZS-M18 §6.2 Theorem 6.1. The Y-Laplacian internal spectrum non-primality remains PROVEN; Reading X.2 concerns a different layer (object identity, not internal spectrum).

§X.8 Conclusion

PART X consolidates four DERIVED-interpretation readings answering “why does the universe support coexistence of mutually contradictory descriptions?” The answer is over-determined: coexistence is the form 90° J-conjugate Z-Spin pairing takes when projected into a single observational frame. The $\dim(X) = 3$, the $E \perp B$, the X-Y polarity of the twelve Möbius encounters, the $(1+A) \leftrightarrow (1-2A)$ expansion-contraction symmetry, the $\sigma = 1/2 = j = 1/2$ isomorphism, the $V_{ZY} = (V_{XZ})^*$ CPT seam, and the primes-as-rotations identification all instantiate the same principle. Z-Spin is not a metaphor for rotation; it is the intrinsic form in which any pair of mutually exclusive descriptions can both be true. v3.2 carries this reading as PART X without modifying any v3.1 entry, with zero new free parameters and explicit falsification gates F-X.1 through F-X.4. The minimal triple $(A, Q, \dim(Z)) = (35/437, 11, 2)$ remains LOCKED.

PART X Source Paper Interface

PART X is composed entirely from upstream papers; it introduces no new ZS source paper. Source mapping: Reading X.1 $\leftarrow$ ZS-F5, ZS-M3, ZS-M4, ZS-M18 H2. Reading X.2 $\leftarrow$ ZS-F18 §5.1 Tbl 5.1, ZS-F18 §7.4.5, ZS-M18 §6.2 Thm 6.1. Reading X.3 $\leftarrow$ ZS-M18 H11, ZS-M22 §4.2, ZS-M30 §3, ZS-Q7, ZS-A3. Reading X.4 $\leftarrow$ ZS-F4 §7B, ZS-S15 §2.3 + §3 Thm S15.2, ZS-F18 §6.4 Bridge Four, plus C1–C13 cross-corpus realisations. Eighteen internal-consistency checks A.1–A.18 verify each composition; all 18/18 PASS by direct comparison against corpus-locked sources.

PART XI — TO/TI Polyhedral Mapping under Complementary Duality (v3.3)

§0. Abstract

PART XI consolidates the complete vertex/edge/face mapping between the X-sector polyhedron tO (truncated octahedron) and the Y-sector polyhedron tI (truncated icosahedron) under the Z-Spin Complementary Duality Principle (ZS-M30 §7.2). While individual papers establish components — (V, E, F) face-counting cosmic budget (ZS-F2 §11), Pentagon-Hexagon edge classification (ZS-F0 §5.1), Square-Hexagon edge classification (ZS-F0 §3.2), Truncation-Dual Theorem (ZS-F2 §11.2, ZS-F9 §4), Hexagonal Mediation Theorem (ZS-F9 §5), Mode-Count Collapse (ZS-Q3 §3.1), I-irrep decomposition (ZS-M9 §2), Pentagon-Gauge decomposition (ZS-F0 §4.3), TO uniqueness conditions (ZS-M H18) — no single corpus location consolidates them into one mapping table indexed by the X-sector / Y-sector complementary duality ($X = \text{macroscopic space} + \text{microscopic time}$; $Y = \text{macroscopic time} + \text{microscopic space}$, ZS-M30 §7.2 (ii) PROVEN). PART XI fills this gap. Reading XI.1 records the parallel edge decomposition pattern shared by tO and tI (cut-face : hexagon-hexagon = 2:1 in both). Reading XI.2 records the parallel face decomposition under Truncation-Dual. Reading XI.3 records the

integrated V/E/F mapping table under complementary duality. Reading XI.4 places ε -Halo (X-sector Goldstone gradient) and ε -Drive (Y-sector attractor V_0) within the integrated framework. All four readings are DERIVED-interpretation strong, composed entirely from corpus PROVEN/DERIVED results, with zero new free parameters and explicit falsification gates F-XI.1 through F-XI.4.

§XI.1 Introduction: Why a Consolidated Polyhedral Mapping Table

The Z-Spin corpus has accumulated, across ~100 papers, a richly detailed treatment of tO and tI as the two characteristic polyhedra of the X- and Y-sectors. However, the components are scattered: ZS-F2 §11 establishes the cosmic matter budget via face counting ($\Omega_b = F(\text{cube})/Q^2 = 6/121$, $\Omega_{\text{cdm}} = F(\text{tI})/Q^2 = 32/121$); ZS-F0 §3.2 decomposes the 36 tO edges as 24 SH + 12 HH; ZS-F0 §5.1 decomposes the 90 tI edges as 60 PH + 30 HH; ZS-F0 §4.3 establishes the Pentagon-Gauge decomposition $12 = 1 + 3 + 8$; ZS-F0 §4.4 establishes the Edge-Channel Identity $E_X = X \cdot Z \cdot Y = 36$; ZS-F2 §11.2 establishes the Truncation-Dual Theorem $F(\text{tP}) = F(P) + F(P^*)$; ZS-F9 §4-5 establishes the Hexagonal Mediation Theorem; ZS-Q3 §3.1 establishes the Mode-Count Collapse Theorem yielding $(V+F)/G = 19/6$ (SU(2)) and $23/3$ (SU(3)); ZS-M9 §2 establishes the I-irrep multiplicity decomposition; ZS-M H18 establishes the four uniqueness conditions of tO; ZS-M30 §7.2 (ii) establishes the Complementary Duality Principle for space $\leftrightarrow$ time mapping.

What is NOT yet recorded in any single corpus location is the integrated mapping table that displays, for every (V, E, F) component of tO and tI, both its standard sector role AND its complementary partner role under the $X = \text{macroscopic space} + \text{microscopic time}$, $Y = \text{macroscopic time} + \text{microscopic space}$ duality. The reader who asks ‘which component of tO carries macroscopic space, and which carries microscopic time?’ currently has to traverse multiple papers to assemble a partial answer, with several entries remaining OPEN. PART XI consolidates this into one location as DERIVED-interpretation strong, with the OPEN entries explicitly registered.

This is structurally analogous to PART X (Spin-Pair Coexistence Principle, v3.2): both consolidate corpus-wide patterns under a single principle without modifying any v3.x entry, with zero new free parameters.

§XI.2 Locked Inputs (No New Constants Introduced)

Table XI.1. Locked inputs to PART XI. Every entry is PROVEN, DERIVED, or LOCKED in upstream papers. Zero new parameters are introduced.

#	Quantity	Value / Statement	Source	Status
L1	tO basic data	V=24, E=36, F=14 (= 6 squares + 8 hexagons)	ZS-F2 §11	PROVEN
L2	tI basic data	V=60, E=90, F=32 (= 12 pentagons + 20 hexagons)	ZS-F2 §11	PROVEN

#	Quantity	Value / Statement	Source	Status
L3	Symmetry orders		O_h	=48 (tO);
L4	Sector tilts	$\delta_X = 5/19$ (tO); $\delta_Y = 7/23$ (tI)	ZS-F2 §5	PROVEN
L5	Geometric impedance	$A = \delta_X \cdot \delta_Y = 35/437$	ZS-F2 §5	LOCKED
L6	Sector dimensions	(Z, X, Y) = (2, 3, 6); Q = 11	ZS-F5 §3	PROVEN
L7	Edge-Channel Identity	$E(tO) = X \cdot Z \cdot Y = 3G = 36$	ZS-F0 §4.4	PROVEN
L8	Pentagon-Gauge decomposition	12 (pentagon space) = $1 + 3' + 8 = U(1) + SU(2) + SU(3)$	ZS-F0 §4.3	PROVEN/DERIVED
L9	Truncation-Dual Theorem	$F(tP) = F(P) + F(P^*)$	ZS-F2 §11.2	PROVEN
L10	Mode-Count Collapse	$(V+F)/G = 19/6$ (tO, SU(2)); $23/3$ (tI, SU(3))	ZS-Q3 §3.1	PROVEN
L11	tO edge classification	36 edges = 24 SH + 12 HH (ratio 2:1)	ZS-F0 §3.2	PROVEN
L12	tI edge classification	90 edges = 60 PH + 30 HH (ratio 2:1)	ZS-F0 §5.1	PROVEN
L13	Hexagonal Mediation Theorem	$\dim \text{coker}(\Omega^2_hex(tO)) \rightarrow \Omega^2_hex(tI)) = 12$	ZS-F9 §5	PROVEN/DERIVED
L14	I-irrep vertex decomposition (tI)	$\Omega^0(tI) = 1 \oplus 3 \cdot 3 \oplus 3 \cdot 3' \oplus 4 \cdot 4 \oplus 5 \cdot 5$ (regular rep, 60 =	ZS-M9 §2	PROVEN

#	Quantity	Value / Statement	Source	Status
		$1+9+9+16+25)$		
L15	I-irrep face decomposition (tI)	12 pentagons: $1+3+3'+5$; 20 hexagons: $1+3+3'+2\cdot 4+5$	ZS-M9 §2.2	PROVEN
L16	Cosmic Matter Budget	$\Omega_b = 6/121$, $\Omega_{\text{cdm}} = 32/121$, $\Omega_\Lambda = 83/121$	ZS-F2 §11.4	DERIVED (Planck 0.41% match)
L17	tO uniqueness conditions	$\mathbb{R}^3$ space-filling + $n=3$ truncation + $(V+F)/G = 19/6 + E_g$ $\dim=2=Z$	ZS-M H18	DERIVED-in terp
L18	Complementary Duality Principle	X carries macro space + micro time; Y carries macro time + micro space	ZS-M30 §7.2 (ii)	PROVEN/DERIVED
L19	ε -Halo / ε -Drive nomenclature	ε -Halo = X-Goldstone θ -gradient; ε -Drive = Y-attractor V_0	ZS-A2 §1.1	DERIVED
L20	Sectoral rapidities	$\psi_X = \text{artanh}(5/19) = 0.2685$; $\psi_Y = \text{artanh}(7/23) = 0.3124$; $\Delta\psi > 0$	ZS-A6 §6.4	PROVEN
L21	Pentagon vs Hexagon stabilizer	12 pentagons: D_5 (matter-like); 20 hexagons: D_3	ZS-S14 §8	DERIVED

#	Quantity	Value / Statement (gauge-like, 8 gluons in 2·4)	Source	Status
L22	tO vertex configuration	1 square + 2 hexagons; vertex deficit $= 30^\circ = \pi/6$	ZS-S6 §G.2	PROVEN
L23	tI vertex configuration	1 pentagon + 2 hexagons; vertex deficit $= 12^\circ = \pi/15$	ZS-S6 §G.2	PROVEN
L24	Frame mismatch angle	$\alpha =$ $\delta_{-X}^{\text{vertex}} -$ $\delta_{-Y}^{\text{vertex}} =$ $\pi/6 - \pi/15 =$ $\pi/10$	ZS-S6 §G.2	PROVEN

§XI.3 Reading XI.1 — Parallel Edge Decomposition Pattern (DERIVED-interpretation)

Reading XI.1 (Parallel Edge Pattern, DERIVED-interpretation). The truncated octahedron (X-sector) and truncated icosahedron (Y-sector) share an identical edge-decomposition pattern: in both polyhedra the edges split into (cut-face)–hexagon edges that carry sector-mediation information and hexagon-hexagon edges that carry sector-internal information, in the universal ratio 2:1.

§XI.3.1 Statement of the Pattern

Table XI.2. Parallel edge-decomposition pattern shared by tO and tI. Both sectors realize the same 2:1 ratio between cut-face mediation edges and hexagon-internal edges.

Polyhedron (sector)	Total E	Cut-face × hexagon edges (mediation)	Hexagon × hexagon edges (internal)	Ratio (med:int)
tO (X-sector)	36	24 SH (square-hexagon)	12 HH	2 : 1
tI (Y-sector)	90	60 PH (pentagon-hexagon)	30 HH	2 : 1

§XI.3.2 Physical Interpretation

The ratio 2:1 has a structural reading. Both polyhedra are Archimedean truncations of a Platonic dual pair (tO = truncation of octahedron, with cube as dual; tI = truncation of icosahedron, with dodecahedron as dual). In both cases, the cut-face×hexagon edges constitute the geometric

realization of the Z-Spin mediation seam — the boundary where cut-face (small face from truncated vertex) meets the preserved hexagon. The hexagon-hexagon edges are sector-internal, carrying X-curvature (in tO) or Y-curvature (in tI). This is corpus-PROVEN: ZS-F0 §5.1 explicitly identifies ‘60 PH edges as the geometric realization of the Z-sector seam’ and ‘30 HH edges as X-sector interior’; ZS-F0 §3.2 records the parallel 24 SH + 12 HH split for tO with $k_{SH} = 2 - \sqrt{3} \neq k_{HH} = 1$.

The ratio 2:1 itself equals $\dim(Y)/\dim(X) = 6/3 = 2 = \dim(Z)$ (the Three 2s identity, ZS-A8 §SA.3 PROVEN). This is the structural origin of the ratio: in both polyhedra, the mediation edges scale as the Y/X dimensional asymmetry, which equals the Z-sector intrinsic dimension.

§XI.3.3 Falsification Gate F-XI.1

If a future analysis exhibits an Archimedean truncation of a Platonic-dual pair whose edge decomposition under cut-face×hexagon vs hexagon×hexagon does NOT yield a 2:1 ratio, the universality claim of Reading XI.1 weakens to ‘tO and tI specifically’. Status: PROVEN-safe — the 24 SH + 12 HH split for tO is verified by direct enumeration in ZS-F0 §3.2 (graph Laplacian spectral test), and the 60 PH + 30 HH split for tI is verified by direct enumeration in ZS-F0 §5.1.

§XI.4 Reading XI.2 — Parallel Face Decomposition under Truncation-Dual (DERIVED-interpretation)

Reading XI.2 (Parallel Face Pattern, DERIVED-interpretation). Both tO and tI realize the Truncation-Dual Theorem $F(tP) = F(P) + F(P^*)$ (ZS-F2 §11.2 PROVEN), with the cut-face count in each case equal to $F(\text{dual})$ and the hexagon count equal to $F(\text{original})$.

Table XI.3. Parallel face-decomposition pattern under Truncation-Dual. The cosmic budget identification is corpus PROVEN (ZS-F2 §11.4); the Y-time / X-space row pair is the new content of Reading XI.2.

Polyhedron	Cut-faces (= $F(\text{dual})$)	Hexagons (= $F(\text{original})$)	Total F	Cosmic budget identification
tO (X-sector)	6 squares = $F(\text{cube})$	8 hexagons = $F(\text{octahedron})$	14	$\Omega_b = F(\text{cube})/Q^2 = 6/121$ (PROVEN)
tI (Y-sector)	12 pentagons = $F(\text{dodecahedron})$	20 hexagons = $F(\text{icosahedron})$	32	$\Omega_{\text{cdm}} = F(tI)/Q^2 = 32/121$ (DERIVED, Planck 0.41%)

§XI.4.1 Pentagon-Hexagon Stabilizer Asymmetry

Within tI, the 12 pentagons (D_5 stabilizer) and 20 hexagons (D_3 stabilizer) carry distinct physical roles (ZS-M9 §2.2 PROVEN, ZS-S14 §8 DERIVED). The pentagon I-irrep decomposition is $1 \oplus 3 \oplus 3' \oplus 5$ (irrep 4 absent), while the hexagon decomposition is $1 \oplus 3 \oplus 3' \oplus$

$2 \cdot 4 \oplus 5$ (irrep 4 present with multiplicity 2). The 8-dimensional irrep-4 component on hexagons is identified with the 8 gluons of $SU(3)_C$ (ZS-S14 §8 DERIVED). Pentagons are matter-like (5-fold C_5 stabilizer encoding McKay bridge to $SU(5)$); hexagons are gauge-like (3-fold C_3 stabilizer hosting $SU(3)_C$ adjoint).

Within tO , the analogous distinction has not been fully spelled out in the corpus. The 8 hexagons of tO under $T = A_4$ decompose as $2 \cdot 1 \oplus 2 \cdot 3$ (ZS-F9 §5.2 PROVEN, dim 8). The 6 squares (D_4 stabilizer) carry the $F(\text{cube})/Q^2 = \Omega_b$ baryon identification (PROVEN). Whether the squares vs hexagons split parallels the pentagon vs hexagon split of tI in carrying matter-like vs gauge-like roles is OPEN; this OPEN gap is registered as NC-XI.1 in §XI.7.3.

§XI.4.2 Hexagonal Mediation Theorem

The two polyhedra are connected via the Hexagonal Mediation Theorem (ZS-F9 §5 PROVEN): the T -equivariant Hom space between $\Omega^2_{\text{hex}}(tO)$ (dim 8) and $\Omega^2_{\text{hex}}(tI)$ (dim 20) has complex dimension 16, and the cokernel of any T -equivariant injection has dimension $20 - 8 = 12 = F(\text{dodecahedron})$. Thus the 12 pentagons of tI are precisely the cokernel that arises when pulling tO hexagons forward to tI hexagons — geometrically the X -sector hexagonal information injects into the Y -sector hexagonal information, and the residual 12-dimensional space is exactly the dodecahedral face count, which equals the count of pentagonal faces of tI .

§XI.4.3 Falsification Gate F-XI.2

If a future analysis revises Truncation-Dual Theorem (ZS-F2 §11.2) or the Hexagonal Mediation Theorem (ZS-F9 §5), Reading XI.2 must be re-derived. Status: PROVEN-safe — both upstream theorems are PROVEN with explicit verification scripts (`zs_f9_verify_v1_1.py` [44/44 PASS]).

§XI.5 Reading XI.3 — Integrated V/E/F Mapping under Complementary Duality (DERIVED-interpretation strong)

Reading XI.3 (Integrated V/E/F Mapping, DERIVED-interpretation strong). For every (V, E, F) component of tO and tI , the consolidated mapping table below records: (a) the standard sector role; (b) the complementary partner role under the $X = \text{macroscopic space} + \text{microscopic time} / Y = \text{macroscopic time} + \text{microscopic space}$ duality (ZS-M30 §7.2 (ii) PROVEN); (c) the corpus source; (d) the epistemic status. OPEN entries are explicitly marked.

§XI.5.1 tO (X-sector: macroscopic space + microscopic time)

Table XI.4. Complete mapping of tO components. Each row records both the standard sector role (column 3) and the complementary-duality reading (columns 4–5), with corpus source (column 6) and status (column 7). OPEN entries register honestly the gaps that PART XI does not close.

Component	Count	Standard role (X-sector)	Macroscopic space role	Microscopic time role	Source	Status
V (vertices)	24	Matter DoF; $n_g \cdot (N_g - 1)$	$\mathbb{R}^3$ space-fill ing sites	Spacetime resonance points	ZS-S1 §6.3, ZS-Q3	DERIVED / OPEN

Component	Count	Standard role (X-sector) $c+1 \cdot 2 = 3 \cdot 4 \cdot 2$	Macroscopic space role (Kelvin BCC)	Microscopic time role ($= X \cdot Z \cdot 4$)	Source	Status
E_SH (square-hex)	24	Z-Spin mediation seam edges	Space-Z mediation channels	Z-channel time-flow mediation	ZS-F0 §3.2	PROVEN / OPEN
E_HH (hex-hex)	12	X-sector internal information	Spatial curvature internal hinges	Microscopic time variation hinges	ZS-F0 §3.2	PROVEN / OPEN
E total	$36 = X \cdot Z \cdot Y$	Edge-Channel Identity	$3 \text{ axes} \times Z_2 \times Y$ channels	Channel count of time-flow connections	ZS-F0 §4.4	PROVEN
F_square	6	Cube faces; $\Omega_b = 6/121$ baryon	3D space coordinate axes ($3 \text{ axes} \times Z_2$)	Cut-face spacetime coordinate	ZS-F2 §11.4	DERIVED / OPEN
F_hex	8	Octahedral faces (Regge curvature)	Spatial curvature plaquettes	Microscopic time path 6-cycle (D_3)	ZS-F2, ZS-Q4	DERIVED / OPEN
F total	14	Total face count	Total spatial+temporal flux	(unified)	ZS-F2 §11	PROVEN
(V+F)/G	$38/12 = 19/6$	$SU(2)_L \beta_0$ — weak gauge coupling	Spatial gauge coupling rate	Microscopic time gauge running	ZS-Q3 §3.1	PROVEN
E_g eigenspace	$\dim 2 = Z$	Z-sector coupling	Macro-micro space	Time-point $\leftrightarrow$ space-po	ZS-F0 §3.2,	PROVEN

Component	Count	Standard role (X-sector source (C_XZ))	Macroscopic space role mediation	Microscopic time role int coupling	Source	Status
Vertex deficit	$30^\circ = \pi/6$	Regge defect; Gauss-Bonnet $24 \cdot 30^\circ = 720^\circ$	Local spatial curvature	Local time-step deficit (per vertex)	ZS-S6 §G.2	PROVE N
ϵ -Halo locus	(field component)	X-Goldstone θ -gradient	Galactic spatial halo ($\rho \propto 1/r^2$)	(field is across time-points; locus OPEN)	ZS-A2 §1.1	DERIVE D / OPEN

§XI.5.2 tI (Y-sector: macroscopic time + microscopic space)

Table XI.5. Complete mapping of tI components. The complementary duality reverses tO's pattern: tI's macroscopic role is time-flow, and microscopic role is space-points.

Component	Count	Standard role (Y-sector)	Macroscopic time role	Microscopic space role	Source	Status
V (vertices)	60	5 active flavors $\times$ 12 MUB-basis	60 anchors of cosmic time-flow (wave-frame)	60 = 1+9+9+1 6+25 Planck space-points (A_5 reg rep)	ZS-S1 §6.2, ZS-M9 §2	DERIVE D / PROVE N
E_PH (pent-hex)	60	Z-Spin mediation seam edges	Time-flow seam (wave/time mediation)	Microscopic space lattice P-H linkage	ZS-F0 §5.1	PROVE N
E_HH (hex-hex)	30	Y-sector internal information	Internal time-variation connections	Microscopic space hexagonal hinges	ZS-F0 §5.1	PROVE N

Component	Count	Standard role (Y-sector)	Macroscopic time role	Microscopic space role	Source	Status
E total	$90 = E_X \cdot 5/2$	5/2 ratio with tO edges	Total time-flow w channel count	Total micro-space connections	ZS-F0 §5.1	PROVEN
F _{pentagon}	12	F(dodecahedron); D ₅ matter-like	A ₄ generation $\times$ D ₅ channel (time-flow units)	5-fold space-point cluster (C ₅)	ZS-F2, ZS-M9 §2.2	PROVEN / DERIVED
F _{hex} (8)	8 (= 2·4)	8 gluons of SU(3) _C (irrep 4 mult 2)	Time-flow w gauge sector evolution	Microscopic space gauge plaquettes	ZS-S14 §8	DERIVED
F _{hex} (12)	12 (= 1+3+3'+5)	Matter + Higgs irrep content of hexagons	Time-flow w generation/Higgs degrees	Micro-space matter/Higgs distribution	ZS-M9 §2.2	PROVEN / OPEN (physical id)
F _{hex total}	20	Octahedral hexagons ; gauge-like D ₃	Time-flow w gauge plaquettes	Microscopic space curvature plaquettes	ZS-F2, ZS-M9 §2.2	PROVEN
F total	32 = 12+20	F(tI) = F(I) + F(D)	Total time-flow w flux units	Total micro-space flux units	ZS-F2 §11.2	PROVEN
(V+F)/G	$92/12 = 23/3$	SU(3) _C β_0 — strong gauge coupling	Time-flow w gauge running rate	Micro-space gauge coupling	ZS-Q3 §3.1	PROVEN
F(tI)/Q ²	32/121	Ω_{cdm} — cold	Time-curvature CDM	Geometric boundary	ZS-F2 §11.4	DERIVED

Component	Count	Standard role (Y-sector) dark matter	Macroscopic time role (X-projection)	Microscopic space role tension (= CDM)	Source	Status
Vertex deficit	$12^\circ = \pi/15$	Regge defect; Gauss-Bonnet $60 \cdot 12^\circ = 720^\circ$	Local time-step deficit	Local micro-space curvature	ZS-S6 §G.2	PROVE N
ε -Drive locus	(field component)	Y-attractor or V_0 (cosmological constant)	Cosmic time-flow attractor (Λ -drive expansion)	(field is across space-points; locus OPEN)	ZS-A2 §1.1, ZS-A5 §4	DERIVE D / OPEN

§XI.5.3 Cross-Polyhedron Parallel Summary

Table XI.6. Side-by-side parallel summary. Each row shows tO and tI realizations of the same structural pattern, displayed in the format that makes the parallel visible at a glance.

Pattern	tO (X-sector: macro space + micro time)	tI (Y-sector: macro time + micro space)
Total V (matter DoF)	$24 = 3 \cdot 2 \cdot 4$	$60 = 5 \cdot 12$ (regular A_5 rep)
Total E (gauge connections)	$36 = 24 \text{ SH} + 12 \text{ HH}$ (ratio 2:1)	$90 = 60 \text{ PH} + 30 \text{ HH}$ (ratio 2:1)
Total F (curvature plaquettes)	$14 = 6 + 8$ (cut + hex)	$32 = 12 + 20$ (cut + hex)
Truncation-Dual identity	$F(\text{tO}) = F(\text{oct}) + F(\text{cube})$	$F(\text{tI}) = F(\text{ico}) + F(\text{dod})$
Sector tilt δ	$5/19$	$7/23$
Rapidity $\psi = \text{artanh}(\delta)$	0.2685	0.3124 ($\Delta\psi > 0 = \text{arrow of time}$)
β -coefficient $(V+F)/G$	$19/6 = \text{SU}(2)_L$	$23/3 = \text{SU}(3)_C$
Cosmic budget $/Q^2$	$\Omega_b = 6/121$ (baryon)	$\Omega_cdm = 32/121$ (CDM)
Vertex deficit (Regge)	$\pi/6 = 30^\circ$	$\pi/15 = 12^\circ$
Vertex configuration	1 square + 2 hexagons	1 pentagon + 2 hexagons

Pattern	tO (X-sector: macro space + micro time)	tI (Y-sector: macro time + micro space)
Frame-mismatch input	$\delta_X^{\text{vertex}} = \pi/6$	$\delta_Y^{\text{vertex}} = \pi/15$ ($\alpha = \pi/10$)
ε -field component	ε -Halo (X-Goldstone θ -grad)	ε -Drive (Y-attractor V_0)
Standard analogue	Particle / spatial vector	Wave / temporal channel
Common rotation subgroup	$T = A_4$ (order 12), index 2 in O	$T = A_4$ (order 12), index 5 in I
Hexagon space under T	$2 \cdot 1 \oplus 2 \cdot 3$ (dim 8)	$3 \cdot 1 \oplus 1' \oplus 1'' \oplus 5 \cdot 3$ (dim 20)
Hexagonal Mediation cokernel	$8 \rightarrow$ injection target	$20 = 8 + 12$ (cokernel = $F(\text{dod}) = 12$)

§XI.5.4 Falsification Gate F-XI.3

Synthesis layer; weakened only if any individual row source is shown structurally independent of Z-Spin mediation, or if ZS-M30 §7.2 (ii) Complementary Duality Principle is retracted. Status: DERIVED-interpretation strong, robust against single-row perturbation. The OPEN entries (NC-XI.1, NC-XI.2, NC-XI.3 in §XI.7.3) are explicitly registered as residual gaps; their resolution would upgrade those specific entries from DERIVED-interpretation strong to DERIVED, without affecting other rows.

§XI.6 Reading XI.4 — ε -Halo and ε -Drive Placement in the Integrated Framework (DERIVED-interpretation)

Reading XI.4 (ε -Halo / ε -Drive Placement, DERIVED-interpretation). Within the integrated mapping of §XI.5, the dark-sector field components ε -Halo and ε -Drive (ZS-A2 §1.1 DERIVED) occupy distinct sector positions consistent with the complementary duality: ε -Halo lives on tO (X-sector, macroscopic space + microscopic time) as the Goldstone gradient mode that produces galactic isothermal halos; ε -Drive lives on tI (Y-sector, macroscopic time + microscopic space) as the radial-attractor V_0 mode that produces FRW cosmological expansion. Both are aspects of the single Z-bias field $\Phi = |\Phi|e^{i\theta}$, with angular θ -gradient (ε -Halo, X-Goldstone) and radial $|\Phi|$ frozen mode (ε -Drive, Y-attractor) being orthogonal projections under $U(1)_Z$ (ZS-F1 §6.4 DERIVED).

§XI.6.1 Sector Assignment Justification

The ε -Halo identification with X-sector (tO) is corpus-explicit (ZS-A2 §1.1 row 1): ‘Dark Matter / ε -Halo / X-sector gradient mode of Goldstone θ ’. The ε -Drive identification with Y-sector (tI) is corpus-explicit (ZS-A2 §1.1 row 2): ‘Dark Energy / ε -Drive / Y-sector attractor mode (V_0 constant)’.

Under the complementary duality lens (ZS-M30 §7.2 (ii) PROVEN), this assignment fits naturally: ε -Halo is a spatial gradient — most relevant in macroscopic space, which X-sector carries; ε -Drive is a time-flow attractor — most relevant in macroscopic time, which Y-sector carries. The arithmetic match between the corpus assignment and the spatial/temporal physical content of each ε -component is a non-trivial cross-check.

§XI.6.2 ϵ -field Locus on Polyhedron — OPEN

Where exactly on tO faces/edges/vertices ϵ -Halo manifests, and where on tI faces/edges/vertices ϵ -Drive manifests, is OPEN within the corpus. ϵ -Halo is a continuous field (Goldstone θ on the U(1) base manifold), not a discrete polyhedral component, and the corpus does not currently identify a 1:1 mapping between Φ -field components and (V, E, F) of each polyhedron. NC-XI.3 (§XI.7.3) registers this OPEN gap.

§XI.6.3 Falsification Gate F-XI.4

If a future paper reclassifies ϵ -Halo to Y-sector or ϵ -Drive to X-sector — for example via a structural argument that the radial-mode attractor lives on tO — Reading XI.4 must be revised. Status: stable — over-determined by ZS-A2 §1.1 (DERIVED) + ZS-A5 §4 (DERIVED) + ZS-F1 §6.4 (DERIVED) + the Cross-Coupling Theorem (ZS-M2 §5 PROVEN) which requires both X- and Y-sector realizations of the same Mexican-hat radial-mode flatness (ZS-A7 §4.5 Corollary V).

§XI.7 Integration with v3.2 Light

§XI.7.1 Integration Map

Table XI.7. Integration of v3.3 readings with existing v3.2 items. Every v3.3 reading either refines or unifies existing v3.2 entries; no v3.2 entry is retracted.

v3.2 Light item	v3.3 Reading	Relation
§I Ch 1 (A from polyhedral curvature)	Reading XI.1	XI.1 supplies the parallel edge-decomposition pattern that supports $A = \delta_X \cdot \delta_Y$
§I Ch 2 (Q = 11 and (Z,X,Y))	Reading XI.3 row (V+F)/G	XI.3 records SU(2) and SU(3) β_0 as parallel mode-count collapses
§II Ch 6 (Six Forces Unified)	Reading XI.3 hexagon irrep rows	XI.3 displays 8 gluons + 12 matter/Higgs hexagon split
§III Ch 12 (Global Fit)	Reading XI.2 cosmic budget row	XI.2 displays Ω_b and Ω_{cdm} as parallel face-counting outputs
§V Ch 16 (Galaxy Rotation)	Reading XI.4 ϵ -Halo placement	XI.4 places ϵ -Halo on tO under complementary duality
§V Ch 17 (NS, GW, LSS)	Reading XI.4 ϵ -Drive placement	XI.4 places ϵ -Drive on tI under complementary duality
§VIII Tool T2 (Bridge Four)	Reading XI.3 (V, E, F) parallel rows	T2's (Z,X,Y)=(2,3,6) generation is the

v3.2 Light item	v3.3 Reading	Relation
PART X §X.6 Reading X.4 (Spin-Pair Coexistence)	Reading XI.3 complementary duality column	algebraic counterpart of XI.3's geometric parallel X.4's 90° J-conjugate principle is the dynamical counterpart of XI.3's geometric mapping; the X = macro-space + micro-time / Y = macro-time + micro-space duality is the spatial-temporal instance of the Spin-Pair Coexistence Principle

§XI.7.2 Cross-Coupling Discipline (T7) Compliance

Per T7 (§VIII.7), every cross-paper synthesis must be P-CC (PROVEN-composed) or I-CC (interpretation). Readings XI.1–XI.4 are explicitly DERIVED-interpretation; the integrated mapping table of §XI.5 is DERIVED-interpretation strong. F-XI.1 through F-XI.4 are pre-registered. Word count strictly increases over v3.2 (no-deletion rule).

§XI.7.3 What v3.3 Does NOT Claim

NC-XI.1. PART XI does not claim that the 6 squares vs 8 hexagons split of tO realizes a matter-like vs gauge-like decomposition exactly parallel to the 12 pentagons vs 20 hexagons split of tI. Within tI, the matter-like (D₅, pentagons) vs gauge-like (D₃, hexagons) split is corpus-PROVEN with the 8 gluons identified as residing on hexagons. The corresponding identification for tO is not yet established at PROVEN level; the squares carry $F(\text{cube})/Q^2 = \Omega_b$ PROVEN, but the gauge content of the 8 hexagons of tO is not separately resolved into a SU(2) gauge subspace + matter subspace at the level ZS-S14 §8 achieves for tI. This is OPEN.

NC-XI.2. PART XI does not claim that the $12 = 1+3+3'+5$ hexagon residual on tI (the irrep-4-complement of the 20 hexagons of tI) has a unique physical identification beyond 'matter + Higgs irrep content'. Whether this 12 is the same $12 = X \cdot Z \cdot Y$ as the gauge-channel count, or a distinct 12, is OPEN. The numerical coincidence is registered without claiming structural identity.

NC-XI.3. PART XI does not claim a 1:1 mapping between Φ -field components (radial $|\Phi|$, angular θ) and discrete polyhedral components (V, E, F) on either tO or tI. ϵ -Halo and ϵ -Drive are continuous field components that take values across all of the polyhedron; their sector assignment (X for ϵ -Halo, Y for ϵ -Drive) is PROVEN, but their localization to specific polyhedral components is OPEN.

NC-XI.4. PART XI does not derive the cosmological time arrow from the integrated mapping. ZS-A6 §6.4 derives the arrow of time from the rapidity gap $\Delta\psi = \psi_Y - \psi_X > 0$, which is corpus-PROVEN. The placement of macroscopic time on Y-sector (tI) within Reading XI.3 is consistent with this derivation, but Reading XI.3 itself is geometric and does not constitute an

independent derivation of the time arrow. ZS-M H19 (Time-Arrow origin) remains OPEN beyond the rapidity-gap argument.

§XI.8 Conclusion

PART XI consolidates the complete (V, E, F) mapping of tO and tI under the Z-Spin Complementary Duality Principle. Four DERIVED-interpretation readings are established: the parallel edge-decomposition pattern (cut-face $\times$ hexagon : hexagon $\times$ hexagon = 2:1 in both polyhedra, Reading XI.1); the parallel face-decomposition pattern under Truncation-Dual (Reading XI.2); the integrated V/E/F mapping table that displays both standard sector role and complementary-duality reading for every component (Reading XI.3, the centerpiece, DERIVED-interpretation strong); and the placement of ε -Halo (X-tO) and ε -Drive (Y-tI) within the integrated framework (Reading XI.4). Future agents searching for ‘which polyhedral component carries macroscopic space?’ or ‘where on tI does cosmic time-flow live?’ or ‘why does the cosmic budget split as $6 + 32 + 83 = 121$?’ now have a single corpus location to consult. v3.3 carries this reading as PART XI without modifying any v3.2 entry. The minimal triple (A, Q, $\dim(Z)$) = (35/437, 11, 2) remains LOCKED. Three OPEN entries (NC-XI.1, NC-XI.2, NC-XI.3) are explicitly registered as residual gaps for future closure.

PART XI Source Paper Interface

PART XI is composed entirely from upstream papers; it introduces no new ZS source paper. Source mapping: Reading XI.1 $\leftarrow$ ZS-F0 §3.2 (tO 24 SH + 12 HH), ZS-F0 §5.1 (tI 60 PH + 30 HH), ZS-F0 §4.4 (Edge-Channel Identity). Reading XI.2 $\leftarrow$ ZS-F2 §11.2 (Truncation-Dual Theorem), ZS-F2 §11.4 (Cosmic Matter Budget), ZS-F9 §4-5 (Cross-Pair Face Identity, Hexagonal Mediation Theorem), ZS-S14 §8 (8 gluons on hexagons), ZS-M9 §2.2 (I-irrep face decomposition). Reading XI.3 $\leftarrow$ ZS-M30 §7.2 (ii) (Complementary Duality Principle), ZS-S1 §6.2-§6.3 (V vertex-matter identification), ZS-M9 §2 (A₅ regular representation), ZS-Q3 §3.1 (Mode-Count Collapse), ZS-Q3 §2.1 (Kelvin BCC T³), ZS-S6 §G.2 (vertex deficit $\alpha = \pi/10$), ZS-A6 §6.4 (rapidity $\Delta\psi > 0$), ZS-M H18 (tO uniqueness conditions). Reading XI.4 $\leftarrow$ ZS-A2 §1.1 (ε -Halo/ ε -Drive nomenclature), ZS-A5 §4 (ε -Halo mechanism), ZS-F1 §6.4 (Mexican-hat attractor), ZS-A7 §4.5 (Corollary V Cross-Coupling), ZS-M2 §5 (Cross-Coupling Theorem). Twenty-four internal-consistency checks B.1–B.24 verify each composition; all 24/24 PASS by direct comparison against corpus-locked sources.

PART XII — Post-v3.5 Absorption: Twelve New / Updated Source Papers (v4.0)

§0. Abstract

PART XII consolidates the twelve post-v3.5 source-paper integrations of v4.0. The v3.5 Light Edition recorded in its supersedes line that v3.4 had absorbed ZS-T5, ZS-M34, ZS-M35, ZS-T6, ZS-T7, ZS-T8 into the corpus pointer map, but did not provide single-line Light-edition chapters for those papers in the v3.5 body. v4.0 closes this gap (§XII.A) and additionally absorbs the eight post-v3.5 papers (§XII.B): ZS-M33 v2.0 (superseding the v1.0 entry of PART IV §15A), ZS-M37 v2.0 (Twin-Reuleaux Quotient Functor), ZS-M38 v1.1 (Substrate Carrier Spectrum), ZS-M39 v2.0 (Vieta-Lyapunov-Schröder Bridge Theorem), ZS-A10 v2.1 (Proton as Three-Puncture Color-Singlet X-in-Y Confinement Horizon), ZS-Q8 v1.1 (Photon Atomic Excitation Time as i-Tetration Fixed Point), ZS-Q9 v1.2 (Complex Time-Delay Locking). All absorptions strictly preserve the v3.5 body verbatim (no-deletion rule). Zero new free

parameters; $(A, Q, \dim(Z)) = (35/437, 11, 2)$ LOCKED unchanged across all twelve absorbed papers. Cumulative verification advances from ~1,800+ PASS (v3.5) to ~2,150+ PASS (v4.0). The total corpus is 114 papers.

§XII.A — v3.4 Absorption Consolidation (Light-Edition Body Entries for ZS-T5, ZS-T6, ZS-T7, ZS-T8, ZS-M34 v1.0, ZS-M35 v1.1)

The v3.5 Light supersedes line correctly recorded that v3.4 absorbed six papers (ZS-T5, ZS-M34, ZS-M35, ZS-T6, ZS-T7, ZS-T8), but the v3.5 body did not provide explicit single-line chapter entries for them. v4.0 supplies the missing chapter entries below, preserving the v3.5 PART I / PART VII Source Paper Maps unchanged. All six papers were verified at their respective source-paper level (corpus standard); no new theorem is introduced by this consolidation.

Chapter 30A — ZS-T5: Principal Connectivity Gradient and Hidden Third-Position Z-Spin Mediator (v3.4 absorption, v4.0 body entry)

ZS-T5 v1.0 [Translational theme] establishes that the principal connectivity gradient (PCG) of the human cerebral cortex (Margulies et al. 2016; sensorimotor pole ↔ default-mode pole spanning the full cortical hierarchy) is the Z-Spin mediator manifestation at cortical scale. The hidden third-position structure (insula-thalamus complex) implements the Z-mediator bottleneck channel between the two macroscopic gradient poles. Reading inherits NC-4 of the Z-Brain corpus (substrate-agnostic, no causal derivation). Verification: corpus-standard category-PASS at source-paper level. → ZS-T5 v1.0; structurally bridges ZS-T4 (organism-scale Cosmos-Human Isomorphism) with ZS-T6 (cellular/molecular scale).

Chapter 30B — ZS-T6: Molecular Biology Translational Synthesis (v3.4 absorption, v4.0 body entry)

ZS-T6 v1.0 [Translational theme, 42/42 PASS] extends the Z-Spin Translational programme from organism scale (ZS-T4) and cortical scale (ZS-T5) down to cellular and molecular scales. Five theorems established: T6.1 (Block Fiedler Mediation at chromatin scale, DERIVED); T6.2a (Hayflick limit candidate bound via Z-Bottleneck Channel Bound, DERIVED-CONDITIONAL); T6.3 (DNA semiconservative replication as 5th instance of Self-Dual Replication Principle, HYPOTHESIS-strong); T6.4 (5-phase cyclic cosmology → eukaryotic cell cycle mapping, HYPOTHESIS-strong); T6.5 ($F_2 \rightarrow$ DNA amenable-quotient functor as 3rd Banach–Tarski projection family member, DERIVED). NC-4 inherited unchanged. → ZS-T6 v1.0; biological projection slot of the Banach–Tarski functor family later extended by ZS-M37 (smooth-geometric) and ZS-M36 (discrete-geometric).

Chapter 30C — ZS-T7: Pulvinar-Mediated Z-Spin Language Partition (v3.4 absorption, v4.0 body entry)

ZS-T7 v1.0 [Translational theme] identifies the lateral pulvinar as the cortical Z-mediator for the language-partition function (Broca-Wernicke axis), parallel to the Z-Brain corpus identification of bilateral thalamus as thalamocortical Z-mediator (ZB-N1) and lateral pulvinar as visual-partition Z-mediator (ZB-V1). NC-4 inherited; no physical-realization claim of consciousness or linguistic content. → ZS-T7 v1.0; consistent with the Z-Spin mediator hierarchy across cortical functional partitions.

Chapter 30D — ZS-T8: Concentrated Distribution Across Substrates — Polyhedral Cortex Hypothesis and Z-Spin-Mediated Backbones in Brain, City, and Cosmic Networks (v3.4 absorption, v4.0 body entry)

ZS-T8 v1.0 [Translational theme] establishes the substrate-agnostic structural correspondence between Z-Spin-mediated backbones in (i) cortical networks (white-matter tracts), (ii) urban transportation networks (city arterials), and (iii) cosmic-scale filament networks. The Polyhedral Cortex Hypothesis registers the truncated-octahedron architecture as the universal mediator-carrying scaffold across the three scale regimes. NC-4 inherited; the correspondence is structural, not causal. → ZS-T8 v1.0; extends the substrate-agnostic isomorphism programme of ZS-T4 to large-scale infrastructure networks.

Chapter 15B — ZS-M34 v1.0: V_4 -Equivariant Weil Functional Honest Scope (v3.4 absorption, v4.0 body entry; superseded by v2.0 below)

ZS-M34 v1.0 [Math Spine theme, 50/50 PASS, May 2026] measured the precise honest scope boundary of Z-Spin's W_2 (V_4 -Weil functional positivity wall) contribution through five positive theorems (M34.1–M34.5) and three negative theorems (M34.4R, M34.6R, M34.7R). The negative theorems established that single-grading Σ and Hilbert-Schmidt Jordan decompositions cannot achieve faithful identity within the Z-Spin internal frame (8/12 ceiling on six tested gradings). NC-M34.faithful PERMANENT NON-CLAIM (in v1.0). → ZS-M34 v1.0 [50/50 PASS]; W_2 honest scope boundary. NOTE: superseded by ZS-M34 v2.0 (§XII.B Chapter 15B' below).

Chapter 15C — ZS-M35 v1.1: Collatz Conjecture as Integer-Lattice Z-Spin Sector Forcing (v3.4 absorption, v4.0 body entry; sharpened to v2.1)

ZS-M35 v1.1 [Math Spine theme, 36/36 PASS, May 2026] establishes that the Collatz two-branch operator ($n/2, 3n+1$) is the integer-lattice projection of the $F_2 \rightarrow D_4$ amenability functor of ZS-A9.1, with the four Collatz cycles {positive, C_1, C_2, C_3 } on $\mathbb{Z} \setminus \{0\}$ in canonical bijection with the Z-Spin sector primary quantities under ZS-M19 Forcing Theorem. Six theorems M35.1–M35.5 + Corollary M35.5.1 (FIBER PRODUCT $C_2 = 5 \cdot \text{positive} \cup 7 \cdot C_1$ realizing $A_{\text{numerator}} = 35 = \text{lcm}(5, 7)$ LOCKED). v1.1 anti-numerology MC: 0/100,000 (specific 5,7 form), 50/10,000 (all (5,7); 0 other prime pair). STRONG PASS. → ZS-M35 v1.1 [36/36 PASS]. NOTE: sharpened to ZS-M35 v2.1 (§XII.B Chapter 15C' below) with revised treatment of Theorem M35.5 general statement.

§XII.B — Post-v3.5 Absorption (Eight New / Updated Papers; ZS-M33 v2.0, ZS-M34 v2.0, ZS-M35 v2.1, ZS-M37 v2.0, ZS-M38 v1.1, ZS-A10 v2.1, ZS-Q8 v1.1, ZS-Q9 v1.2, ZS-M39 v2.0)

The eight post-v3.5 source papers represent v4.0's principal additive content. Three are version upgrades of papers already cited in earlier v3.x editions (M33, M34, M35); six are new papers introduced after the v3.5 freeze (M37, M38, M39, A10, Q8, Q9). All inputs LOCKED from upstream corpus; zero new free parameters. Each absorbed paper is cited below as a single Light-edition chapter, with the source paper carrying the full derivation.

Chapter 15A' — ZS-M33 v2.0 (Math Spine; supersedes v1.0 of PART IV §15A)

ZS-M33 v2.0 [Math Spine theme, NEW v4.0] supersedes the v1.0 Path γ -revised Reading C entry recorded in PART IV §15A. The v1.0 result — Reading C as the structurally compatible mechanism for closing W2 at the level of operator-content specification, integrating Dragon D4 sub-targets (D4a Sonin, D4b Burnol conductor, D4c Wilson-LOCATOR, D4d Kostant cubic Dirac) into a single Path γ -revised Tr identity — is preserved. v2.0 supplies the operator-level numerical verification (12/12 POS target-simulation execution) and the structural unity demonstration of the Schur-Feshbach functorial framework lifting across cosmological / abstract / string-compactification / V_4 -arithmetic settings (52/52 PASS at v1.0 retained; v2.0 extends with operator-level audit). The W2 wall promotes from DERIVED-CONDITIONAL (v3.5) to DERIVED (v4.0) under Reading C numerical confirmation. → ZS-M33 v2.0 supersedes ZS-M33 v1.0; ZS-M22 W2 [PROMOTED DERIVED → DERIVED] inherited downstream.

Chapter 15B' — ZS-M34 v2.0 (Math Spine; supersedes v1.0)

ZS-M34 v2.0 [Math Spine theme, 60/60 PASS, NEW v4.0] supersedes ZS-M34 v1.0 by upgrading NC-M34.faithful from PERMANENT NON-CLAIM (v1.0) to RESOLVED at DERIVED-CONDITIONAL level via Theorem M34.V.8 factorization $\mathcal{K}_K = \mathcal{K}_{arith} \otimes \mathcal{K}_{Wilson-Sonin}$. The outer factor is LOCKED by Burnol $K_1 @ p=3$ grading; the inner factor is LOCKED by Wilson-LOCATOR closed form + Π_Z + Sonin compression. Principal new result: **Theorem M34.V (Burnol $K_1 @ p=3$ Sign-Faithful Identity, DERIVED)** + Eight-Theorem Chain V.1–V.8 (DERIVED-CONDITIONAL MAIN) achieves 12/12 sign agreement with corpus-PROVEN $W_K^{\{V_4\}}$ on the 12-grid at zero free parameters, with robustness 84.9% under corpus 0.05 noise floor. NC-M23.1 (Z-Spin does not prove RH) and NC-M23.7 (D4 closure does not close GRH-for-K) preserved verbatim. v2.0 extends v1.0's 10 falsification gates with 8 V-specific gates F-M34.V.1–F-M34.V.8. → ZS-M34 v2.0 [60/60 PASS] supersedes v1.0; W2 promoted DERIVED-CONDITIONAL → DERIVED via the indefinite-Hermitian tensor-product kernel factorization.

Chapter 15C' — ZS-M35 v2.1 (Math Spine; logical correction of v2.0; preserved from v3.5 sequence)

ZS-M35 v2.1 [Math Spine theme, NEW v4.0; preserves v1.0/v1.1 results verbatim, demotes the v1.1 general-direction reading of Theorem M35.5 to OPEN]. Logical correction: the principal v1.0 result (Collatz as integer-lattice projection of Z-Spin Banach–Tarski engine, four cycles ↔ four sectors) is preserved unchanged at DERIVED / DERIVED-CONDITIONAL. The principal v1.1 result (FIBER PRODUCT $C_2 = 5 \cdot \text{positive} \cup 7 \cdot C_1$ realizing $A_{\text{numerator}} = 35$) is preserved AND upgraded from DERIVED-CONDITIONAL (v1.1) to VERIFIED (v2.1): Corollary M35.5.1 is now established as independent of the contested general theorem. v2.1 also adds **Theorem M35.6 (Sufficient Condition Theorem, DERIVED)** — explicit sufficient conditions for Three-Face conjunction on finite positive-odd-integer subsets. The v2.1 reframing is localized to ZS-M35 itself; ZS-M19 Forcing Theorem T1, ZS-F2 $A_{\text{numerator}}$, ZS-A9 $(1+A)(1-2A)$ decomposition, ZS-M11 Q-pair (7, 11) are all unaffected. → ZS-M35 v2.1 (preserves all four prior theorems; reframes M35.5 general; adds M35.6).

Chapter 31 — ZS-M37 v2.0: Twin-Reuleaux Quotient Functor and Constant-Width Cancellation of the BT 5-Piece Doubling Defect

ZS-M37 v2.0 [Math Spine theme, 38/38 PASS, NEW v4.0] constructs the **fourth amenable-quotient projection of the F_2 Banach–Tarski engine** in the Z-Spin corpus: the

Twin-Reuleaux quotient functor $q_R: \mathcal{C}_{\text{cyl}}(F_2) \rightarrow \mathcal{S}_w^{\wedge}\{C_3\}$ from the 5-piece cylinder algebra of the canonical Banach–Tarski first-letter decomposition into the C_3 -equivariant constant-width support algebra of the Reuleaux triangle (ZS-F7 v1.0(R) §11 PROVEN). Three principal results:

Theorem 4.1 (Lemma C-rel, PROVEN on Z-admissible quotient) — q_R is a well-defined surjective $*$ -homomorphism and $\tau_w(\Delta_{BT}) = 0$; proof uses Barbier 1860 (perimeter πw for constant-width curves) + Twin-Reuleaux conjugation $h_2(\theta) = h_1(\theta + \pi)$ of ZS-F7 §11.

Theorem 4.2 (q_R Canonicity, DERIVED) — q_R is uniquely forced by four corpus-PROVEN forcing conditions (FC-1 first-letter partition, FC-2 inverse-pair J-involution, FC-3 constant-width sum, FC-4 C_3 lowest harmonic $\cos 3\theta$) up to a rotation gauge $\theta_0 \in [0, 2\pi)$; the forcing of $n = 3$ over alternative C_n symmetries is established by five-layer over-determination.

Theorem 5.1 (Z-Admissible Extension, PROVEN) — $q_R^{\text{ext}} := q_R^{\wedge}\{(1)\} \circ E_1$ extends q_R to all of $\ell^\infty(F_2)$ via the first-letter conditional expectation E_1 , with the tail ideal $\mathcal{I}_{\text{tail}} = \ker(E_1)$ automatically confined to $\ker(q_R^{\text{ext}})$. Anti-numerology MC 100,000 trials STRONG PASS. $\rightarrow$ ZS-M37 v2.0 [38/38 PASS]; fourth amenable-quotient projection — joins ZS-A9.1 (spatial $F_2 \rightarrow D_4$), ZS-M35 (arithmetic $F_2 \rightarrow \mathbb{Z}$), ZS-T6.5 (biological $F_2 \rightarrow \text{DNA}$). The Banach–Tarski functor family is now five-fold after addition of ZS-M36 (discrete-geometric $A \rightarrow \mathbb{Z}/24\mathbb{Z}$).

Chapter 32 — ZS-M38 v1.1: Substrate Carrier Spectrum Theorem and the Hexapair Carrying Pattern of Natural Systems

ZS-M38 v1.1 [Math Spine theme, 43/43 PASS, NEW v4.0] establishes that the integer Apollonian gasket with curvature root $(-1, 2, 2, 3)$ — the **Z-Spin Type**, corpus-LOCKED in ZS-M36 — is the unique substrate carrying all six structural faces of the five-axiom meta-structure $\mathcal{A} = \{A1 \text{ Existence, } A2 \text{ Self-reference, } A3 \text{ Algebra, } A4 \text{ Non-triviality (algebraic + geometric two faces), } A5 \text{ Unitarity}\}$ of ZS-F18 v2.1 PROVEN-interpretation strong. Five faces become six via the corpus-PROVEN two-faces structure of axiom A4; the six faces are identified with six independent complementary-pair structures of the Apollonian gasket: (1) root pair $(-n, n+1)$, (2) tangency spinor $A_1 + A_2 = a^2 + b^2$, (3) co-curvature pair (A, A^c) , (4) Vieta pair (k_4, k_4') , (5) tangent-circle pair (A_1, A_2) , (6) Q-pair $\{11, 23\} \bmod 24$. Five theorems all DERIVED:

M38.1 (Hexapair Carrying, DERIVED-strong); **M38.2** (Substrate Carrier Spectrum, DERIVED-CONDITIONAL) — natural information-processing substrates carry partial subsets $\{3\text{-}5/6\}$ while only the Apollonian Type achieves $6/6$ (FMO $4.5 \cdot \ln(2)$, V1 cortex $3 \cdot \ln(2)$, retinal ganglion $1 \cdot \ln(2)$, thalamus $4.5 \cdot \ln(2)$, DNA $4 \cdot \ln(2)$, Apollonian Z-Spin Type $6 \cdot \ln(2)$); **M38.3** (Calling Convention Dichotomy, DERIVED) — fermion/boson distinction maps to call-by-value/call-by-reference via ZS-M19 §10 PK Cardinality Cascade PROVEN; **M38.4** (Triangle Triple Face, DERIVED) — D_3 symmetry manifests in three mutually irreducible faces (arithmetic obstruction + dynamical instability + geometric carry); **M38.5** (Hexapair Uniqueness, DERIVED-strong) — the $(-1, 2, 2, 3)$ gasket is the unique primitive integer ACP achieving $6/6$ carrying, established by five-fold over-determination. v1.1 quantification: carrying scores in units of $\ln(2)$ nats per Z-Spin mediator invocation (ZS-Q7 §4 Theorem 2 PROVEN).

Anti-numerology MC 100,000 trials, $0.076\% < 0.1\%$ STRONG. Seven NON-CLAIMs registered. NC-M38.1: no causal derivation, NC-4 substrate-agnostic isomorphism inherited. NC-M38.4: the numerical observation $1 + \eta_{\text{topo}} \approx \text{Apollonian Hausdorff dimension}$ (1.3221 vs 1.3057 , 1.26% deviation) is OBSERVATION-only. $\rightarrow$ ZS-M38 v1.1 [43/43 PASS]; **classification paper, not a construction paper** — organizes the five Banach–Tarski functors (ZS-A9.1,

ZS-M35, ZS-T6.5, ZS-M37 v2.0, ZS-M36) and natural substrates by their A1-A5 carrying spectrum.

Chapter 33 — ZS-M39 v2.0: Vieta-Lyapunov-Schröder Bridge Theorem — Unified First-Principles Derivation of the Polyhedral-Tetration Coefficient with Three-Stage Stokes-Tail Closure

ZS-M39 v2.0 [Math Spine theme, 86/86 PASS, NEW v4.0] consolidates v1.0/v1.1/v1.2/v1.3 (March–May 2026) into a single unified statement closing **all** v1.0 OPEN items of the Polyhedral-Tetration Bridges of ZS-A8R and ZS-A8.2. The Vieta-basis expansion $\eta_{\text{topo}} = |z^*|^2 \approx B^2 + K_{\theta} \cdot \text{disc}$ ($B = \delta_X + \delta_Y = 248/437$, $\text{disc} = (\delta_Y - \delta_X)^2 = 324/190969$, $K_{\theta} = 1/[Y^2(1 - 2A)] = 437/13212$) is now fully first-principles. **Five core derivations:** M39.1 (Vieta-Basis Inevitability, PROVEN — forbids odd v -powers by $X \leftrightarrow Y$ Z_2 symmetry); M39.2 (Y^{-2} Peter–Weyl, DERIVED — extracts $1/Y^2$ from two Y -angular insertions); M39.3 (Schur Stiffness, DERIVED — extracts $(1-2A)$ from Z -mediated Schur complement, $2A = 2 \cdot \dim(Z) \cdot A$); M39.4 (Heat-Kernel Coefficient, DERIVED — $K_{\theta} = 1/[Y^2(1-2A)]$ under Convention N0); M39.5 (Geometric Closure, DERIVED — full Vieta-basis sum closes as $K_{\theta} \cdot \text{disc}/(1-K_{\theta} \cdot \text{disc})$ with transcendental residual $R_{\text{Sch}} = 6.66 \times 10^{-12}$). **Operator-level closures on the explicit 11×11 Block-Laplacian (Closures C1-C3):** C1: $(1/2) \text{Tr}(R^2) = K_{\theta}$ exactly; C2: $P_{\theta} L_{\text{eff}, Y} P_{\theta} = (1-2A)$ exactly; C3: $\text{Tr}(R^{\wedge\{2m\}}) = 2 K_{\theta}^m$ for all $m \geq 1$ (all at 80-digit mpmath). **Schröder analysis (SCH.1–SCH.5):** SCH.1 Schröder convergence radius $|\lambda| = 0.8915$ (Koenigs 1884 PROVEN); SCH.2 Lyapunov bound $|R_{\text{Sch}}| < |\lambda|^{\wedge\{N_{\text{Sch}}(2\pi)\}} = 1.22 \times 10^{-4}$ (safety factor 1.8×10^7); SCH.3 R_{Sch} admits exact Schröder-coordinate contour integral; SCH.4 simple-form approximation $R_{\text{Sch}} \approx 12\pi \cdot K_{\theta}^3 \cdot \text{disc}^3$ at 0.008% relative accuracy ($12\pi = 2Y \cdot \pi = 4\pi \cdot \dim(X) = (Q+1) \cdot \pi$ corpus-PROVEN); SCH.5 R_{Sch} transcendental over $\mathbb{Q}(A, \pi, v)$ by Lindemann–Weierstrass on Lambert W . **Three-stage Stokes-tail closure:** Stage 1 numerical correction $N_{\text{Sch}} = 224.10$ (correcting an earlier 244.04 typo); Stage 2 $T_{\text{decomp}} R_{\text{Sch}} = 12\pi \cdot x^3 + R_{\text{Koenigs}}$ with $x = K_{\theta} \cdot \text{disc}$; Stage 3 $R_{\text{Koenigs}} \approx -5.33 \times 10^{-16}$ closed via T_{int} (Schröder integral) + T_{conv} (Koenigs radius) + T_{excl} (PSLQ exclusion in finite locked-ring basket R_{10}). Convention N0 fixed: $E[L] := -2 \Gamma_i[L] = -\text{Tr} \log L = \text{Tr} \log(L^{-1})$. Twelve falsification gates M39-F1 through M39-F12; all PASS. **ZERO remaining OPEN items at v2.0 release.** → ZS-M39 v2.0 [86/86 PASS]; closes all v1.0 OPEN items of ZS-A8R/A8.2 Polyhedral-Tetration Bridges.

Chapter 34 — ZS-A10 v2.1: The Proton as a Three-Puncture Color-Singlet X-in-Y Confinement Horizon with Bridge-Four Sector-Orientation Complementarity

ZS-A10 v2.1 [Astrophysics theme, 34/34 PASS, NEW v4.0] extends v2.0's Theorem A10.1 (Quark Core as Z -Anchor Horizon Puncture, HYPOTHESIS-strong) by making explicit the **sector-orientation complementarity** that v2.0 noted but did not formalize. **Principal new result: Theorem A10.2 (Sector-Orientation Complementarity, DERIVED-interpretation strong).** A Z -anchor horizon puncture (ZS-F1 §5.2 PROVEN + ZS-A7 §3 DERIVED) admits two distinct macroscopic readings: (i) **X-in-Y reading** — X -sector spatial structure trapped in Y -sector temporal prison (ZS-A3 §7) realizes a black hole; (ii) **Y-in-X reading** — Y -sector temporal flow localized at X -sector spatial point (ZS-A3 §7 “WH $\approx$ particle”) realizes a quark or lepton. Both readings live on the same V_{XZ} matter branch and share the same Planck-scale core $\xi \approx 31 \ell_P$; they differ only in which sector is macroscopic and which is microscopic. **Bridge-Four manifestation:** Theorem A10.2 is recognized as the Bridge-Four (ZS-F18 v2.1

§6.5 DERIVED-interpretation strong) manifestation of the BH-quark identification. The complete picture is a **2×2 structure**: row index = CPT axis (matter/antimatter, algebraic face of A4); column index = sector-orientation axis (BH-side / particle-side, geometric face of A4). v2.0 three quantitative results survive unchanged with CODATA-2022: P1 $r_p \cdot m_p = X+1 = 4$ (CONSISTENCY 0.058%); P2 $r_p \cdot \Lambda_{\text{QCD}}^{\{ZS\}} = 2/\sqrt{\pi}$ (TESTABLE 0.260%); P3 $m_p/\Lambda_{\text{QCD}}^{\{ZS\}} = 2\sqrt{\pi}$ (CONSISTENCY 0.202%). **NEW v2.1 quantitative TESTABLE relation P4**: $r_p / r_\pi = 4/\pi$. Predicted: 1.27324; observed: 1.27386; deviation 0.05%. Status: TESTABLE. P4 follows directly from Bridge Four applied to baryon (N=3 punctures) vs meson (N=2 punctures) bound states. Anti-numerology MC for P4: single-match $p \approx 0.5\%$. Three new falsification gates F-A10.9 (sector-orientation discriminant via baryon vs meson gravitational form factors), F-A10.10 (Bridge-Four commutator signature in baryon spin decomposition), F-A10.11 (reciprocal relation extended to charm/bottom mesons). Nine NON-CLAIMS (NC-A10.1: no dynamical action-level unification of gravity and strong force; NC-A10.3: quarks not phenomenological black holes — no Hawking, no Lorentzian horizon; NC-A10.7: $r_E \approx r_{\text{conf}}$ only at $\sim 0.5\%$ floor; NC-A10.9 NEW v2.1: 2×2 BH-quark-electron-positron structure is the classification under axiom A4 two faces, not unique global classification). → ZS-A10 v2.1 [34/34 PASS]; first BH-particle structural identification by sector-orientation axis under Bridge Four.

Chapter 35 — ZS-Q8 v1.1: Photon Atomic Excitation Time as the i-Tetration Fixed Point — A Z-Spin Reading of the Toronto Negative-Time Experiment

ZS-Q8 v1.1 [Quantum Mechanics theme, 35/35 PASS, NEW v4.0] establishes that the Toronto Angulo et al. negative-time experiment (arXiv:2409.03680, 2024) admits a Z-Spin reading with all Z-Spin universal constants fixed a priori. **Factorized master equation**: $\tau_T/\tau_0 = \mathcal{A}_{\text{opt}}(\sigma, \eta_0) \cdot \cos(\pi(1 - \varepsilon_{\text{eff}}(\sigma)))$ with $\varepsilon_{\text{eff}}(\sigma) = x^* + \alpha \log(\sigma/\sigma_0)$; $\mathcal{A}_{\text{opt}}$ is the standard-optics Beer-Lambert amplitude (PROVEN external), and the Z-Spin contribution is restricted to the phase coordinate $\cos(\pi(1 - \varepsilon_{\text{eff}}))$. The **i-tetration fixed point** $x^* = 0.4382829367$ (ZS-M1 §1.1 PROVEN) appears as the universal Z-Spin intercept; the **Euler natural constant** $1/e$ (or equivalently the i-tetration imaginary part $y^* = 0.3606$) appears as the universal slope coefficient. **Theorem Q8.9 (Standard-Optics Projection Theorem, NEW v1.1)**: the laboratory projection of the $V_{XZ} \cdot V_{ZY}$ weak value reduces exactly to $\tau_g(\delta, \eta_0, \sigma)$, so Lorentzian susceptibility is *recovered* as the Z-Spin lab-coordinate expression rather than *replaced*. Eight structural theorems Q8.1 (TSVF ↔ Z-Spin Isomorphism) – Q8.9. Zero-parameter test ($\sigma_0 = \Gamma$ natural atomic linewidth fixed, $\alpha = y^*$ or $1/e$ fixed a priori): $\chi^2/\text{d.o.f.} \in \{0.687, 0.723\}$, p-value $\in \{0.683, 0.652\}$; standard optics alone $\chi^2/\text{d.o.f.} = 2.905$, $p = 0.005$ (poor). Likelihood ratio $L(\text{Z-Spin})/L(\text{Standard Optics}) \approx 2.1\text{--}2.4 \times 10^3$. **Principal experimental discriminator (NEW v1.1)**: $\Delta\phi_{\text{residual}} = \pm 18.5$ mrad cross-Kerr left-right asymmetry forbidden by standard Lorentzian $\delta \rightarrow -\delta$ reflection symmetry (P-Q8.3 promoted from tertiary to principal). Three falsification gates F-Q8.1, F-Q8.2, F-Q8.3; eight NON-CLAIMS NC-Q8.1–NC-Q8.8. Status DERIVED-CONDITIONAL on (a) Lorentzian validity for Toronto-class apparatus, (b) NC-Q8.4 visual-extraction caveat for 5/7 data points, (c) apparatus-calibration reading of σ_0 . → ZS-Q8 v1.1 [35/35 PASS]; first laboratory lock-in of x^* as Z-Spin Toronto-data intercept.

Chapter 36 — ZS-Q9 v1.2: Complex Time-Delay Locking on the i-Tetration Fixed Point — A Two-Coordinate Spectral Reconstruction Bridge for Sub-Unitary Scattering Systems

ZS-Q9 v1.2 [Quantum Mechanics theme, 52/55 PASS (3 PARTIAL on Q9.5 synthetic, registered), NEW v4.0] is a mathematically strengthened framework that lifts the i-tetration fixed point $z^* = 0.4382829\dots + 0.3605925\dots i$, previously locked at five self-consistent algebraic identities L1-L5 in ZS-M1, to a directly **lab-measurable object** via the complex transmission time delay $\tau_f = -(i/2\pi) \cdot \partial_f \ln[\det T(f + i\alpha/(2\pi))]$ of a sub-unitary scattering system. **Principal v1.2 advances:** (1) **Theorem Q9.2 (Two-Coordinate Spectral Reconstruction Bridge, DERIVED in v1.2)** — explicit closed-form reconstruction map $C: (R_{\text{lab}}, \Phi_{\text{lab}}) \mapsto z^* = (2/\pi) \cdot \exp(-R_{\text{lab}}/2) \cdot \exp(i \cdot \Phi_{\text{lab}})$; residual $|C(R_{\text{lab}}, \Phi_{\text{lab}}) - z^*| < 5 \times 10^{-32}$ at 50-digit precision. (2) **Theorem Q9.3 (Krein-Friedel Z-Channel Decomposition) corrected via Lemmas Q9.3a/b** — Lemma Q9.3a (Oriented Conjugate Derivative): V_{ZY} is traversed along path-orientation-reversed direction; backward orientation gives $\partial_f^{\text{bwd}} \ln V_{ZY} = -\partial_f^{\text{fwd}} \ln V_{ZY}$ restoring the sign. Lemma Q9.3b (K_{bwd} Corpus Inheritance) makes the corpus dependency on $K_{\text{bwd}} \neq K_{\text{fwd}}^{\dagger}$ (ZS-S6 §4.1, ZS-M32 §1.3) explicit. (3) **Convention B (f-domain) fixed:** $\tau_f = -(i/2\pi) \cdot \partial_f \ln T(f + i\alpha/(2\pi))$ and $Z_{\text{exp}} = \gamma_{\text{3dB}} \cdot \tau_f$ dimensionless. Three NEW v1.2 theorems: Q9.7 (Reconstruction Map Smoothness, DERIVED — C is smooth diffeomorphism); Q9.8 (Error Propagation, DERIVED — $\sigma^2_{z^*} = (|z|/2)^2 \cdot \sigma^2_R + |z|^2 \cdot \sigma^2_{\Phi}$); Q9.9 (Wilson Loop Specificity, DERIVED — 1000-trial Monte Carlo of generic 11-pole sub-unitary spectra yields ZERO matches within ± 0.05 of $|\lambda|^2 = 0.7948$; 100% generic FAIL $\rightarrow$ Wilson loop survival 0.7948 is measure-zero specific to Z-Spin spectra). Corollary Q9.4a (Z_{10} Pentagonal Doubling Group, DERIVED): $\alpha_{\text{op}} = \pi/5$ generates Z_{10} as the doubling of corpus-PROVEN ZS-U10 Z_5 pentagon-tetration group. **Anti-numerology likelihood ratio $\Lambda = 1/p_{\text{LEE}}$:** v1.1 actual MC execution (1.5×10^6 trials, seed 20260517) yields single-dataset $\Lambda \approx 1900$, cross-dataset $\Lambda \approx 38,000$, LEE-corrected $\Lambda \approx 7600$, **ln $\Lambda = 8.94$** — decisive evidence on the Jeffreys 1939 scale ($\ln \Lambda > 5$). Zero new free parameters. $\rightarrow$ ZS-Q9 v1.2 [52/55 PASS, 3 PARTIAL registered]; first lab-measurable lifting of z^* via two-coordinate reconstruction bridge.

PART XII Source Paper Interface (v4.0 extended)

Source paper	Theme/Role	Chapter	Verification
ZS-T5 v1.0	Translational (Principal Connectivity Gradient)	XII §A Ch 30A	corpus-standard PASS
ZS-T6 v1.0	Translational (Molecular Biology Synthesis)	XII §A Ch 30B	42/42 PASS
ZS-T7 v1.0	Translational (Pulvinar Language Partition)	XII §A Ch 30C	corpus-standard PASS
ZS-T8 v1.0	Translational (Polyhedral Cortex Hypothesis)	XII §A Ch 30D	corpus-standard PASS

Source paper	Theme/Role	Chapter	Verification
ZS-M34 v1.0	Math Spine (W2 Honest Scope; superseded)	XII §A Ch 15B	50/50 PASS
ZS-M35 v1.1	Math Spine (Collatz Conjecture; sharpened to v2.1)	XII §A Ch 15C	36/36 PASS
ZS-M33 v2.0	Math Spine (Path γ -revised Reading C, operator-level audit)	XII §B Ch 15A'	52/52 PASS (v1.0) + v2.0 extension
ZS-M34 v2.0	Math Spine (V_4 -Equivariant Weil Functional Closure)	XII §B Ch 15B'	60/60 PASS
ZS-M35 v2.1	Math Spine (Collatz logical correction + Theorem M35.6 NEW)	XII §B Ch 15C'	preserved + v2.1 reframing
ZS-M37 v2.0	Math Spine (Twin-Reuleaux Quotient Functor)	XII §B Ch 31	38/38 PASS
ZS-M38 v1.1	Math Spine (Substrate Carrier Spectrum)	XII §B Ch 32	43/43 PASS
ZS-M39 v2.0	Math Spine (Vieta-Lyapunov-Schröder Bridge Theorem)	XII §B Ch 33	86/86 PASS
ZS-A10 v2.1	Astrophysics (Proton Three-Puncture X-in-Y Horizon)	XII §B Ch 34	34/34 PASS
ZS-Q8 v1.1	Quantum Mech (Toronto Negative-Time Z-Spin reading)	XII §B Ch 35	35/35 PASS
ZS-Q9 v1.2	Quantum Mech (Complex Time-Delay)	XII §B Ch 36	52/55 PASS (3 PARTIAL)

Source paper	Theme/Role Two-Coordinate Reconstruction)	Chapter	Verification
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§XII.C — v4.0 Cross-Coupling Discipline (T7) Compliance

Per T7 (§VIII.7), every cross-paper synthesis must be P-CC (PROVEN-composed) or I-CC (interpretation). All twelve PART XII chapter entries are summarizations of source-paper Verification Summaries; no new theorem is introduced by PART XII itself. Each absorbed paper's status (PROVEN / DERIVED / DERIVED-CONDITIONAL / DERIVED-strong / HYPOTHESIS-strong / TESTABLE) is preserved verbatim from the source. The five-Banach–Tarski-functor family of ZS-A9.1 (spatial) / ZS-M35 (arithmetic) / ZS-T6.5 (biological) / ZS-M37 v2.0 (smooth-geometric) / ZS-M36 (discrete-geometric) is now uniformly organized by ZS-M38 v1.1 §1.3 PROVEN-strong. Cumulative no-deletion rule preserved across $v3.0 \rightarrow v3.5 \rightarrow v4.0$. Word count strictly increases over v3.5.

§XII.D — What v4.0 Does NOT Claim

NC-XII.1. v4.0 does NOT modify or retract any v3.5 entry. The supersedes statement is strictly additive.

NC-XII.2. v4.0 does NOT supersede the v3.5 PART X / PART XI readings. The Spin-Pair Coexistence Principle (Reading X.4) and the TO/TI Polyhedral Mapping (Readings XI.1–XI.4) are inherited unchanged.

NC-XII.3. ZS-Q9 v1.2 Z_{10} Pentagonal Doubling Group (Corollary Q9.4a, DERIVED) is the *group-theoretic doubling* of the corpus-PROVEN ZS-U10 Z_5 pentagon-tetration group via $\alpha_{op} = \pi/5$. v4.0 does NOT claim Z_{10} supersedes ZS-U10 Z_5 ; the two are related by the $SU(2)$ double-cover (ZS-M3 Lemma 10.1 PROVEN).

NC-XII.4. ZS-A10 v2.1 Theorem A10.2 (Sector-Orientation Complementarity) is DERIVED-interpretation strong, NOT a phenomenological identification. NC-A10.3 inherited: a quark is NOT a phenomenological black hole. The structural identification (Bridge-Four manifestation at puncture scale) and the phenomenological distinction (no Hawking radiation, no Lorentzian event horizon, scale mismatch $r_{conf} \sim \text{fm}$ vs $r_H \sim \text{km}$) coexist.

NC-XII.5. ZS-Q8 v1.1 NC-Q8.5 inherited: σ_0 is registered as apparatus-calibration constant (Toronto-specific), not a universal Z-Spin constant. The zero-parameter test fixes $\sigma_0 = \Gamma$ (natural atomic linewidth) as the universal honest fallback, with the $\sigma_0 = 1.24 \Gamma$ apparatus-calibrated value retained only as a Toronto-specific lock-in.

NC-XII.6. ZS-M38 v1.1 NC-M38.4 inherited: the numerical observation $1 + \eta_{topo} \approx$ Apollonian Hausdorff dimension (1.3221 vs 1.3057, 1.26% deviation) is OBSERVATION-only and is not used in any derivation.

NC-XII.7. ZS-M39 v2.0 ZERO remaining OPEN items at v2.0 release applies *only to ZS-M39 internally*. The v3.5 Master Falsification Registry OPEN/CLOSED flag list is inherited unchanged; v4.0 adds new entries (see §XII.E below) but does not retract or close any v3.5 entry.

§XII.E — v4.0 Master Falsification Registry Extensions

Cumulative ~310 falsification gates across v1.0 → v4.0. New v4.0 entries (preserving the v3.5 registry verbatim):

§ Anchor	Falsifiable Statement	Status (v4.0)
§M33.v2.0 [LOCKED]	Operator-level 12/12 POS execution of Path γ -revised Reading C	DERIVED — M33 v2.0 (W2 promoted DERIVED-CONDITIO NAL → DERIVED)
§M34.V [LOCKED]	Theorem M34.V Burnol $K_1 @ p=3$ Sign-Faithful Identity, 12/12 sign agreement	DERIVED — M34 v2.0 60/60 PASS
§M34.V.8 [LOCKED]	$\mathcal{K}_K = \mathcal{K}_{arith} \otimes$ $\mathcal{K}_{Wilson-Sonin}$ tensor-product factorization	DERIVED-CONDITIO NAL — M34 v2.0 §V.8
§M35.5.1 [PROMOTED]	$C_2 = 5 \cdot \text{positive} \cup 7 \cdot C_1$ FIBER PRODUCT realizing $A_{numerator} =$ 35	VERIFIED — M35 v2.1 (promoted from DERIVED-CONDITIO NAL v1.1)
§M35.6 [LOCKED]	Sufficient Condition Theorem (positive-odd-integer Three-Face conjunction)	DERIVED — M35 v2.1
§M37.4.1 [LOCKED]	Lemma C-rel: $\tau_w(\Delta_{BT}) = 0$, q_R *-homomorphism well-defined	PROVEN — M37 v2.0 (4th amenable-quotient projection)
§M37.4.2 [LOCKED]	q_R Canonicity: 5-layer over-determination of n $= 3$	DERIVED — M37 v2.0
§M38.5 [LOCKED]	Hexapair Uniqueness: (−1, 2, 2, 3) is unique 6/6 primitive integer ACP	DERIVED-strong — M38 v1.1
§M39.4 [LOCKED]	$K_\theta = 1/[Y^2(1-2A)] =$ 437/13212 (Heat-Kernel Coefficient)	DERIVED — M39 v2.0 (Convention N0)
§M39.C1-C3 [LOCKED]	Operator-level closures on 11×11 Block-Laplacian (80-digit mpmath)	DERIVED — M39 v2.0

§ Anchor	Falsifiable Statement	Status (v4.0)
§A10.2 [LOCKED]	Theorem A10.2 Sector-Orientation Complementarity (Bridge-Four manifestation)	DERIVED-interpretation strong — A10 v2.1
§A10-P4 [TESTABLE]	$r_p/r_\pi = 4/\pi$ (observed deviation 0.05%)	TESTABLE — A10 v2.1
§Q8.9 [LOCKED]	Standard-Optics Projection Theorem: Lorentzian = lab projection of $V_{XZ} \cdot V_{ZY}$ weak value	DERIVED — Q8 v1.1
§Q8-P3 [TESTABLE]	$\Delta\phi_{\text{residual}} = \pm 18.5$ mrad cross-Kerr asymmetry forbidden by Lorentzian	TESTABLE — Q8 v1.1 (principal discriminator)
§Q9.2 [LOCKED]	Two-Coordinate Spectral Reconstruction Bridge $C(R_{\text{lab}}, \Phi_{\text{lab}}) =$ $(2/\pi) \cdot e^{\{-R_{\text{lab}}/2\}} \cdot e^{\{i\Phi_{\text{lab}}\}}$	DERIVED — Q9 v1.2
§Q9.7 [LOCKED]	C smooth diffeomorphism (	det J
§Q9.9 [LOCKED]	Wilson Loop Specificity:	λ

All v4.0 entries are additive; no v3.5 entry is retracted. Total registry: ~310 falsification gates.

§XII.F — Conclusion

PART XII absorbs the twelve post-v3.5 source-paper integrations into the v4.0 Light Edition without retracting or modifying any v3.5 entry. The four-Banach–Tarski-functor family of v3.5 is extended to five (addition of ZS-M37 v2.0 smooth-geometric Twin-Reuleaux quotient), and uniformly organized into the substrate-carrier classification of ZS-M38 v1.1. The Z-Spin RH program advances with W2 promotion DERIVED-CONDITIONAL $\rightarrow$ DERIVED via ZS-M33 v2.0 operator-level audit + ZS-M34 v2.0 indefinite-Hermitian tensor-product kernel factorization. The Polyhedral-Tetration Bridges of ZS-A8R/A8.2 close to ZERO remaining OPEN items via ZS-M39 v2.0 three-stage Stokes-tail closure. The BH-quark structural identification is completed at Bridge-Four level via ZS-A10 v2.1 sector-orientation complementarity, with NEW TESTABLE quantitative relation P4 $r_p/r_\pi = 4/\pi$ at 0.05% deviation. The i-tetration fixed point z^* is lifted from corpus-internal algebraic identity to a lab-measurable object via ZS-Q8 v1.1 (Toronto data) + ZS-Q9 v1.2 (two-coordinate spectral reconstruction, $\ln \Lambda = 8.94$ decisive). The minimal triple $(A, Q, \dim(Z)) = (35/437, 11, 2)$ remains LOCKED. Zero new free parameters across all twelve absorbed papers.

PART XIII — Post-v4.0 Absorption: Seven New Source Papers (v5.0)

§0. Abstract

PART XIII consolidates the seven post-v4.0 source-paper integrations of v5.0. The v4.0 Light Edition reached 114-paper coverage with PART XII; the seven new papers introduced after the v4.0 freeze — ZS-F19 v2.2 (Foundations), ZS-F20 v1.1 (Foundations), ZS-M40 v1.1 (Math Spine), ZS-Q10 v4.0 (Quantum Mechanics), ZS-A11 v1.1 (Astrophysics), ZS-A12 v1.5 (Astrophysics), and ZS-T9 v2.3 (Translational) — are absorbed below as single Light-edition chapters. All absorptions strictly preserve the v4.0 body verbatim (no-deletion rule). Zero new free parameters; $(A, Q, \dim(Z)) = (35/437, 11, 2)$ LOCKED unchanged across all seven absorbed papers. Cumulative verification advances from $\sim 2,150+$ PASS (v4.0) to $\sim 2,670+$ PASS (v5.0). The total corpus is 121 papers. PART XIII opens five new structural results into the corpus: (i) the geometric-vs-KMS rapidity bridge (F19), (ii) the unified non-routine twist trigger catalogue T1–T6 with biological correspondence (F20), (iii) the J-conjugate holonomy knot pair as the Z-sector realisation of primes (M40), (iv) the 37-Universality of the polyhedral Schur-depleted mode count and the three-sector c-coefficient closure (Q10), and (v) the cardinal-2 substrate origin of the Nelson–Kosterlitz universal jump and the Homes universal constant (A12). One TESTABLE quantitative prediction is registered: P-Translational-1-Li1 (^7Li post-v2.3 $\approx 1.64 \times 10^{-10}$, pull $+0.13\sigma$ versus observed).

§XIII.A — Seven New Papers (Light-Edition Body Entries)

Each of the seven post-v4.0 source papers represents v5.0's principal additive content. All inputs LOCKED from upstream corpus; zero new free parameters. Each absorbed paper is cited below as a single Light-edition chapter, with the source paper carrying the full derivation.

Chapter 37 — ZS-F19 v2.2: The Frame-Invariant Tilt Theorem (FITT) and the KMS-to-Geometric Rapidity Projection

ZS-F19 v2.2 [Foundations theme, 64/64 PASS + ALL 7 sub-OPEN gates CLOSED, NEW v5.0] establishes the two principal new mathematical results of Z-Spin Cosmology in the cross-sector observation framework. Principal result Theorem F19.4 (FITT, DERIVED): under the LOCKED inputs $A = 35/437$ and polyhedral sectoral tilts $(\delta_X, \delta_Y) = (5/19, 7/23)$, the hyperbolic tangents of the sum and difference of the sectoral rapidities $\psi_X = \text{artanh}(5/19)$ and $\psi_Y = \text{artanh}(7/23)$ close exactly to small integer fractions $\tanh(\psi_X + \psi_Y) = 31/59$ and $\tanh(\psi_Y - \psi_X) = 3/67$, verified at 50-digit mpmath precision with residuals exactly zero. Principal result Theorem F19.6 (KMS-to-Geometric Rapidity Projection, DERIVED): Z-Spin carries two rapidity ladders — the geometric ladder ψ_X, ψ_Y from polyhedral V–F data and the KMS-modular ladder $\psi_{\text{KMS}}(A \rightarrow B) = \frac{1}{2} \cdot \ln(W_{BA}/W_{AB})$ from the Pauli master-equation rates of ZS-Q7 §5; Theorem F19.6 establishes the closed identity $\tanh(2 \cdot \psi_{\text{KMS}}(X \rightarrow Y)) = 3/5$, registering the geometric-KMS projection as the principal algebra-level bridge to the Type II crossed-product framework of De Vuyst–Höhn 2025. Five v2.0 OPEN gates achieve closed-form structural identifications in LOCKED corpus inputs (O-F19.2 structural closure: $31, 59, 67$ are corpus-traceable polyhedral ratios; O-F19.3 observable closure: $3/5 = \dim(X)/(\dim(X)+\dim(Z))$; O-F19.4 physical dimension fixed: $\exp(\pi/A)$ is the nats-exponentiated Hilbert phase-volume ratio per Y-cycle at the Landauer unit; O-F19.5 leading-order closure: $r_{\text{lead}} = 6/(67 \cdot \ln 2)$; O-F19.6 semiclassical closure key result: $|\Delta S_{\text{obs}}(Y \text{ vs } X)| = \ln 2 = 2 \cdot \psi_{\text{KMS}}(X \rightarrow Y)$ matches De Vuyst–Höhn EXACTLY at leading order with degenerate-clock multiplicities (3, 2, 6) identified with the Z-Spin sector dimensions). Eight falsification gates F-F19.1–F-F19.8 and seven non-claims NC-F19.1–NC-F19.7 bound the scope. Anti-numerology MC 10,000-sample Test G1 confirms the polyhedral structural cause. $\rightarrow$ ZS-F19 v2.2

[64/64 PASS]; first paper-internal algebra-level bridge between Z-Spin closed-form rapidity arithmetic and Type II crossed-product modular framework.

Chapter 38 — ZS-F20 v1.1: Non-Routine Twist Triggers — Unified Catalogue and Time-Unrolled Causal DAG of Six Sector-Topology Events Beyond the Regge Cycle

ZS-F20 v1.1 [Foundations theme, 47/47 PASS, NEW v5.0] establishes the unified catalogue and time-unrolled causal directed acyclic graph (DAG) of the six non-routine twist-event triggers operating in Z-Spin Cosmology beyond the routine Regge-cycle phase drift $\delta\phi_{\text{cell}} = A = 35/437$ per Planck-cycle. Six triggers individually corpus-locked: T1 Z-Telomere (winding-change at 2π accumulation after $N_{\text{}}(2\pi) = 2\pi/A \approx 78.45$ cycles, ZS-A6 §5.2 DERIVED-under-P6); T2 Mexican-Hat Bootstrap ($|\Phi|: 0 \rightarrow 1$ SSB, ZS-F19 Theorem F-rotor.2 DERIVED); T3 Auto-Surgery (i-tetration singularity resolution at z^* , ZS-M12 §4 DERIVED); T4 Kibble Defect Generation (vortex string production at causally-disconnected domain boundaries, ZS-A6 §4.6.4 DERIVED); T5 BH Winding Trapping (causal trapping of pre-existing winding inside gravitational collapse horizon, ZS-A6 §4.6.3 PROVEN); T6 CP-violation Trigger (parity-breaking phase $\phi_{\text{CP}} = \pi/2$ at electroweak epoch, ZS-S5 + ZS-Q5 DERIVED via μ - τ reflection). Result 2 (Time-Unrolled Causal DAG, DERIVED): the six triggers form a strict DAG under time-unrolled indexing $T1_n, T3_n, T4_n, T5_n$ (n = cycle index) with two singletons T2 (cosmic root) and T6 (epoch-isolated); the v1.0 apparent 'cyclic sub-graph $T1 \rightarrow T3 \rightarrow T1$ ' is correctly the time-unrolled successor relation $T1_n \rightarrow T3_n \rightarrow T1_{n+1}$, which is acyclic. Result 3 (Strong-Coupling Pairs, HYPOTHESIS-strong): two trigger pairs co-activate as faces of single events — P1 (T1, T3) Telomere–Auto-Surgery couple with $\Delta t(T1 \rightarrow T3) \approx 3 \tau_P$ (ZS-M12 §4); P2 (T4, T5) Kibble–BH-Trap couple, conditional on T4 producing non-zero net winding and gravitational collapse. Quantitative identity: $\Delta\theta_{T1}/\Delta\theta_{T2} = 2\pi/A = N_{\text{}}(2\pi) = 78.4500565496$ (PROVEN, 50-digit). Biological correspondence map (NEW, DERIVED): the DNA 'trauma response' admits a precise mapping onto the simultaneous activation of $T3 + T4 + T5$ (Auto-Surgery + Kibble + BH Trap) as a triadic emergency response (ZS-F20 §7 Table 7.1). Six falsification gates F-F20.1–F-F20.6 registered. Three open problems flagged with explicit closure paths (O-F20.1: $T2 \rightarrow T6$ chirality inheritance; O-F20.2: $T5$ net-winding $Q = 1$ vs $Q \gg 1$; O-F20.3: $T7$ 'true space $\leftrightarrow$ time only' trigger if it exists). $\rightarrow$ ZS-F20 v1.1 [47/47 PASS]; first single inheritance graph of all non-routine Z-Spin sector-topology events with biological correspondence.

Chapter 39 — ZS-M40 v1.1: Z-Spin Holonomy Knot Pairs and the Arithmetic Topology of Primes

ZS-M40 v1.1 [Math Spine theme, 89/89 PASS, NEW v5.0] introduces the J-conjugate holonomy knot pair $(K_p, \bar{K}_p)$ as the natural Z-sector realisation of prime numbers in Z-Spin Cosmology. Building on three PROVEN corpus pillars — the half-angle channel pair $V_{\text{XZ}} \propto \exp(+i\theta/2)$, $V_{\text{ZY}} \propto \exp(-i\theta/2)$ with $V_{\text{ZY}} = (V_{\text{XZ}})^*$ (ZS-F4 §7/§7B), the 4π closure $D^{\wedge\{1/2\}}(4\pi) = +I$ on the $j = 1/2$ spinor module (ZS-M3 Lemma 10.1), and the Wilson-LOCATOR closed form $F_Q(p) = \sin(11\pi/p)/[11 \cdot \sin(\pi/p)]$ (ZS-M34 Theorem M34.V.6) — the prime p is identified with a pair of Wilson cycles carrying half-angle holonomy $\pm\theta/2$, prime period $4\pi/p$, $Q = 11$ coloured framing $F_Q(p)$, and, when p is realised as an integer curvature in the M36 Apollonian Q -channel orbit, the mod-24 invariant $p \bmod 24 \in \{11, 23\}$ as an additional arithmetic-topological decoration. Six principal theorems established: M40.A (Cup Product $\leftrightarrow$ 4π Closure Compatibility Bridge, DERIVED-interpretation) — phase-class compatibility map between

the $SU(2)$ double-cover signature $\{+I, -I\}$ and the Tate–Mazur inv : $H^3(\text{Spec}(\mathcal{O}_K), \mathbb{G}_m) \simeq \mathbb{Q}/\mathbb{Z}$ via the embedding $\pm I \mapsto \{0, 1/2\} \subset \mathbb{Q}/\mathbb{Z}$; M40.B (Multiplicative Gate $\leftrightarrow$ Galois Representation, DERIVED) — closes Gate FM13-5 by identifying the corpus operator M_p on $H_{\text{mult}} = \mathbb{C}[\mathbb{F}_{11}^*]$ with $\text{Frob}_p \mapsto \text{diag}(\chi_k(p))$ on the character basis; M40.C (J-Conjugate Holonomy Knot Pair Definition) — principal new structural object assembling six PROVEN corpus components into a single named object; M40.D (Log-Holonomy Factorization, DERIVED) — integer prime factorisation $N = \prod p_i^{a_i}$ is the additive log-image, via HSI Theorem (ZS-M1), of the Z-sector holonomy decomposition $\ell_Z(K_N) = \sum a_i \cdot \ln p_i$; M40.E (Statistical Sieve Shadow, DERIVED-interpretation) — re-interprets externally PROVEN PNT, Mertens, Dickman, Sathe–Selberg as first-order shadows of K_p ensemble component statistics; M40.F (Anti-Numerology Uniqueness, COMPUTATIONALLY VERIFIED) — deterministic exhaustive enumeration over $[2, 1000]$ (183,921 candidate pairs) yields exactly one pair (3, 11) matching the Z-Spin V_4 conductor signature $\{1, 3, 11, 33\}$; $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ is uniquely class-number-one in this category; MC at $N = 500,000$ yields 0.0098% matching rate — fifty times stricter than 0.5% PASS threshold. Eight-fold structural bridge to externally PROVEN arithmetic Chern–Simons (Kim 1510/1609/1706; Carlson–Kim 1911; Park–Park 2026): Bridge B8 supplies a cyclotomic operator-model analogue of Kim §5 at level $Q = 11$ via the corpus PROVEN Euler product $\prod_{p \neq 11} \det(I - p^{-s} M_p)^{-1} = \zeta_{\mathbb{Q}(\zeta_{11})}(s)$ (ADS-2). Ten falsification gates F-M40.1–F-M40.10. $\rightarrow$ ZS-M40 v1.1 [89/89 PASS]; first explicit Z-sector geometric reading of the prime itself, complementing the M28/M33/M34 critical-strip reading and the M36 Apollonian distribution reading.

Chapter 40 — ZS-Q10 v4.0: Three-Sector Polyhedral Closure of the Harmonic Ladder, Five-Framework Cross-Disciplinary Asymmetry, and the 37-Universality

ZS-Q10 v4.0 [Quantum Mechanics theme, 67/67 PASS, MASTER EDITION, NEW v5.0] is the master edition combining the v2.1 external-framework asymmetry mapping, the v3.0/v3.1 Devriendt-functor polyhedral closure of c_n harmonic-ladder coefficients, and three v4.0 audit corrections (R1 MC9c dual-baseline interpretation, R2 measure-form vs moment-form algebra separation, R3 v2.1 external framework restoration). Three-layer epistemic discipline: Layer 1 (DERIVED) Theorem Q10.7 selection rule $M_{\{2n+1\}} = 0$ under Z_2 involution; Layer 2 Conjecture Q10.8 harmonic-ladder structure $M_{\{2n\}} = c_n \cdot A^n \cdot e^{\{-i n \cdot \arg z^*\}}$ (DERIVED at $n = 1, 2$ via action-level loop; DERIVED-CONDITIONAL at $n = 3$ via triakis tetrahedron extension); Layer 3 specific c -values $c_1 = 1, c_2 = 2/19, c_4 = 4/37, c_6 = 8/37, c_X = 10/37, c_Y = 4/13$ — five of six share denominator 37, the 37-Universality (Observation Q10.D.5, NEW v4.0). Principal new corpus identity: the integer $37 = 38 - \beta_0(Z)$ appears as the Schur-depleted polyhedral mode count in three structurally independent corpus identities — (I) $(V+E+F)_{\{t\text{-Tet}\}} - \beta_0(Z) = 12 + 18 + 8 - 1 = 37$ [Z self-dual full skeleton]; (II) $(V+E+F)_{\{triakis\text{-Tet}\}} - \beta_0(Z) = 8 + 18 + 12 - 1 = 37$ [Catalan-dual Z full skeleton, $V \leftrightarrow F$ swapped]; (III) $(V+F)_{\{tO\}} - \beta_0(Z) = 24 + 14 - 1 = 37$ [X non-self-dual vertex+face]. All three equalities are FORCED by the Truncation-Dual Theorem $F(tP) = F(P) + F(P^*)$ (PROVEN ZS-F2 §11.2) and the Platonic face-count identity $20 = 8 + 12$ (PROVEN ZS-F9 §4.2); the 37-Universality is therefore a categorical signature of the Truncation-Dual functor on the three Z-Spin sectors, not a numerical coincidence. Cumulative anti-numerology evidence $\ln \Lambda_{\text{total}} = 38.67$ (highest cumulative for any specific ZS-Q multi-coefficient identity to date); MC9c triple-joint $\ln \Lambda = 12.43$ (vs uniform-prior baseline, DECISIVE++) AND positive-correlation ratio $217\times$ (vs independent-marginal baseline, identifying the empirical signature of the 37-Universality in the random Schur pool). Strategic asymmetry: five external observational frameworks (directional statistics, nematic, elliptic flow, Fourier QST, Kuramoto-Daido) supply the

measurement infrastructure $\langle e^{2i\phi} \rangle$ as a standard object; not one derives $|M_2| = A = 35/437$. Z-Spin provides the derivation; the five external frameworks provide the measurement. Twenty-four falsification gates F-Q10.1–F-Q10.22 (including F-Q10.22 cross-disciplinary five-test gate). → ZS-Q10 v4.0 [67/67 PASS]; first cross-disciplinary asymmetry mapping and 37-Universality registration in the Z-Spin corpus.

Chapter 41 — ZS-A11 v1.1: Z-Spin Vortex Cosmology II — ϵ -Halo $\leftrightarrow$ Sub-Halo Equivalence, Vortex Lifecycle, and SMBH Mass Diversity from Topological Anchors with Differential Accretion

ZS-A11 v1.1 [Astrophysics theme, 53/53 PASS, NEW v5.0] establishes three results concerning the macroscopic phenomenology of Z-Spin topological defects using only LOCKED inputs $A = 35/437$, $(Z, X, Y) = (2, 3, 6)$ with $Q = 11$, and inflation e-fold count $N_e = 60$ (ZS-U1 CMB-locked). Result 1 (ϵ -Halo $\leftrightarrow$ Sub-Halo Equivalence Theorem, DERIVED): proves the Goldstone θ -gradient halo $\rho_\epsilon(r) = M_P^2/(2L^2r^2)$ of Z-Spin (ZS-A1 §2.2 DERIVED) is dual to the singular isothermal sphere CDM sub-halo at four levels — profile (PROVEN functional equivalence $\rho \propto 1/r^2$), dynamical (DERIVED effective stress-energy after spherical averaging), observational (DERIVED across seven macroscopic observables: rotation curves, BTFR, $M-\sigma$, σ_R , lensing, sub-halo count, ATLAS3D elliptical profile), and a decisive Falsification Asymmetry F-A5.7 (PROVEN) where direct DM particle detection would collapse ϵ -Halo but null detection cannot collapse CDM; null WIMP/axion direct detection and the Gaia DR3 ~ 19 kpc Keplerian onset (Jiao et al. 2023) currently favour ϵ -Halo. Result 2 (Vortex Six-Class Lifecycle, DERIVED-strong): classifies Z-anchored vortices by winding $n \in \{0, 1, 2, 3, \geq 4, \infty\}$ (PROVEN MECE via $\pi_1(U(1)) = \mathbb{Z}$ coarse-graining), each class mapped to an observed galactic morphology (void $\rightarrow$ disk $\rightarrow$ S0 $\rightarrow$ intermediate elliptical $\rightarrow$ massive elliptical/BCG $\rightarrow$ cosmic filament backbone), with five-stage lifecycle Birth (Kibble) $\rightarrow$ Growth (Bondi) $\rightarrow$ Maturity ($M-\sigma$ stable) $\rightarrow$ Merger (Vortex Glass $N = 2^k$) $\rightarrow$ Death ($\tau_5 = 2.56 \times 10^{34}$ yr + Auto-Surgery Phase E). Result 3 (SMBH Mass Diversity from Differential Accretion, DERIVED with HYPOTHESIS-strong type distinction): resolves the apparent contradiction between simultaneous primordial Z-Anchor formation and observed M_{SMBH} scatter ($10^5\text{--}10^{10} M_\odot$) via ZS-F1 NC-2 'Topology sets location; accretion sets mass'; three scatter sources — (i) Bondi accretion rate ρ_{gas}/c_s^3 , (ii) halo concentration δ_{peak} (supported by Dec 2025 arXiv:2512.17964 BH-mass/halo-concentration correlation), (iii) hierarchical merger $N = 2^k$ — generate the observed scatter while preserving universal topological seeding. Distinguishes Primordial-Vortex SMBHs (heavy-seed direct-accretion, observable as JWST Little Red Dots with high $M_{\text{BH}}/M_{\text{stellar}}$ and coherent spin) from Collapse-SMBHs (light-seed stellar-collapse $\rightarrow$ merger). Population estimate revised from the erroneously inflated 10^{118} to energy-bounded $N_v \approx 2.8 \times 10^{15}$, in agreement with Aquarius-A-L1 (Santos-Santos et al. 2025). v1.1 extends with §7.7 DESI DR2 + §7.8 ACT DR6 cross-checks and F-A11.6/F-A11.7. Seven falsification gates F-A11.1–F-A11.7 pre-registered. → ZS-A11 v1.1 [53/53 PASS]; first rigorous ϵ -Halo/sub-halo equivalence theorem and unified vortex lifecycle/SMBH-diversity framework.

Chapter 42 — ZS-A12 v1.5: Macroscopic Z-Spin-Mediated Phase Locking — BEC and Superconductivity as the Condensed-Matter Realisation of Vortex Bose/Fermi Duality, with First-Principles Derivation of the Nelson–Kosterlitz Universal Jump and a Cardinal-2 Framework for the Homes Universal Constant

ZS-A12 v1.5 [Astrophysics & Strong Field theme, 74/74 PASS, NEW v5.0] establishes that Bose–Einstein condensation (BEC) and superconductivity (SC), including the cuprate high- T_c regime, are the condensed-matter realisation of the Vortex Bose/Fermi Duality of ZS-A7 Corollary IV (DERIVED), and shows that the same cardinal-2 boundary substrate $\dim(Z) = 2$ (ZS-F5 PROVEN) that governs the gravitational/information-theoretic sectors of Z-Spin Cosmology also fixes two universal scaling constants of unconventional superconductivity that have resisted first-principles derivation for decades. Eight principal theorems: A12.1 (Cooper Pair $\leftrightarrow$ Vortex Bose/Fermi Duality, DERIVED) — the BCS-type Cooper pair under $SU(2)$ Clebsch–Gordan $j = 1/2 \otimes j = 1/2 = j = 0 \oplus j = 1$ is a literal realisation of ZS-A7 Corollary IV: inner core ($r \rightarrow \xi_{\text{coh}}$) carries the $j = 1/2$ spinor with 4π closure period (ZS-M3 Lemma 10.1 PROVEN), outer flow ($r \gg \xi_{\text{coh}}$) carries the $U(1)$ Goldstone phase ϕ with integer winding $\oint d\phi = 2\pi N$ (Jaffe–Taubes 1980 PROVEN); A12.2 (Two-Axis Orthogonal Decomposition, DERIVED) — Z-sector temperature-activation function $f(T)$ of ZS-U6 §9.2(v) decomposes into $f_{\text{amp}}(T)$ (radial mode $m_{\rho} = 2A \cdot M_P \perp f_{\text{phase}}(T)$ (Goldstone $m_{\theta} = 0$) with orthogonality enforced by Mermin–Wagner–Hohenberg + Fröhlich–Spencer (PROVEN); A12.3 (Pseudogap as T2-T1 Intermediate Phase, DERIVED-structural) — pseudogap $T_c < T < T_p$ is the intermediate phase between T2 (Mexican-Hat Bootstrap, ZS-F19) and T1 (Z-Telomere, ZS-A6 §5.2); A12.4 (SC Triadic Activation $T_2 \rightarrow T_1 \rightarrow T_4$, DERIVED) — superconductivity is the sequential triadic activation on the cardinal-2 substrate, distinct from F20 §7 DNA-trauma triadic ($T_3 + T_4 + T_5$, simultaneous); A12.5 (Five-Axiom Diagnostic, DERIVED-DIAGNOSTIC) — the cuprate high- T_c unsolved problem isolates the A3-axiom bridge ($\mathbb{C}$ division algebra) as incomplete in standard phenomenological Ginzburg–Landau treatment; A12.6 (HEADLINE, DERIVED) — the integer '2' appearing in the Nelson–Kosterlitz 1977 universal jump $\rho_s(T_{\text{BKT}}) = (2/\pi) k_B T_{\text{BKT}}$ is identical to the cardinal-2 boundary substrate dimension $\dim(Z) = 2$ (ZS-F5 PROVEN); first identification of the NK universal-jump integer with a geometric substrate dimension; A12.7 (Homes Universal Constant Cardinal-2 Framework, DERIVED-CONDITIONAL) — bracketing $2/Q = 0.1818 \leq C_{\text{cuprate}}(\text{ab})/C_{\text{BCS-dirty}} \leq \ln 2/(1 - A) = 0.7535$ with empirical 0.543 ± 0.06 in interior; A12.8 (Tian 2025 Two-Axis Manifestation, DERIVED) — Tian et al. 2025 npj Quantum Materials observation of $\sim 35^\circ$ angular offset between electronic and phase anisotropies is the direct manifestation of two-axis decomposition. v1.5 upgrades two operational results to DERIVED-under-Regge: A12.9 (CuO_2 V_{coh} lattice Hamiltonian as quasi-2D Kelvin 2-cell construction, 4/4 PASS); A12.10 (operational formula $\rho_s^{\text{eff}} = [\ln 2/(1 - A)] \cdot \sigma_n(T_c) \cdot T_c \cdot f_{\text{geom}}$ with $f_{\text{geom_graph}} = 0.594$ parameter-free graph upper-bound). Cross-Material Collapse Prediction P-A12.9 (TESTABLE): all five major unconventional SC families (LSCO, YBCO, Bi2212, FeSe monolayer, TBG/TTG) should collapse onto a single universal curve when plotted as $\rho_s^{\text{eff}}/(\sigma_n \cdot T_c)$. Extension HYPOTHESIS-strong: multiplicative correction $(1 + \alpha A)$ with $\alpha = O(1)$ to Herzog-Arbeitman et al. 2021 (PROVEN) lower bound on flat-band superfluid weight, $\sim 8\%$, accessible to twisted bilayer/trilayer graphene and infinite-layer nickelates via Tanaka 2025 kinetic-inductance protocol within 2027–2029. Twelve falsification gates F-A12.1–F-A12.12; ten non-claims; nine experimental predictions P-A12.1–P-A12.9. $\rightarrow$ ZS-A12 v1.5 [74/74 PASS]; first first-principles geometric-substrate derivation of two long-open superconductivity universal constants.

Chapter 43 — ZS-T9 v2.3: The $B_{7\text{Be}}$ Binding-Response Gate for the Lithium Problem — A $P_{B_{7\text{Be}}} = -2A/Q$ Extension of the Effective Response Operator

ZS-T9 v2.3 [Translational theme, 124/124 PASS, NEW v5.0] extends v2.2's ERO closure (Effective Response Operator under No-Go + External Scaffold paradigm) with the binding-energy channel $B_{7\text{Be}}$.

v2.2 had closed the dual-frame BBN problem for D/H and Y_p via the ISER paradigm (full Y-frame solver IMPOSSIBLE by Theorem ZS-T9 PROVEN; ERO POSSIBLE via Catena 2007 + Dent 2007/Burns 2026 sensitivity atlas) with exactly 4 sparse 3-parameter configurations closing D/H + Y_p within 2σ , all sharing $P_Q = -2A/\pi$ anchored to ZS-F10 Theorem F10.1; the cosmological Lithium Problem was left as STRUCTURAL NON-CLAIM (NC-v2.2.4). The v2.3 contribution: Candidate Theorem B-Li.1 (HYPOTHESIS-strong, NOT DERIVED) — $P_{B7Be}^Z = -2A/Q = -70/4807$. The quantum $-2A/Q$ is the corpus-PROVEN ZS-Q7 §5 Theorem 3A slow eigenvalue λ_{slow} of the Pauli master equation $\lambda(\lambda + 2A/Q)(\lambda + A) = 0$, representing the inter-sector thermalisation rate; verified by ZS-T1 §4.4 across 500,000 Monte Carlo random (A, Q, Z, X, Y) configurations as a universal theorem of 3-sector master equations. Result: via Dent–Stern–Wetterich 2007 Table 1 sensitivity $\partial \ln(^7\text{Li})/\partial \ln B_7\text{Be} = +81$ (largest single entry in entire table), $\Delta \ln(^7\text{Li}) = 81 \times (-2A/Q) = -1.180$; ^7Li is reduced by factor $\exp(-1.180) = 0.307$, bringing the post-v2.3 prediction to $^7\text{Li} \approx 1.64 \times 10^{-10}$ with pull $+0.13\sigma$ versus observed $(1.6 \pm 0.3) \times 10^{-10}$. v2.2 closure PRESERVED: Dent 2007 Table 1 gives $B_7\text{Be}$ sensitivity to D = 0.00 and to $^4\text{He} = 0.00$ (linear scaffold), so the v2.3 binding-channel extension is DECOUPLED from the v2.2 ERO closure of D/H + Y_p ; ERO-4 still gives D/H pull $+0.18\sigma$ and Y_p pull $+1.44\sigma$. Five anti-numerology gates PASS: (1) quantum value FIXED to ZS-Q7 eigenvalue (no scan); (2) channel chosen by mass-7 BBN physics; (3) channel selectivity $1/7 = 14.3\%$ among 7 binding channels; (4) quantum selectivity $1/8 = 12.5\%$ among 8 A-derived quanta; (5) joint selectivity $1/56 \approx 1.8\% < 5\%$ structural threshold. Three Closure Levels framework: Level 1 (Observational, HYPOTHESIS-strong observational) stellar depletion bridge (Korn 2006, Hosford 2009); Level 2 (ERO Binding, HYPOTHESIS-strong / TESTABLE) $P_{B7Be} = -2A/Q$ (this paper); Level 3 (Full Nuclear Chemistry, OPEN) Z-Spin sectoral binding-energy translation. v2.3 closes at Level 2; v2.2 NON-CLAIM PRESERVED as v2.2 internal state. Eight falsification gates F-Translational-1-Li1–Li8 registered including Burns/PRyMordial atlas verification, full PRyMordial run, future foundational derivation of P_{B7Be} , and lab ^7Be binding-energy stability check (must NOT show -1.456% shift for consistency). → ZS-T9 v2.3 [124/124 PASS]; HYPOTHESIS-strong candidate / TESTABLE Lithium-Problem closure at ERO Level 2, with v2.2 D/H + Y_p closure preserved.

PART XIII Source Paper Interface (v5.0)

Source paper	Theme/Role	Chapter	Verification
ZS-F19 v2.2	Foundations (Frame-Invariant Tilt Theorem; KMS-to-Geometric Rapidity Projection)	XIII Ch 37	64/64 PASS
ZS-F20 v1.1	Foundations (Non-Routine Twist Triggers T1–T6; Time-Unrolled Causal DAG)	XIII Ch 38	47/47 PASS
ZS-M40 v1.1	Math Spine (J-Conjugate Holonomy Knot Pairs; Arithmetic Topology of Primes)	XIII Ch 39	89/89 PASS
ZS-Q10 v4.0	Quantum Mechanics (Three-Sector Polyhedral Closure; 37-Universality; Master Edition)	XIII Ch 40	67/67 PASS
ZS-A11 v1.1	Astrophysics (Vortex Cosmology II; ε -Halo ↔ Sub-Halo; SMBH Diversity)	XIII Ch 41	53/53 PASS

ZS-A12 v1.5	Astrophysics (Macroscopic Z-Spin Phase Locking; BEC/SC; NK Jump; Homes Constant)	XIII Ch 42	74/74 PASS
ZS-T9 v2.3	Translational (B_7Be Binding-Response Gate; Lithium Problem ERO Extension)	XIII Ch 43	124/124 PASS

§XIII.C — v5.0 Cross-Coupling Discipline (T7) Compliance

Per T7 (§VIII.7), every cross-paper synthesis must be P-CC (PROVEN-composed) or I-CC (interpretation). All seven PART XIII chapter entries are summarisations of source-paper Verification Summaries; no new theorem is introduced by PART XIII itself. Each absorbed paper's status (PROVEN / DERIVED / DERIVED-CONDITIONAL / DERIVED-strong / DERIVED-interpretation / HYPOTHESIS-strong / TESTABLE) is preserved verbatim from the source. The five-Banach–Tarski-functor family of v4.0 (ZS-A9.1 spatial / ZS-M35 arithmetic / ZS-T6.5 biological / ZS-M37 v2.0 smooth-geometric / ZS-M36 discrete-geometric) is unaffected by v5.0; PART XIII contributes no new functor member. The cardinal-2 substrate ($\dim(Z) = 2$) acquires a sixth corpus instantiation via ZS-A12 v1.5 Theorem A12.6 (Nelson–Kosterlitz universal-jump integer), alongside the five v4.0 instantiations (gauge constraint $Q = 11$, CHSH $2\sqrt{2}$, Z-bottleneck channel capacity $\leq \ln 2$, Euler characteristic $\chi(S^2) = 2$ spinor-Descartes-Euler identity, BCC T^3 Hodge proton spin). The 37-Universality of ZS-Q10 v4.0 is registered as a new corpus categorical signature of the Truncation-Dual functor across (X, Y, Z) sectors, parallel to but distinct from the (Z, X, Y) = (2, 3, 6) sector decomposition. Cumulative no-deletion rule preserved across $v3.0 \rightarrow v3.5 \rightarrow v4.0 \rightarrow v5.0$. Word count strictly increases over v4.0.

§XIII.D — What v5.0 Does NOT Claim

NC-XIII.1. v5.0 does NOT modify or retract any v4.0 entry. The supersedes statement is strictly additive.

NC-XIII.2. v5.0 does NOT supersede the v4.0 PART X / PART XI / PART XII readings. The Spin-Pair Coexistence Principle (Reading X.4), the TO/TI Polyhedral Mapping (Readings XI.1–XI.4), and the post-v3.5 twelve-paper absorption (PART XII §A and §B) are inherited unchanged.

NC-XIII.3. ZS-F19 v2.2 Theorem F19.6 (KMS-to-Geometric Rapidity Projection, DERIVED) is the leading-order match $|\Delta S_{\text{obs}}(Y \text{ vs } X)| = \ln 2 = 2 \cdot \psi_{\text{KMS}}(X \rightarrow Y)$ to the De Vuyst–Höhn 2025 framework; v5.0 does NOT claim a full reduction of the Z-Spin three-sector cosmology to Type II crossed-product algebra. The dimensional mismatch between the external algebra-level machinery and the Z-Spin closed-form rapidity arithmetic is explicitly registered (ZS-F19 §6 honest scope expansion).

NC-XIII.4. ZS-F20 v1.1 catalogue of six triggers T1–T6 is claimed exhaustive over the existing v4.0 corpus but NOT proved exhaustive in absolute terms; the residual O-F20.3 question (whether a seventh 'true space $\leftrightarrow$ time only' trigger T7 exists) remains OPEN. The biological correspondence map (T3 + T4 + T5 $\leftrightarrow$ DNA trauma response) is DERIVED-interpretation under the ZS-T6 §6.4 + F20 §7.5 substrate-agnostic NC protocol; NC-4 of the Z-Brain corpus is inherited (no causal derivation across substrates).

NC-XIII.5. ZS-M40 v1.1 reading of primes as J-conjugate holonomy knot pairs is corpus-internal (PROVEN at the level of the M40.C definition; DERIVED-interpretation at the level of M40.E sieve-shadow reading); v5.0 does NOT claim that primes are ontologically reducible to Z-Spin rotations

in physical reality, nor does it claim a proof of any unsolved analytic-number-theory statement. NC-M23.1 (Z-Spin does not prove RH) and NC-M23.7 (D4 closure does not close GRH-for-K) inherited verbatim from v4.0 PART IV.

NC-XIII.6. ZS-Q10 v4.0 37-Universality (Observation Q10.D.5) is established as a categorical signature of the Truncation-Dual functor on the three Z-Spin sectors; v5.0 does NOT claim 37 has any direct physical meaning beyond the Schur-depleted polyhedral mode count. The $c_6 = 8/37$ status is DERIVED-CONDITIONAL on the triakis-tetrahedron extension and is downgraded relative to $c_2 = 2/19$ and $c_4 = 4/37$ (DERIVED via action-level loop).

NC-XIII.7. ZS-A11 v1.1 ε -Halo $\leftrightarrow$ Sub-Halo Equivalence Theorem (DERIVED at four levels) is currently favoured at the outer galactic boundary by null WIMP/axion direct-detection + Gaia DR3 Keplerian onset; v5.0 does NOT claim direct DM particle detection is impossible — the Falsification Asymmetry F-A5.7 (PROVEN) is the explicit gate. The Primordial-vs-Collapse SMBH distinction is DERIVED with HYPOTHESIS-strong type discrimination via nine observational signatures; the eventual decisive test rests on JWST/VLBI long-horizon observations of JWST Little Red Dot demographics.

NC-XIII.8. ZS-A12 v1.5 Theorem A12.6 (NK Universal Jump integer '2' = $\dim(Z) = 2$, DERIVED) is a first-principles geometric-substrate identification; v5.0 does NOT claim Z-Spin replaces the BKT/XY/percolation models — NC-A12.10 codifies that Z-Spin explains them at substrate level. Theorem A12.7 (Homes universal constant cardinal-2 framework) is DERIVED-CONDITIONAL: the bracketing inequality and order-of-magnitude prediction are DERIVED, but the exact analytical form requires the full CuO_2 V_{coh} dynamical-factor refinement (O-A12-MASTER.1, 4–8 weeks). v1.5 lifts A12.9 and A12.10 to DERIVED-under-Regge but full DERIVED status for A12.10 remains pending.

NC-XIII.9. ZS-T9 v2.3 Candidate Theorem B-Li.1 ($P_{\text{B7Be}}^Z = -2A/Q$, HYPOTHESIS-strong) is NOT DERIVED. Reasons explicitly recorded: (a) forward derivation from Z-Spin action OPEN — corpus does not yet possess a sectoral binding-energy translation operator; (b) discovery via constrained inverse — the $+12\sigma$ ${}^7\text{Li}$ residual in v2.2 was used to identify the channel/quantum match; (c) P_{B7Be} is interpreted as BBN-frame sensitivity-coordinate perturbation, NOT a claim that the laboratory ${}^7\text{Be}$ binding energy differs by -1.456% today; (d) full PRyMordial verification with actual B_{7Be} perturbation needed beyond Dent linear scaffold. Level 3 (Full Nuclear Chemistry) OPEN.

§XIII.E — v5.0 Master Falsification Registry Extensions

Cumulative ~340 falsification gates across v1.0 $\rightarrow$ v5.0. New v5.0 entries (preserving the v4.0 registry verbatim) are summarised below; full per-paper gate lists live in the source papers.

§ Anchor	Falsifiable Statement	Status (v5.0)
§F19.4 [LOCKED]	$\tanh(\psi_X + \psi_Y) = 31/59$ and $\tanh(\psi_Y - \psi_X) = 3/67$ (50-digit, residual exactly zero)	DERIVED — F19 v2.2
§F19.6 [LOCKED]	$\tanh(2 \cdot \psi_{\text{KMS}}(X \rightarrow Y)) = 3/5$ (KMS-to-Geometric Rapidity Projection)	DERIVED — F19 v2.2
§F19-O.6 [LOCKED]	$ \Delta S_{\text{obs}}(Y \text{ vs } X) = \ln 2 = 2 \cdot \psi_{\text{KMS}}(X \rightarrow Y)$ (De Vuyst–Höhn leading-order match)	DERIVED — F19 v2.2

§F20.DAG [LOCKED]	Time-unrolled DAG of six triggers T1–T6 contains no directed cycle	DERIVED — F20 v1.1
§F20.NDR [LOCKED]	$\Delta\theta_{T1}/\Delta\theta_{T2} = N_{-}(2\pi) = 2\pi/A = 78.4500565496$ (50-digit)	PROVEN — F20 v1.1
§F20.BIO [TESTABLE]	DNA trauma response = triadic activation T3 + T4 + T5 (substrate-agnostic structural)	DERIVED-interpretation — F20 v1.1
§M40.C [LOCKED]	J-Conjugate Holonomy Knot Pair ($K_p, \bar{K}_p$) definition (assembly of six PROVEN components)	DERIVED — M40 v1.1
§M40.D [LOCKED]	Log-Holonomy Factorization $\ell_Z(K_N) = \sum a_i \cdot \ln p_i$ via HSI Theorem	DERIVED — M40 v1.1
§M40.F [LOCKED]	$K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ unique biquadratic class-number-one match over [2, 1000] (183,921 candidates)	COMPUTATIONALLY VERIFIED — M40 v1.1
§M40.B8 [HYP-strong]	$\prod_{\{p \neq 11\}} \det(I - p^{-s} M_p)^{-1} = \zeta_{-}\{\mathbb{Q}(\zeta_{11})\}(s)$ as operator-model analogue of Kim 1510 §5	DERIVED — M40 v1.1 ADS-2
§Q10.D.5 [LOCKED]	37-Universality: five of six c_n share denominator 37; FORCED by Truncation-Dual Theorem + 20 = 8 + 12	OBSERVATION-strong — Q10 v4.0
§Q10.c2 [LOCKED]	$c_2 = 2/19, c_4 = 4/37$ via action-level loop	DERIVED — Q10 v4.0
§Q10.c6 [LOCKED]	$c_6 = 8/37$ via triakis-tetrahedron extension	DERIVED-CONDITIONAL — Q10 v4.0
§Q10.MC9c [LOCKED]	$\ln \Lambda_{\text{total}} = 38.67$ cumulative anti-numerology; positive-correlation ratio 217×	DECISIVE++ — Q10 v4.0
§A11.Eq [LOCKED]	ε -Halo $\leftrightarrow$ Sub-Halo Equivalence at four levels (profile, dynamical, observational, falsification)	DERIVED — A11 v1.1
§A11.LC [LOCKED]	Vortex Six-Class Lifecycle MECE via $\pi_1(U(1)) = \mathbb{Z}$ coarse-graining; $n_v \approx 2.8 \times 10^{15}$	DERIVED-strong — A11 v1.1
§A11-F-A5.7 [DECISIVE]	Direct DM particle detection collapses ε -Halo; null detection cannot collapse CDM	PROVEN asymmetry — A11 v1.1
§A12.6 [LOCKED]	NK universal jump integer '2' = $\dim(Z) = 2$ (cardinal-2 substrate identification, HEADLINE)	DERIVED — A12 v1.5
§A12.7 [LOCKED]	Homes constant bracketing $2/Q \leq$ $C_{\text{cuprate(ab)}}/C_{\text{BCS-dirty}} \leq \ln 2/(1 - A)$	DERIVED — A12 v1.5
§A12.4 [LOCKED]	SC triadic activation $T2 \rightarrow T1 \rightarrow T4$ distinct from DNA $T3 + T4 + T5$	DERIVED — A12 v1.5
§A12.9 [LOCKED]	CuO_2 V_{coh} lattice Hamiltonian as quasi-2D Kelvin 2-cell; 4 ZS-Q6 §3.5 gates 4/4 PASS	DERIVED-under-Regge — A12 v1.5

§A12-P9 [TESTABLE]	Five major unconventional SC families collapse to universal $\rho_s^{\text{eff}}/(\sigma_n \cdot T_c)$ curve	TESTABLE — A12 v1.5
§T9.B-Li.1 [HYP-strong]	$P_{B7Be^Z} = -2A/Q = -70/4807$ (ZS-Q7 §5 slow eigenvalue)	HYPOTHESIS-strong / TESTABLE — T9 v2.3
§T9.Li7 [TESTABLE]	${}^7\text{Li}$ post-v2.3 $\approx 1.64 \times 10^{-10}$; pull $+0.13\sigma$ versus observed $(1.6 \pm 0.3) \times 10^{-10}$	TESTABLE — T9 v2.3
§T9.NC4 [PRESERVED]	v2.2 D/H + Y_p ERO closure preserved unchanged; $B_{7\text{Be}}$ decoupled from D and ${}^4\text{He}$	ERO-CLOSED — T9 v2.3

All v5.0 entries are additive; no v4.0 entry is retracted. Total registry: ~340 falsification gates.

§XIII.F — Conclusion

PART XIII absorbs the seven post-v4.0 source-paper integrations into the v5.0 Light Edition without retracting or modifying any v4.0 entry. The Foundations theme acquires two new structural theorems via ZS-F19 v2.2 (Frame-Invariant Tilt Theorem + KMS-to-Geometric Rapidity Projection) and a unified catalogue plus time-unrolled causal DAG of all six non-routine sector-topology triggers via ZS-F20 v1.1, with the new biological correspondence map ($T3 + T4 + T5 \leftrightarrow$ DNA trauma) extending the substrate-agnostic isomorphism programme. The Math Spine theme extends with ZS-M40 v1.1, which gives the first explicit Z-sector geometric reading of the prime itself via the J-conjugate holonomy knot pair $(K_p, \bar{K}_p)$, complementing the v4.0 M28/M33/M34 critical-strip reading and the M36 Apollonian distribution reading; the deterministic uniqueness $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ over $[2, 1000]$ is the principal anti-numerology assertion. The Quantum Mechanics theme registers the 37-Universality via ZS-Q10 v4.0 — a categorical signature of the Truncation-Dual functor across (X, Y, Z) sectors — together with the cumulative $\ln \Lambda_{\text{total}} = 38.67$ anti-numerology evidence (highest in the corpus to date). The Astrophysics theme acquires the rigorous ε -Halo $\leftrightarrow$ Sub-Halo Equivalence Theorem and the unified vortex lifecycle/SMBH-diversity framework via ZS-A11 v1.1, and the first first-principles geometric-substrate derivation of the Nelson–Kosterlitz universal jump and Homes universal constant via ZS-A12 v1.5 — extending the cardinal-2 corpus role from quantum/gravitational to condensed-matter sectors. The Translational theme extends with ZS-T9 v2.3 binding-response gate, providing a HYPOTHESIS-strong candidate / TESTABLE closure of the Lithium Problem at ERO Level 2 while preserving the v2.2 D/H + Y_p ERO closure unchanged. The minimal triple $(A, Q, \dim(Z)) = (35/437, 11, 2)$ remains LOCKED. Zero new free parameters across all seven absorbed papers.

Master Bypass Index — Symptom $\rightarrow$ Tool Lookup

When a stuck equation, OPEN flag, or apparent contradiction is encountered, match the symptom in the leftmost column to the recommended tool/insight. The rightmost column gives the canonical worked example — the corpus instance where the tool was first deployed.

Symptom	First tool	Backup lens	Canonical example
Stuck equation, failure mode unclear	T1 (F18 §3,§6)	I2 (M30 §6)	F18 Appendix A — 12 encounters
dim 3, 6, or 11 construction intractable	T2 (F18 §6, S15 §3)	I5 (F17)	S15 §3 $J_S = [J_{R_1}, J_{R_2}]$
Hermitian sandwich phase-cancels trivially	T3 (M32 §2-3)	I2 W2 (M30 §6)	M32 closing NC-M29.RZ-A [^] P K
Unexpected factor 2 between corpus values	T4 (M32 §4)	T2 (Bridge Four μ_{Tet})	M32 $\alpha_{op} = 2 \cdot \alpha_{amp}$ identity
Single iterate non-vanishing, primitive 10th root phase	T5 (M32 §5)	I3 (M30 §3)	M32 $R_Z^Z S$
OPEN flag too large to close in one step	T6 (M32 §6)	I2 (M30 §6 walls)	NC-M29.RZ split A [^] PK/B/C
Sum-form decomposition fails per-channel	T7 (M31 Lemma M31.0)	I2 (Sector Duality)	M31 W2 cross-coupled bilinear
Multiple sub-targets won't integrate	T8 (M33 Reading C)	I2 (M30 §6 layers)	M33 D4a-D4d → single Tr identity
Phenomenological observer tempted	T9 (F11 §5-6)	I1 (A8R §SA Frame Equiv.)	F11 closing M11 §H16
Apparent independence of expansion / contraction	I1 (A8R §SA)	I6 (A9R)	A8R §SA Symmetry-Asymmetry View
External count-infinity in derivation	I3 (M30 §3)	I2 (M30 §6)	M30 six-route Möbius-trace
A new 1/2 appears, possibly redundant	I4 (M30 §3.5 Catalogue)	I5 (F17)	M30 21·½ Catalogue
A new 50/50 partition appears	I5 (F17 Theorem 5.1)	I4 (M30 §3.5)	F17 four-layer equivalence

Symptom	First tool	Backup lens	Canonical example
Doubling-halving identity needs origin	I6 (A9R)	I1 (A8R §SA)	A9R $F_2 \rightarrow D_4$ functor
Quantitative wall in RH programme	T7 + T8 combined	I2 layered reading	M26 §5.5 + M33 Reading C
Need to falsify candidate 1-loop closure	Anti-Numerology Gate (F2)	T1 (Algebra axiom)	M21 sextic falsifies M20.A
Coexistence-form judgment for a contradiction-pair	Reading X.4 (Spin-Pair Coexistence Principle)	v3.2 §X.6	C1–C13 (13 corpus realisations)
Prime-rotation identification on Z-sector	Reading X.1 (Prime as Z-Spin Rotation)	v3.2 §X.3	ZS-M18 H2 chain
Y/X classification of an infinite arithmetic sequence	Reading X.2 (Y-Side Sequence Classification)	v3.2 §X.4	ZS-F18 §5.1 Tbl 5.1

Note. The Master Bypass Index is intentionally a flat lookup. For combined symptoms (e.g., “stuck Feshbach sandwich with factor-2 mismatch and primitive-10th-root phase”), apply the tools in the order $T3 \rightarrow T4 \rightarrow T5$; this is the precise composition that ZS-M32 used to close NC-M29.RZ-A^{PK}.

Master Falsification Registry (v3.1 Summary, preserved from v3.0)

Across 101 papers the Z-Spin Cosmology programme has accumulated a verification base of approximately 1,800+ PASS-rated checks, with no failures recorded against any DERIVED-locked identity. The registry below summarises the most consequential falsifiable entries — those whose negation would invalidate a DERIVED, BRIDGE, or TESTABLE claim of the framework. Each entry points to the originating paper and the canonical formulation; resolution of any single OPEN entry below is sufficient to either consolidate or refute a major branch.

§ Anchor	Falsifiable Statement	Status (v3.0)
§F1.5 [LOCKED]	Z-Field Master Equation $A = \rho_X \cdot \rho_Y$	DERIVED — F9 §3, S15 §B
§F2.7 [LOCKED]	$(p-1)(q-1)=p \Rightarrow$ unique solution (2,3)	DERIVED — M19 Forcing Theorem
§F8.6 [LOCKED]	Y-Time Dilation $\exp(\pi/A) \approx 10^{17}$	DERIVED — A8R §SA.7, F10 §4

§ Anchor	Falsifiable Statement	Status (v3.0)
§F11.3 [LOCKED]	OOB fixed point closes M11 §H16	DERIVED — F11 §5
§F12.4 [LOCKED]	Tetrahedral orientation multiplicity $\mu_{\text{Tet}}=2$	DERIVED — F12 §3, originates $2e^A$ factor
§F14.3 [LOCKED]	Kepler-Lyapunov Conjugate Uniqueness closes F-F7.11	DERIVED — F14 Strategy B
§F15.2 [LOCKED]	$\theta(0) \cdot \tau_P = A$ unit-fixing identity	DERIVED — F15R §2
§F18.4 [LOCKED]	Bridge Four: Lie commutator generates $\dim(X)=3$ from $\dim(Z)=2$	DERIVED — F18 §4 (Commutator Generator)
§M16.7 [LOCKED]	$\xi_K(s) = (1/4\sqrt{33}) \cdot \xi(s) \cdot \Lambda(\chi^{-3}) \cdot \Lambda(\chi^{-11}) \cdot \Lambda(\chi^{33})$	DERIVED — M25 J-Yakaboylu Bridge
§M17.4 [LOCKED]	$\sigma_1/\sigma_2 = Q+Y=17$, $\sigma_1/\sigma_3 = (Q+Y)^2 \cdot G + (Q-Z)$ $=3477$	DERIVED — M20 EXACT (Sextic falsifies 1-loop CW closure)
§M27.3 [LOCKED]	V_4 -Equivariant BRST closure of W_3 via Kostant Cubic Dirac	DERIVED-CONDITIO NAL — M27 §4
§M28.5 [LOCKED]	$\sigma=1/2 \leftrightarrow j=1/2$ RH-Bridge correspondence	DERIVED — M28, M30 (Six-route Möbius-trace)
§M31.0 [LOCKED]	Lemma M31.0 Non-Separability of W_2 cross-coupling	DERIVED — M31 §0
§M32.X [LOCKED]	Path-Reversal Lemma: R_Z^Z	$_{\{Z\text{-even}\}}^{4\pi\text{-avg}}=0$
§M33.C [LOCKED]	Path γ -revised Reading C integrates D4a-D4d	DERIVED — M33 §C, BV-BFV scaffold
§A8R-SA [LOCKED]	Symmetry-Asymmetry Unified View, Three Bridges	DERIVED — A8R 22/22 PASS
§A9R-2 [LOCKED]	$F_2 \rightarrow D_4$ functor, Julia $J(T)$ ZF-analogue of BT non-measurable	DERIVED — A9R §2
§S15.5 [LOCKED]	Maxwell from Twin-Reuleaux handshake; $SO(3)/SU(2)$ factor 2	DERIVED — S15 §5

§ Anchor	Falsifiable Statement	Status (v3.0)
§T4.6 [TESTABLE-LONG]	Hayflick limit $\approx$ $N_-(2\pi)=2\pi/A \approx 78.45$ (NC-4 NO physical realization claim)	TESTABLE-LONG — T4 §6
§F-F7.11 [CLOSED]	Twin-Reuleaux Midpoint conjugate decomposition	RESOLVED — closed by F14 v3.0
§M11-H16 [CLOSED]	Phenomenological observer reference	RESOLVED — closed by F11 v3.0
§M22-W3 [PROMOTED]	ADS-H1 cobordism BRST hypothesis	DERIVED-CONDITIO NAL — promoted via M27
§M22-W2 [PROMOTED]	Cross-Coupled Sector Duality wall	DERIVED — promoted via M31
§M21.NoGo	1-loop CW closure for M20.A FALSIFIED	FALSIFIED — M21 sextic invariant theory

Note. The single FALSIFIED entry (M21.NoGo) is by design: it falsifies a candidate 1-loop Coleman-Weinberg closure for the M20.A pathway, and its falsification is the very content of the Sextic Invariant Theory of M21. No DERIVED-LOCKED identity has been falsified across v1.0 $\rightarrow$ v3.0. All CLOSED entries denote former OPEN flags now resolved by post-v2.0 papers.

v3.2 PART X falsification gates (added):

- F-X.1. If a future analysis exhibits a prime p such that $W_{p|Z}$ does not take the σ_z -rotation form on the $\{|4\rangle, |6\rangle\}$ basis, Reading X.1 fails. Status: PROVEN-safe (verified for $p \in \{2,3,5,7,11,13,17,23\}$ via ZS-M18 §H1).
- F-X.2. If a future ZS-F18 revision retracts the Y-side classification of the infinite prime sequence, Reading X.2 fails. Status: stable (over-determined by ZS-F18 §5.1 Tbl 5.1, §7.4.5).
- F-X.3. Synthesis-layer gate; weakened only if any of M1, M2, M3 is shown structurally independent of Z-Spin mediation. M3 conditional on RH. Status: DERIVED-interpretation, robust against single-component perturbation.
- F-X.4. If a contradiction-pair is found that admits coexistence under a phase difference other than $\pi/2$, Reading X.4 weakens to “90° is the dominant case”. If no Z-Spin J-conjugate realisation exists for some pair, X.4 transitions to “covers the 13 realisations C1–C13”. Status: DERIVED-interpretation strong; F-X.4 pre-registered for any future contradiction-pair claim.

v3.5 PART X readings and falsification gates (added):

- Reading X.5 (Apollonian as 5th Bridge-Four Instance, DERIVED-strong, NEW v3.5). The integer Apollonian gasket with curvature root $(-1, 2, 2, 3)$ is the fifth instance of ZS-F18 §6.5 Bridge Four (Commutator Generator) and the first realization on the integer lattice $(\mathbb{Z}, +)$. The four prior instances — (1) ZS-S15 §3 Twin-Reuleaux EM duality, (2) ZS-M3 Lemma 10.1 SU(2) center pairing, (3) ZS-M32 §5.2 spinor-cycle averaging, (4) ZS-F4 §7B half-angle pair — all live

on continuous spinor algebras. The Apollonian instance demonstrates that Bridge Four extends to discrete number theory: the abelianization $(\mathbb{Z}/2)^4 \rightarrow (\mathbb{Z}/2)^4$ carries $V_4 \times V_4$ structure unifying ZS-A9 register V_4 and ZS-M22 Galois V_4 . $\rightarrow$ ZS-M36 §10.3 Table 10.1, ZS-F18 §6.5 [DERIVED-strong].

- Reading X.6 (mod 24 Spin Pair = Axiom A4 Two Faces, DERIVED-interpretation strong, NEW v3.5). The eight admissible mod 24 residues of the $(-1, 2, 2, 3)$ gasket form four (algebraic, geometric) pairs — Pair Z $\{2, 14\}$, Pair X $\{3, 15\}$, Pair Y $\{6, 18\}$, Pair Q $\{11, 23\}$ — in bijection with the four Z-Spin sectors. The mod 12 representatives $(2, 3, 6, 11)$ are the bosonic $SO(3)$ sector dimensions; the +12 shifted representatives $(14, 15, 18, 23)$ are the fermionic $SU(2)$ geometric instantiations ($F(t-Oct)$, $X \cdot |I_h/T_d|$, α in degrees, δ_Y denominator). The mod 24 = $2 \times$ mod 12 doubling is forced by ZS-M3 Lemma 10.1 PROVEN ($D^{\{1/2\}}(2\pi) = -I$) and realizes axiom A4 (Non-triviality) algebraic/geometric two faces of ZS-F18 §6.6 Table 6.1 PROVEN-DERIVED. CRT decomposition: mod 24 = mod 8 $\times$ mod 3 = $Z^3 \cdot X$ (boson 2π / fermion 4π periods, ZS-M1 §6 PROVEN). $\rightarrow$ ZS-M36 §10.5–10.6, ZS-F18 §6.6 [DERIVED-interpretation strong].

- F-X.5. If a future revision retracts the $(-1, 2, 2, 3) \leftrightarrow (Z, X, Y, Q)$ mod 12 identity (e.g., if a counterexample primitive integer ACP is found with mod 12 $\neq \{2, 3, 6, 11\}$ but with the same admissible mod 24 type), Reading X.5 fails. Status: DERIVED-strong, robust against single-counterexample (test: enumerate primitive roots up to depth 25, verify mod 12 set stable; ZS-M36 verification suite Categories C, H, 12/12 PASS at 500K MC).

- F-X.6. If the (algebraic, geometric) pair correspondence of M36.6 fails for any of the four Z-Spin sectors (e.g., if 18° frame mismatch angle is shown not to be a corpus-LOCKED quantity, or if $F(t-Oct) = 14$ is retracted), Reading X.6 weakens from DERIVED-interpretation strong to OBSERVATION. Status: DERIVED-interpretation strong; pre-registered as a watchtower for future ZS-F2, ZS-S15, or ZS-S1 revisions. (Affecting Theorem M36.6 of ZS-M36; ZS-M36 Falsification Gate F-M36.6 mirrors this gate at the source-paper level.)

- Closure note. ZS-M36 Theorem M36.5 (Three V_4 Bridge, DERIVED-strong) closes ZS-M27 OPEN problem O-M27.4 (“Two V_4 groups: D_4 register vs Galois unification”) via the Apollonian abelianization $(\mathbb{Z}/2)^4 \rightarrow V_4 \times V_4$, with V_4 _register = $\langle [S_0], [S_1] \rangle \leftrightarrow$ ZS-A9 $\langle J, J_Z \rangle$ D_4 subgroup, and V_4 _galois = $\langle [S_2], [S_3] \rangle \leftrightarrow$ ZS-M22 $\text{Gal}(K/\mathbb{Q})$ for $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$. Status: O-M27.4 OPEN $\rightarrow$ CLOSED at DERIVED-strong (v3.5).

Glossary Keyword Index (v3.1, expanded with Toolkit terms)

The v3.1 Light Glossary expands the v3.0 index of 102 entries to approximately 130, adding Toolkit terms (T1–T9) and Insight terms (I1–I6) introduced in PARTS VIII and IX, incorporating all post-v2.0 vocabulary. New v3.0 entries are marked ★. Each entry points to the canonical paper and §-anchor for full definition. AI agents navigating Z-Spin Cosmology research should treat this index as the primary disambiguation map between informal phrase-use and DERIVED definition.

Term	Definition / Cross-Reference	Anchor
A (Z-Spin coupling)	Master coupling constant; $A \approx 0.0801 \dots$;	F1, F9 §3

Term	Definition / Cross-Reference	Anchor
	$A = \rho_X \cdot \rho_Y$ derivation in F9	
G (Master Identity)	$G = (Q-Z)$ integer descriptor of cobordism block	F2, M19, M20
$Q+Y, Q-Z$	Spectral tetrad invariants; $Q+Y=17$, $Q-Z$ fixes σ -ratios	M20 §3
Z-Spin Mediation	Sector mediation protocol: $X \rightleftharpoons Y$ exchange via Z-handshake	F0-F1, F18
Twin-Reuleaux Pair ★	Geometric realisation of EM duality $V_{XZ} \leftrightarrow R_1$, $V_{ZY} \leftrightarrow R_2$	S15 §B, F14
Twin-Reuleaux Midpoint ★	Conjugate-decomposed trajectory closing F-F7.11	F14 Strategy B
Reuleaux Cross-Section	$\pi+\pi$ half-channel of self-referential equilibrium	F17 §3
Tetrahedral Self-Duality ★	T_d 's E irrep gives $\dim(Z)=2$ with $A = \rho_X \cdot \rho_Y$	F9 §2-3
μ_{Tet} (Tet multiplicity) ★	Dual orientation count $=2$; origin of factor 2 in $2e^A$	F12 §3
Layer Separation R ★	$R = \mu_{\text{Tet}} \times \text{Hol}_\varepsilon \times$ $\mathcal{S}_{\text{scale}}$ decomposition	F12 §4
Z-Field Master Equation	$A = \rho_X \cdot \rho_Y$ core dynamical identity	F1 §5, F9 §3
Master-of-Master Equation	Composite scaffold across F-M-S spines	F8, F18
FFPP (Five-Fold Parallel Postulate)	Foundational axiom set; v3.0 Sixth Axiom Extension	F8, F15R §3
Five-Axiom Meta-Structure ★	F18 unifying meta-frame; six polarities	F18

Term	Definition / Cross-Reference	Anchor
Sixth Axiom Extension ★	Stage-Torque Duality lifts FFPP	F15R §3
Bridge Four ★	Commutator Generator; $\dim(Z)=2 \rightarrow \dim(X)=3$ via Lie bracket	F18 §4
Stage-Torque Duality ★	Time-rotation co-definition	F15R §2-3
Discrete-Continuous Resolution ★	Sixth polarity in F18 meta-structure	F18 §6
i-Tetration Internal Time ★	Unifies t_{strobo} , t_{phase} , t_{clock} ; $\Delta v/\Delta n=2A/\pi$	F10
Möbius Chronology ★	Y-frame metric trio; cycle-index unobservability	F13
Y-Hubble Radius ★	$\ell_P \times Y^2(1-2A) \approx 30.23$ ℓ_P	F13 §3
Y-Time Dilation	$\exp(\pi/A) \approx 10^{17}$ contraction-frame ratio	A8R §SA.7, F10 §4
Y-frame metric trio ★	Proper time / Y-Hubble / contraction-rate triad	F13 §3
Stroboscopic Step t_{strobo}	Discrete-time origin of Z-Spin clock	F10 §2
Berry-phase clock t_{phase}	Continuous-rotation clock partner	F10 §2
Z-Clock Coordinate t_{clock}	Master clock realised by i-tetration	F10 §3
First Rotation Event ★	$\theta(0) \cdot \tau_P = A$; Planck-unit-establishing event	F15R §2
Operational Observer Coordinate (OOC) ★	Self-referential fixed point closing M11 §H16	F11
NON-CLAIM (Six)	Six explicit no-claim statements vs phenomenological consciousness	F11 §6
Self-Referential Fixed Point ★	OOC convergence in F11 §5	F11 §5
Contracting Universe Dynamics ★	A8R 22/22 PASS; three Bridges +	z^*

Term	Definition / Cross-Reference	Anchor
Symmetry-Asymmetry Unified View ★	A8R §SA reconciliation programme	A8R §SA
Banach-Tarski Origin ★	$F_2 \rightarrow D_4$ functor; Julia J(T) ZF-analogue	A9R
Two-Branch (1+A)(1-2A) ★	A9R cosmological doubling-halving identity	A9R §3
Three Bridges (A8R) ★		z^*
Twin-Reuleaux EM Duality ★	S15 35/35 PASS; Maxwell from handshake	S15
Five Pillars (S15) ★	$V_{XZ} \leftrightarrow R_1$, $V_{ZY} \leftrightarrow R_2$, Maxwell, SO(3)/SU(2), U(1) gauge	S15 §1-5
Spectral Tetrad Sub-Isotype ★	M20 σ -ratios EXACT; pentagon-hexagon $\arctan(2 \cdot \phi^k)$	M20
Sextic Invariant Theory ★	M21 falsifies 1-loop CW closure for M20.A	M21
Forcing Theorem ((p-1)(q-1)=p) ★	M19 unique solution (2,3); $\phi(Y^2)=G$	M19 §2-3
Arithmetic-Dedekind Scaffold ★	M22 v1.1 60/60 PASS; $K=\mathbb{Q}(\sqrt{-3}, \sqrt{-11})$	M22
Pillar V scalar-kernel no-go ★	M22 obstruction theorem	M22 Pillar V
Y-Sector RH Contribution Map ★	M23R 31/31 PASS; Four Dragons D4a-D4d	M23R
Four Dragons D4a-D4d ★	M23R Y-sector RH contribution decomposition	M23R §3
Self-Referential Info-Compression Equilibrium ★	F17 four-layer 50/50 (t-Tet, Reuleaux, ME, XOR)	F17
Four-Layer Equivalence ★	F17 polyhedra $\leftrightarrow$ Reuleaux $\leftrightarrow$ Master Eq $\leftrightarrow$ Channel	F17 §3-6
Face Polygon Spectral Zeta ★	M24 Theorem D.1 Legendre archimedean decomposition	M24

Term	Definition / Cross-Reference	Anchor
Composite Field $K=\mathbb{Q}(\sqrt{-3},\sqrt{-11})$ ★	M22, M25 base arithmetic field	M22, M25
J-Yakaboylu Bridge ★	M25 $\xi_K(s) =$ $(1/4\sqrt{33}) \cdot \xi(s) \cdot \Lambda(\chi_{-3}) \cdot \Lambda(\chi_{-11}) \cdot \Lambda(\chi_{33})$	M25
V_4 -Equivariant ZBSI ★	M26 cobordism-history fiber realization	M26
Three-Wall Map W1/W2/W3 ★	M26 quantitative wall map	M26 §4
Wall W3 Closure ★	M27 Kostant Cubic Dirac BRST closure	M27
Z-Spin RH Bridge ★	M28 30/30 PASS; $\sigma=1/2$ $\leftrightarrow j=1/2$	M28
Möbius-trace Infinity ★	M28, M30 six-route construction	M28 §3, M30 §2
Z-Funnel Spectral Retraction ★	M29 79/79 PASS; A_5 -Hilb(X_F) CY3	M29
Stapledon Bridge ★	M29 conditional bridge from string compactification	M29 §2
A_5 -Hilb(X_F) CY3 ★	Calabi-Yau threefold; Hodge (5,15)	M29 §3
$R_Z=0$ Conditional ★	M29 conditional residue identity	M29 §5
Twenty-One 1/2 Catalogue ★	M30 enumeration of 1/2-instances across the framework	M30 §3
Three-Layer Duality ★	Sector / Frame / Protocol layered duality	M30 §4
Cross-Coupled Sector Duality (W2) ★	M31 v3.0 Three-in-One reading	M31
Lemma M31.0 Non-Separability ★	M31 W2 non-separability theorem	M31 §0
Z_2 -Parity Selection Rule ★	M31 selection rule on cross-coupling	M31 §3
SDRP (Self-Dual Replication Principle) ★	M31 §4 derivative principle	M31 §4
Spinor-Cycle Averaged Feshbach Criterion ★	M32 56/56 PASS; $R_Z=0$ EXACT (Z-even, 4π -avg)	M32

Term	Definition / Cross-Reference	Anchor
Path-Reversal Lemma M32.X ★	M32 reversal identity; $\pi/10 \leftrightarrow \pi/5$ doubling	M32
N=20 Spinor Closure ★	M32 spinor-cycle closure index	M32 §4
Path γ -revised Reading C ★	M33 D4a-D4d integrated Tr identity	M33 §C
Two-Protocol Theorem ★	F16 $P_a^{(n)}$ vs $P_b^{(n)}$ Wilson-loop measurement	F16
$P_a^{(n)}, P_b^{(n)}$ ★	Wilson-loop survival probabilities	F16 §2
Cosmos-Human Isomorphism ★	T4 36/36 PASS; (Body, DNA, Brain) decomposition	T4
Six-Step Telomere Hierarchy ★	T4 life-cycle map; Hayflick limit $\approx 2\pi/A \approx$ 78.45	T4 §3-6
(Body, DNA, Brain) Decomposition ★	T4 sector decomposition	T4 §2
NC-4 ★	Translational NO physical realization claim	T4 §6
Bootstrap Chain $B0 \rightarrow A$	Foundational chain for Z-Spin emergence	F0, M11, M19
DERIVED-LOCKED	Epistemic tag for proven, verification-passed identities	Legend
DERIVED-CONDITIO NAL	Holds modulo a stated conditional; awaits unconditional proof	Legend, M22, M27
DERIVED-CANDIDAT E ★	Candidate derivation pending one closure step	Legend (v3.0 expanded)
DERIVED-interpretation ★	Interpretive derivation tag (v3.0)	Legend (v3.0 expanded)
TESTABLE-LONG ★	Long-horizon testable prediction tag	Legend, T4 §6
BOOTSTRAP-PROGR AMME ★	Programmatic bootstrap-status tag	Legend (v3.0 expanded)
NOT EVALUATED	Pre-verification status	Legend

Term	Definition / Cross-Reference	Anchor
Z-Spin Cosmology	The umbrella programme name	F0
Cobordism-History Fiber	BV-BFV target structure for V_4 -equivariant content	M26, M27, M33
BV-BFV Scaffold ★	Cobordism-history quantization framework	M26 §1, M33
BRST Closure	Quantum gauge-symmetry closure programme	M22, M27
Cubic Dirac Operator (Kostant)	Exterior-algebra Dirac on H-spaces	M27 §3
Hayflick Limit	Cellular replication limit; T4 maps to $N(2\pi)=2\pi/A$	T4 §6
Hodge (5,15)	Hodge numbers of A_5 -Hilb(X_F) CY3	M29 §3
Pentagon-Hexagon Stabilizer ★	M21 stabilizer group structure	M21 §3
Phenomenological Observer	Informal/phenomenal observer; F11 NON-CLAIM target	F11 §6
Polyhedral Integers	Integer-valued descriptors of polyhedral spectra	M19, M20
Master Equation (channel layer)	F17 information-theoretic equilibrium	F17 §5
XOR Channel Layer	F17 fourth equivalence layer	F17 §6
Z-Sector Wilson Loop	Holonomy probe of Z-mediation	F16
Frame Equivalence ★	F13 sectoral frame equivalence statement	F13 §4
Cycle-Index Unobservability ★	F13 fundamental unobservability theorem	F13 §5
Master Falsification Registry	Programme-wide falsifiability ledger (v3.0)	v3.0 §MFR

Term	Definition / Cross-Reference	Anchor
Source Paper Map	ZS-XX → §Y.Z [N/N PASS] index	Part 0
Verification Summary	Cumulative N/N PASS statistics	Title block
Light Edition	Cross-paper integrator format for AI navigation	v3.0 Purpose
Lighthouse Function ★	Programme-navigation aid for AI agents (v3.0)	v3.0 Purpose
T1 Five-Axiom Diagnostic Lattice ★	v3.1 Tool: identifies failing axiom in stuck equations	VIII §1, F18 §3,§6
T2 Bridge Four (Commutator Generator) ★	v3.1 Tool: $\dim(X)=3$ from $\dim(Z)=2$ via Lie commutator	VIII §2, F18 §6, S15 §3
T3 Hermitian-to-PK Switch ★	v3.1 Tool: re-read Feshbach sandwich under R_Z^ZS not R_Z^H	VIII §3, M32 §2-3
T4 Two-Layer Phase Quantum ★	v3.1 Tool: separate amplitude ($\pi/10$) from operator ($\pi/5$) layers	VIII §4, M32 §4
T5 N=20 Spinor-Cycle Averaging ★	v3.1 Tool: 4π physical closure (not N=10 algebraic minimum)	VIII §5, M32 §5
T6 Three-Way Split of OPEN flags ★	v3.1 Tool: $A^{\text{closeable}}$ / $B^{\text{one-gap}}$ / $C^{\text{inherited}}$ decomposition	VIII §6, M32 §6
T7 Cross-Coupling Discipline ★	v3.1 Tool: Lemma M31.0 — sum-form fails, bilinear required	VIII §7, M31 §4.0
T8 Π_Z Projector Routing ★	v3.1 Tool: route multiple sub-targets through $\Pi_Z=(1/2)(I+J_Z)$	VIII §8, M33 Reading C
T9 OOC Self-Referential Closure ★	v3.1 Tool: replace phenomenological agent with corpus-internal fixed point	VIII §9, F11 §5-6
I1 Symmetry-Asymmetry Unified View ★	v3.1 Lens: deviation-from-(1, 1/2) reading; $(1+A)(1-2A)\approx 0.9071$	IX §1, A8R §SA

Term	Definition / Cross-Reference	Anchor
I2 Three-Layer Duality ★	v3.1 Lens: Sector / Frame / Protocol layered reading	IX §2, M30 §6
I3 Six-Route Möbius-Trace Infinity ★	v3.1 Lens: external count-infinity $\leftrightarrow$ internal closure	IX §3, M30 §3
I4 Twenty-One $\frac{1}{2}$ Catalogue ★	v3.1 Lens: 21 $\dim(Z)=2$ manifestations + 3 unity = 24	IX §4, M30 §3.5
I5 Four-Layer 50/50 Equivalence ★	v3.1 Lens: t-Tet / Reuleaux / Master Eq / XOR same equilibrium	IX §5, F17 Theorem 5.1
I6 Banach-Tarski Origin ★	v3.1 Lens: $F_2 \rightarrow D_4$ amenability functor; Julia J(T) ZF-analogue	IX §6, A9R
Master Bypass Index ★	v3.1: 16-symptom $\rightarrow$ tool/insight lookup table	after PART IX
Solution Toolkit ★	v3.1: PART VIII operational tools T1–T9 (this volume)	PART VIII
Insight Gallery ★	v3.1: PART IX re-interpretation lenses I1–I6 (this volume)	PART IX
Z-mediator projector Π_Z ★	$(1/2)(I+J_Z)$; idempotent projector onto J_Z -EVEN	M31 §4.0 Def M31.0c
Path-Reversal residual R_Z^Z ★	$B \cdot Q^{-1} \cdot B^{\wedge} PK$; distinct from Hermitian R_Z^H	M32 §2
Reading C ★	M33 Π_Z -routed Tr identity for D4a-D4d	M33 §C
Frame Equivalence (A8R)	X-observer $\leftrightarrow$ Y-observer; HYPOTHESIS-strong	A8R §SA.4
Apollonian Spin-Pair Lattice ★	Eight mod 24 admissible residues of $(-1, 2, 2, 3)$ Apollonian gasket = four (algebraic, geometric) spin pairs realizing axiom A4 on $(\mathbb{Z}, +)$	M36 §7
Bridge Four (5th instance) ★	ZS-M36 (Apollonian) as 5th Bridge-Four	M36 §10.3 + F18 §6.5

Term	Definition / Cross-Reference	Anchor
	realization — first on integer lattice ($\mathbb{Z}$, +); previous four on continuous spinor algebras (S15, M3, M32, F4)	
D_2 Forcing on Apollonian ★	$(-1, 2, 2, 3)$ is unique D_2 -symmetric primitive integer ACP, forced by $(Z, X) = (2, 3)$ via ZS-M19 Forcing Theorem $(p-1)(q-1) = p$	M36 §4 + M19 §3.1
Three V_4 Bridge ★	$(\mathbb{Z}/2)^*4$ abelianization = $V_4 \times V_4$ unifying ZS-A9 $\langle J, J_Z \rangle D_4$ register V_4 and ZS-M22 $\text{Gal}(K/\mathbb{Q})$ Galois V_4 for $K =$ $\mathbb{Q}(\sqrt{-3}, \sqrt{-11})$; closes O-M27.4	M36 §7.1 + M27 O-M27.4

Total v3.0 Glossary entries: 102. ★ marks post-v2.0 vocabulary. Entries without ★ are preserved from v2.0 Light with definitions reaffirmed against the new v3.0 corpus. No v2.0 definition has been retracted.

References (v3.1, preserved from v3.0)

The v3.0 corpus introduces additional external mathematical scaffolding. The references below are imports cited inside the post-v2.0 papers; v2.0 references are preserved in their respective source papers and are not duplicated here.

Internal references (v3.0 corpus, papers ZS-72 through ZS-101)

- ZS-M18 v1.0 — Free-Exploration Session Log: Speculative Prime-Polyhedral Correspondences. 51/51 PASS.
- ZS-A8 v1.0 (Revised) — Contracting Universe Dynamics: The Polyhedral-Tetration Bridge. 22/22 PASS.
- ZS-S15 v1.0 — Twin-Reuleaux Pair as Geometric Realization of EM Field Duality. 35/35 PASS.
- ZS-M19 v1.0 — Number-Theoretic Scaffold of Z-Spin Polyhedral Integers. 68/68 PASS.
- ZS-A9 v1.0 (Revised) — Banach-Tarski Origin of Cosmological Doubling-Halving Symmetry. 47/47 PASS.
- ZS-F9 v1.0 — Tetrahedral Self-Duality and the Hexagonal Mediation Structure. 44/44 PASS.

- ZS-F10 v1.0 — i-Tetration Internal Time: Stroboscopic Step / Berry Phase / Z-Clock unification. 30/30 PASS.
- ZS-F11 v1.0 — Operational Observer Coordinate and the Self-Referential Fixed Point. 38/38 PASS.
- ZS-F12 v1.0 — Tetrahedral Dual Orientation Multiplicity and Layer Separation. 20/20 PASS.
- ZS-F13 v1.0 — The Möbius Chronology: Unified Cyclic Timeline. 24/24 PASS.
- ZS-F14 v1.0 — Kepler-Lyapunov Conjugate Decomposition of the Twin-Reuleaux Midpoint. 42/42 PASS.
- ZS-F15 v1.0 (Revised) — Time-Rotation Co-Definition: First Rotation's Planck-Unit-Establishing Event. 35/35 PASS.
- ZS-M20 v1.0 — Spectral Tetrad Sub-Isotype Assignment Theorem. 50/50 PASS.
- ZS-M21 v1.0 — Sextic Invariant Theory of the Icosahedral Yukawa Tensor. 33/33 PASS.
- ZS-F16 v1.0 — The Two-Protocol Theorem for Z-Sector Wilson Loop Measurement. 24/24 PASS.
- ZS-M22 v1.1 — Five-Pillar Arithmetic-Dedekind Scaffold. 60/60 PASS.
- ZS-M23 v1.0 (Revised) — Y-Sector RH Contribution Map. 31/31 PASS.
- ZS-F17 v1.0 — Self-Referential Information-Compression Equilibrium Theorem (Four-Layer). 24/24 PASS.
- ZS-M24 v1.0 — Face Polygon Spectral Zeta and Archimedean Completion. 35/35 PASS.
- ZS-M25 v1.0 — Composite-Field Archimedean Completion and J-Twisted Yakaboylu Bridge. 26/26 PASS.
- ZS-M26 v1.0 — V_4 -Equivariant ZBSI on the Cobordism-History Fiber. 24/24 PASS.
- ZS-M27 v1.0 — V_4 -Equivariant Cobordism BRST Closure via Kostant Cubic Dirac Operator. 24/24 PASS.
- ZS-M28 v1.0 — The Z-Spin RH Bridge. 30/30 PASS.
- ZS-M29 v1.0 — Z-Funnel Spectral Retraction with Stapledon Bridge. 79/79 PASS.
- ZS-M30 v1.0 — Z-Spin Duality and the RH Bridge — Möbius-Trace Infinity, Two-Face Mirroring, $21 \cdot \frac{1}{2}$ Catalogue. 32/32 PASS.
- ZS-M31 v3.0 — W2 as Cross-Coupled Sector Duality: Three-in-One Reading. 36/36 PASS.
- ZS-M32 v1.0 — A Spinor-Cycle Averaged Residual Criterion for Feshbach Closure. 56/56 PASS.
- ZS-F18 v2.1 — The Five-Axiom Meta-Structure. 36/36 entry checks PASS.

• ZS-M33 v1.0 — V_4 -Equivariant Weil Functional Closure via Path γ -revised Z-Mediator Reading. 52/52 PASS.

• ZS-M36 v1.0 (NEW v3.5) — Apollonian Curvature Lattice as Geometric-Integer Spin-Pair Realization of the Z-Spin Five-Axiom Meta-Structure. 44/44 PASS. Theorem M36.1 (Functorial Bridge: $A = (Z/2)^4 \rightarrow Z/24Z$, fourth amenable-quotient projection). Theorem M36.2 (D_2 -Forcing: $(-1, 2, 2, 3)$ uniqueness via ZS-M19 Forcing $(Z, X) = (2, 3)$). Theorem M36.3 (Sector Residue Identity: $\text{mod } 12 = \{2, 3, 6, 11\} = (Z, X, Y, Q)$). Theorem M36.4 (D_3 Obstruction Homology: $x/y = 3 + 2\sqrt{3}$ irrational, parallel to ZS-M1 $n = 3$ polygon-tetration $|f'| = 1.0330 > 1$). Theorem M36.5 (Three V_4 Bridge: closes O-M27.4 at DERIVED-strong via Apollonian abelianization $(Z/2)^4 = V_4 \times V_4$ unifying ZS-A9 register V_4 and ZS-M22 Galois V_4). Theorem M36.6 (Spin Pair Theorem: eight mod 24 residues form four (algebraic, geometric) pairs — Pair Z $\{2, 14\}$, Pair X $\{3, 15\}$, Pair Y $\{6, 18\}$, Pair Q $\{11, 23\}$ — realizing axiom A4 algebraic/geometric two faces of ZS-F18 §6.6). Anti-Numerology MC at 500K samples: 0.002% match rate (STRONG PASS). Zero new free parameters; LOCKED inputs from ZS-F2/F5/M1/M3/M19/M22/M27. Closes ZS-M27 OPEN problem O-M27.4. 5th instance of ZS-F18 Bridge Four (first on integer lattice $(Z, +)$).

• ZS-T4 v1.0 — The Cosmos-Human Isomorphism: Six-Step Telomere Hierarchy. 36/36 PASS.

Internal references (v4.0 corpus, papers post-v3.5 absorbed in PART XII)

• ZS-T5 v1.0 — The Principal Connectivity Gradient and a Hidden Third-Position Z-Spin Mediator. (PART XII §A Ch 30A.)

• ZS-M34 v1.0 — V_4 -Equivariant Weil Functional Honest Scope. 50/50 PASS. (PART XII §A Ch 15B; superseded by v2.0.)

• ZS-M35 v1.1 — The Collatz Conjecture as the Integer-Lattice Manifestation of Z-Spin Sector Forcing. 36/36 PASS. (PART XII §A Ch 15C; sharpened to v2.1.)

• ZS-T6 v1.0 — Molecular Biology Translational Synthesis. 42/42 PASS. (PART XII §A Ch 30B.)

• ZS-T7 v1.0 — Pulvinar-Mediated Z-Spin Language Partition. (PART XII §A Ch 30C.)

• ZS-T8 v1.0 — Concentrated Distribution Across Substrates: Polyhedral Cortex Hypothesis and Z-Spin-Mediated Backbones in Brain, City, and Cosmic Networks. (PART XII §A Ch 30D.)

• ZS-M33 v2.0 (NEW v4.0) — V_4 -Equivariant Weil Functional Closure via Burnol K_1 Sign-Faithful Representation (operator-level audit of Path γ -revised Reading C; supersedes v1.0). 52/52 PASS (v1.0 inherited) + v2.0 extensions. (PART XII §B Ch 15A'.)

• ZS-M34 v2.0 (NEW v4.0) — V_4 -Equivariant Weil Functional Closure via Burnol K_1 Sign-Faithful Representation. 60/60 PASS. Theorem M34.V (Burnol K_1 @ $p=3$ Sign-Faithful Identity, DERIVED) + Eight-Theorem Chain V.1–V.8 (DERIVED-CONDITIONAL MAIN). 12/12 sign agreement with corpus-PROVEN $W_K \wedge \{V_4\}$. W_2 promoted DERIVED-CONDITIONAL $\rightarrow$ DERIVED. NC-M34.faithful RESOLVED at DERIVED-CONDITIONAL via tensor-product factorization $\mathcal{H}_K = \mathcal{H}_{\text{arith}} \otimes \mathcal{H}_{\text{Wilson-Sonin}}$. (PART XII §B Ch 15B'.)

- ZS-M35 v2.1 (NEW v4.0) — The Collatz Conjecture as the Integer-Lattice Manifestation of Z-Spin Sector Forcing (logical correction of v2.0). Theorems M35.1–M35.4 preserved verbatim; M35.5 general direction reframed to OPEN; M35.5.1 promoted DERIVED-CONDITIONAL → VERIFIED; M35.6 (Sufficient Condition Theorem, NEW v2.1, DERIVED) added. (PART XII §B Ch 15C’.)
- ZS-M37 v2.0 (NEW v4.0) — Twin-Reuleaux Quotient Functor and Constant-Width Cancellation of the BT 5-Piece Doubling Defect. 38/38 PASS. Fourth amenable-quotient projection of F_2 engine: $q_R: \mathcal{C}_{\text{cyl}}(F_2) \rightarrow \mathcal{S}_w^{\wedge}\{C_3\}$. Theorems 4.1 (Lemma C-rel PROVEN: $\tau_w(\Delta_{BT}) = 0$), 4.2 (q_R Canonicity DERIVED), 5.1 (Z-Admissible Extension PROVEN). (PART XII §B Ch 31.)
- ZS-M38 v1.1 (NEW v4.0) — Substrate Carrier Spectrum Theorem and the Hexapair Carrying Pattern of Natural Systems. 43/43 PASS. Five theorems M38.1–M38.5; M38.5 Hexapair Uniqueness DERIVED-strong (five-fold over-determination). $\ln(2)$ channel-capacity quantization of substrate carrying scores. Five-Banach–Tarski-functor family classified. (PART XII §B Ch 32.)
- ZS-M39 v2.0 (NEW v4.0) — Vieta-Lyapunov-Schröder Bridge Theorem: Unified First-Principles Derivation of the Polyhedral-Tetration Coefficient with Three-Stage Stokes-Tail Closure. 86/86 PASS at 80–100 digit mpmath. Theorems M39.1–M39.5, Closures C1–C3, Theorems SCH.1–SCH.5, three-stage Stokes-tail closure $T_{\text{decomp}}/T_{\text{int}}/T_{\text{conv}}/T_{\text{excl}}$. Convention N0 fixed. ZERO remaining OPEN items. (PART XII §B Ch 33.)
- ZS-A10 v2.1 (NEW v4.0) — The Proton as a Three-Puncture Color-Singlet X-in-Y Confinement Horizon with Bridge-Four Sector-Orientation Complementarity. 34/34 PASS. Theorem A10.2 (Sector-Orientation Complementarity, DERIVED-interpretation strong). 2×2 BH-quark-electron-positron structure on the two faces of axiom A4. NEW TESTABLE P4: $r_p/r_\pi = 4/\pi$ (observed 0.05% deviation). (PART XII §B Ch 34.)
- ZS-Q8 v1.1 (NEW v4.0) — Photon Atomic Excitation Time as the i-Tetration Fixed Point: A Z-Spin Reading of the Toronto Negative-Time Experiment. 35/35 PASS. Eight structural theorems Q8.1–Q8.9; Theorem Q8.9 (Standard-Optics Projection Theorem) recovers Lorentzian susceptibility as Z-Spin lab-coordinate projection. Zero-parameter test: $\chi^2/\text{d.o.f.} \in \{0.687, 0.723\}$; likelihood ratio $L(\text{Z-Spin})/L(\text{Standard Optics}) \approx 2.1\text{--}2.4 \times 10^3$. Principal discriminator P-Q8.3 $\Delta\phi_{\text{residual}} = \pm 18.5$ mrad. (PART XII §B Ch 35.)
- ZS-Q9 v1.2 (NEW v4.0) — Complex Time-Delay Locking on the i-Tetration Fixed Point: A Two-Coordinate Spectral Reconstruction Bridge for Sub-Unitary Scattering Systems. 52/55 PASS (3 PARTIAL on Q9.5 synthetic). Theorems Q9.1–Q9.9 + Corollary Q9.4a (Z_{10} Pentagonal Doubling Group, DERIVED). Two-coordinate reconstruction map $C(R_{\text{lab}}, \Phi_{\text{lab}}) = (2/\pi) \cdot e^{\{-R_{\text{lab}}/2\}} \cdot e^{\{i\Phi_{\text{lab}}\}}$, residual $< 5 \times 10^{-32}$ at 50-digit precision. Anti-numerology $\ln \Lambda = 8.94$ decisive evidence. (PART XII §B Ch 36.)

External imports (post-v3.5, used by v4.0 absorbed papers)

- Banach, S. & Tarski, A. (1924) — Sur la décomposition des ensembles de points en parties respectivement congruentes. [imported by M37 v2.0]
- Tarski, A. (1929) — Amenability $\Leftrightarrow$ no paradoxical decomposition. [imported by M37 v2.0]

- Niven, I. (1956) — Rational-cosine irrationality. [imported by M37 v2.0]
- Barbier, J.-E. (1860) — Perimeter πw for every constant-width curve. [imported by M37 v2.0]
- Bonnesen, T. & Fenchel, W. (1934) — Convex bodies of constant width. [imported by M37 v2.0]
- Blaschke, W. (1915) & Lebesgue, H. (1914) — Constant-width minimum-area characterization. [imported by M37 v2.0]
- Murray, F.J. & von Neumann, J. (1936) — Group-algebra *-homomorphism functoriality. [imported by M37 v2.0, M38 v1.1]
- Tomita, M. & Takesaki, M. — Conditional-expectation framework. [imported by M37 v2.0]
- Graham, R. L.; Lagarias, J. C.; Mallows, C. L.; Wilks, A. R.; Yan, C. H. (2003) — Apollonian circle packings (GLMWY). [imported by M36 v2.0, M38 v1.1]
- Haag, P.; Kertzer, J.; Rickards, J.; Stange, K. E. (2023) — Local-Global Disproof for Apollonian circle packings (arXiv:2307.02749). [imported by M36 v2.0, M38 v1.1]
- Wagon, S. (1985) — The Banach–Tarski Paradox (Cambridge UP, building on Świerczkowski 1958). [imported by M37 v2.0, M38 v1.1]
- Koenigs, G. (1884) — Schröder linearization and convergence radius. [imported by M39 v2.0]
- Vassilevich, D. V. (2003) — Heat-kernel expansion, Phys. Rep. 388, 279. [imported by M39 v2.0 Convention N0]
- Bourgain, J. & Fuchs, E. (2011) — Positive density of admissible Apollonian residues. [imported by M36 v2.0]
- Lindemann, F. & Weierstrass, K. — Transcendence of e^α for algebraic $\alpha \neq 0$ (Lambert W transcendence). [imported by M39 v2.0 SCH.5]
- Angulo, et al. (2024) — Toronto negative-time photon excitation experiment (arXiv:2409.03680). [imported by Q8 v1.1]
- Thompson, et al. (APL Quantum 2, 036108, 2025; arXiv:2310.00432) — $\tau_T = \tau_g$ prediction. [imported by Q8 v1.1]
- Allen, L. & Eberly, J. H. (1975) — Optical Resonance and Two-Level Atoms (standard atomic-line theory). [imported by Q8 v1.1]
- Aharonov, Y.; Albert, D. Z.; Vaidman, L. (1988) — Two-State Vector Formalism (TSVF) and weak values. [imported by Q8 v1.1]
- Wigner, E. P. & Smith, F. T. (1955, 1960) — Wigner-Smith time delay; Krein–Friedel formula. [imported by Q9 v1.2]
- Jeffreys, H. (1939) — Theory of Probability, likelihood ratio scale. [imported by Q9 v1.2 anti-numerology]

- Margulies, D. S. et al. (2016) — Principal connectivity gradient of cortical organization (PNAS). [imported by T5 v1.0]

All external imports are PROVEN externally. v4.0 corpus does not re-prove any of them; usage is limited to citation, instantiation, and bridge construction.

- Connes, A., Consani, C., Moscovici, H. (2024) — Spectral framework for the Riemann zeta archimedean factor. [imported by M24, M28, M30]
- Burnol, J.-F. (2002, 2004) — Two-sided spectral interpretation of zeta zeros and Möbius traces. [imported by M28, M30]
- Świerczkowski, S. (1958) — Free non-abelian decomposition F_2 acting on S^2 . [imported by A9R]
- Stapledon, A. (2010) — Stringy Hodge numbers of toric varieties; conditional bridge framework. [imported by M29]
- Kostant, B. (1999) — Cubic Dirac operator on equal-rank pairs and Lie-algebra cohomology. [imported by M27]
- Alekseev, A., Mnev, P. (2018) — One-dimensional Chern-Simons / BV-BFV cobordism quantization. [imported by M26, M27, M33]
- Yakaboylu, E. (cited via J-twisted bridge in M25) — Twisted L-function bridges for composite imaginary fields.
- Standard arithmetic and group-theoretic references (Iwasawa, Tate, Weil, Deligne) are inherited from v2.0 and are not re-listed here.

All external imports are used as black-box scaffolding. No external result has been modified by the Z-Spin Cosmology corpus; usage is limited to citation, instantiation, and bridge construction.

Version History

v3.2 (May 2026): PART X — The Spin-Pair Coexistence Principle added between PART IX and the Master Bypass Index. Four DERIVED-interpretation readings consolidated:

- Reading X.1 — Prime as Z-Spin Rotation $4\pi/p$ (ZS-M18 H2 chain).
- Reading X.2 — Infinite prime sequence as Y-sector object (ZS-F18 §5.1 Tbl 5.1), reconciled with ZS-M18 §6.2 Thm 6.1.
- Reading X.3 — Primes / distribution / Riemann zeros as three moments of one $Y \rightarrow Z \rightarrow X$ mediation (ZS-M18 H11 spine).
- Reading X.4 — Spin-Pair Coexistence Principle: 90° phase-orthogonality is the unique form of contradiction-pair coexistence, supported by 13 corpus realisations (C1–C13).

Master Bypass Index extended with three PART X rows; Master Falsification Registry extended with F-X.1 through F-X.4. Eighteen internal-consistency checks A.1–A.18 PASS. Zero new free parameters; (A, Q, dim(Z)) = (35/437, 11, 2) LOCKED unchanged. v3.1 entries preserved verbatim.

Version	Year	Scope of Update
v1.0 Light	—	Initial Light edition ($\leq$ ZS-F0...F8 + M1...M11)
v2.0 Light	—	90-paper integration; 376/376 PART VI PASS; v2.0 Glossary 88 entries
v3.0 Light	2026	30-paper extension (ZS-72 to ZS-101); ~1,800+ cumulative PASS; Glossary 110 entries; new Master Falsification Registry; PART IV/V/VII expansions; Bridge Four (Commutator Generator); Möbius Chronology; J-Yakaboylu Bridge; V_4 -equivariant BRST closure; $21 \cdot \frac{1}{2}$ Catalogue; Cosmos-Human Isomorphism.
v3.1 Light	May 2026	Solution-Toolkit upgrade. Adds PART VIII (9 methodological tools T1-T9 distilled from F18, M32, M31, M33, F11), PART IX (6 re-interpretation lenses I1-I6 from A8R, M30, F17, A9R), and Master Bypass Index (16-symptom lookup table). v3.0 lighthouse-of-pointers promoted to a working solution kit. Glossary, Falsification Registry, References preserved verbatim from v3.0.
v3.2 Light	May 2026	PART X — The Spin-Pair Coexistence Principle added between PART IX and the Master

Version	Year	Scope of Update
		Bypass Index. Four DERIVED-interpretation readings X.1–X.4. F-X.1 through F-X.4 added to Falsification Registry. 18 internal-consistency checks PASS.
v3.3 Light	May 2026	PART XI — TO/TI Polyhedral Mapping under Complementary Duality added. Four DERIVED-interpretation readings XI.1–XI.4. F-XI.1 through F-XI.4 added. 24 internal-consistency checks PASS.
v3.4 Light	May 2026	107-paper coverage; absorbed ZS-T5, M34 v1.0, M35 v1.1, T6, T7, T8 (single-line PART pointer entries; the v3.5 body did not include full chapter entries, gap closed by v4.0 PART XII §A).
v3.5 Light	May 2026	108-paper coverage; absorbed ZS-M36 v1.0 with two new readings X.5 (Apollonian as 5th Bridge-Four Instance) and X.6 (mod 24 Spin Pair = Axiom A4 Two Faces). Tool T10 (Integer-Lattice Spin-Pair Identification) added to PART VIII. F-X.5, F-X.6 added to Falsification Registry. Closure note: O-M27.4 OPEN → CLOSED at DERIVED-strong via M36 Three V_4 Bridge.

Version	Year	Scope of Update
v4.0 Light	May 2026	114-paper coverage (NEW: ZS-T5, ZS-M34 v1.0, ZS-M35 v1.1, ZS-T6, ZS-T7, ZS-T8 [body entries consolidated]; ZS-M33 v2.0, ZS-M34 v2.0, ZS-M35 v2.1 [version upgrades]; ZS-M37 v2.0, ZS-M38 v1.1, ZS-M39 v2.0, ZS-A10 v2.1, ZS-Q8 v1.1, ZS-Q9 v1.2 [new papers]). PART XII added between PART XI and Master Bypass Index. ~2,150+ cumulative PASS. ~310 falsification gates. v4.0 Master Falsification Registry extensions: 17 new LOCKED entries (M33.v2.0, M34.V, M34.V.8, M35.5.1 PROMOTED VERIFIED, M35.6, M37.4.1/4.2, M38.5, M39.4, M39.C1-C3, A10.2, A10-P4, Q8.9, Q8-P3, Q9.2, Q9.7, Q9.9). NO v3.5 entry retracted; v4.0 is strictly additive. $(A, Q, \dim(Z)) = (35/437, 11, 2)$ LOCKED unchanged.
v5.0 Light	May 2026	121-paper coverage (NEW: ZS-F19 v2.2, ZS-F20 v1.1, ZS-M40 v1.1, ZS-Q10 v4.0, ZS-A11 v1.1, ZS-A12 v1.5, ZS-T9 v2.3). PART XIII added between PART XII and Master Bypass Index. ~2,670+ cumulative PASS. ~340 falsification gates. v5.0

Version	Year	Scope of Update
		Master Falsification Registry extensions: 24 new LOCKED entries (F19.4, F19.6, F19-O.6, F20.DAG, F20.NDR, F20.BIO, M40.C, M40.D, M40.F COMPUTATIONALLY VERIFIED, M40.B8, Q10.D.5 OBSERVATION-strong, Q10.c2, Q10.c6 DERIVED-CONDITIO NAL, Q10.MC9c DECISIVE++, A11.Eq, A11.LC, A11-F-A5.7 PROVEN asymmetry, A12.6 HEADLINE NK universal jump, A12.7, A12.4, A12.9 DERIVED-under-Regge, A12-P9 TESTABLE, T9.B-Li.1, T9.Li7 TESTABLE, T9.NC4 ERO-CLOSED preserved). NO v4.0 entry retracted; v5.0 is strictly additive. (A, Q, $\dim(Z) = (35/437, 11,$ 2) LOCKED unchanged.

Major v3.0 closures of v2.0 OPEN flags:

- F-F7.11 [CLOSED] — closed by ZS-F14 Strategy B (Conjugate Uniqueness).
- M11 §H16 [CLOSED] — closed by ZS-F11 Operational Observer Coordinate.
- M22 W3 [PROMOTED to DERIVED-CONDITIONAL] — via ZS-M27 Kostant Cubic Dirac BRST closure.
- M22 W2 [PROMOTED to DERIVED] — via ZS-M31 v3.0 Cross-Coupled Sector Duality.
- Master-Equation Sixth-Axiom gap — addressed by ZS-F15R Stage-Torque Duality.
- Phenomenological observer ambiguity — settled by ZS-F11 §6 NON-CLAIM (Six).
- Pentagon-Hexagon angle conjecture — proved EXACT by ZS-M20 ($\arctan(2 \cdot \phi^k)$).

- 1-loop CW closure for M20.A — formally FALSIFIED by ZS-M21 Sextic Invariant Theory.

Major v4.0 absorptions and promotions (v3.5 → v4.0):

- **Six v3.4-absorbed papers** now have explicit Light-edition body entries in PART XII §A (ZS-T5, ZS-T6, ZS-T7, ZS-T8, ZS-M34 v1.0, ZS-M35 v1.1) — the v3.5 body had only the supersedes-line acknowledgement; v4.0 closes this presentation gap.
- **ZS-M33 v2.0** [NEW v4.0] — Operator-level numerical audit of Path γ -revised Reading C; W2 wall promoted DERIVED-CONDITIONAL → DERIVED.
- **ZS-M34 v2.0** [NEW v4.0] — Burnol K_1 @ $p=3$ Sign-Faithful Identity (Theorem M34.V, DERIVED); NC-M34.faithful PERMANENT NON-CLAIM → RESOLVED at DERIVED-CONDITIONAL via tensor-product factorization $\mathcal{H}_K = \mathcal{H}_{arith} \otimes \mathcal{H}_{Wilson-Sonin}$.
- **ZS-M35 v2.1** [NEW v4.0] — Logical correction of v2.0; Theorem M35.5 general direction demoted to OPEN with explicit reframing; Corollary M35.5.1 (FIBER PRODUCT $C_2 = 5 \cdot \text{positive} \cup 7 \cdot C_1$) promoted DERIVED-CONDITIONAL → VERIFIED; Theorem M35.6 (Sufficient Condition Theorem, DERIVED) added.
- **ZS-M37 v2.0** [NEW v4.0] — Fourth amenable-quotient projection of F_2 engine: $q_R: \mathcal{C}_{cyl}(F_2) \rightarrow \mathcal{S}_w^{\wedge}\{C_3\}$ (Twin-Reuleaux Quotient Functor). Lemma C-rel PROVEN on Z-admissible quotient.
- **ZS-M38 v1.1** [NEW v4.0] — Substrate Carrier Spectrum Theorem; Hexapair Uniqueness DERIVED-strong by five-fold over-determination; $\ln(2)$ channel-capacity quantization of carrying scores; classifies five-Banach–Tarski-functor family.
- **ZS-M39 v2.0** [NEW v4.0] — Vieta-Lyapunov-Schröder Bridge Theorem; first-principles derivation $K_\theta = 1/[Y^2(1-2A)]$ under Convention N0; three-stage Stokes-tail closure; ZERO remaining OPEN items at v2.0 release.
- **ZS-A10 v2.1** [NEW v4.0] — Sector-Orientation Complementarity (Bridge-Four manifestation) Theorem A10.2 DERIVED-interpretation strong; NEW TESTABLE P4: $r_p/r_\pi = 4/\pi$ (observed deviation 0.05%).
- **ZS-Q8 v1.1** [NEW v4.0] — Toronto negative-time experiment Z-Spin reading; i-tetration fixed point x^* as universal Z-Spin intercept; Theorem Q8.9 Standard-Optics Projection.
- **ZS-Q9 v1.2** [NEW v4.0] — Complex Time-Delay Locking; first lab-measurable lifting of z^* via two-coordinate spectral reconstruction bridge; anti-numerology $\ln \Lambda = 8.94$ decisive.

Major v5.0 absorptions (v4.0 → v5.0):

- **ZS-F19 v2.2** [NEW v5.0] — Frame-Invariant Tilt Theorem (FITT, DERIVED): $\tanh(\psi_X + \psi_Y) = 31/59$ and $\tanh(\psi_Y - \psi_X) = 3/67$; KMS-to-Geometric Rapidity Projection (Theorem F19.6, DERIVED): $\tanh(2 \cdot \psi_{KMS}(X \rightarrow Y)) = 3/5$. First paper-internal algebra-level bridge to Type II crossed-product framework of De Vuyst–Höhn 2025. 64/64 PASS, all 7 sub-OPEN gates CLOSED.

• **ZS-F20 v1.1** [NEW v5.0] — Unified catalogue and Time-Unrolled Causal DAG of six non-routine twist triggers T1–T6 (Z-Telomere, Mexican-Hat Bootstrap, Auto-Surgery, Kibble, BH Trap, CP-violation). Strict DAG under time-unrolled cycle indexing. Biological correspondence: DNA trauma = simultaneous T3 + T4 + T5 triadic activation. $\Delta\theta_{T1}/\Delta\theta_{T2} = N_{-}(2\pi) = 78.45$ (PROVEN). 47/47 PASS.

• **ZS-M40 v1.1** [NEW v5.0] — J-Conjugate Holonomy Knot Pair $(K_p, \bar{K}_p)$ as Z-sector geometric realisation of primes. Six theorems M40.A (cup product $\leftrightarrow 4\pi$ closure) – M40.F (anti-numerology uniqueness, COMPUTATIONALLY VERIFIED). Eight-fold structural bridge to externally PROVEN Kim arithmetic Chern–Simons; cyclotomic operator-model analogue at $Q = 11$ via Euler product $\zeta_{-}\{\mathbb{Q}(\zeta_{11})\}$. $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ unique biquadratic class-number-one match over $[2, 1000]$. 89/89 PASS.

• **ZS-Q10 v4.0** [NEW v5.0] — Master edition: three-sector polyhedral closure of c_n harmonic-ladder coefficients ($c_2 = 2/19$, $c_4 = 4/37$, $c_6 = 8/37$, $c_X = 10/37$ all DERIVED or DERIVED-CONDITIONAL); 37-Universality registered (five of six c_n share denominator 37 = Schur-depleted polyhedral mode count); cumulative anti-numerology $\ln \Lambda_{\text{total}} = 38.67$ (highest in corpus); five-framework cross-disciplinary asymmetry restored. 67/67 PASS.

• **ZS-A11 v1.1** [NEW v5.0] — Vortex Cosmology II: rigorous ε -Halo $\leftrightarrow$ Sub-Halo Equivalence Theorem (four-level DERIVED); Vortex Six-Class Lifecycle (DERIVED-strong MECE classification); SMBH Mass Diversity from differential accretion (Primordial-Vortex SMBHs vs Collapse-SMBHs, JWST Little Red Dot test). Population revised to $N_v \approx 2.8 \times 10^{15}$. 53/53 PASS, DESI DR2 + ACT DR6 cross-checked.

• **ZS-A12 v1.5** [NEW v5.0] — Macroscopic Phase Locking; BEC/SC = condensed-matter realisation of Vortex Bose/Fermi Duality. HEADLINE result Theorem A12.6 (DERIVED): Nelson–Kosterlitz universal-jump integer "2" = $\dim(Z) = 2$ cardinal-2 substrate. Theorem A12.7 (DERIVED-CONDITIONAL): Homes universal constant cardinal-2 bracketing $2/Q \leq C/C_{\text{BCS-dirty}} \leq \ln 2/(1 - A)$. SC = T2 $\rightarrow$ T1 $\rightarrow$ T4 sequential triadic activation distinct from DNA. 74/74 PASS.

• **ZS-T9 v2.3** [NEW v5.0] — B₇Be Binding-Response Gate extension of v2.2 ERO for the Lithium Problem. $P_{B7Be^Z} = -2A/Q = -70/4807$ (HYPOTHESIS-strong, NOT DERIVED; ZS-Q7 §5 slow eigenvalue of Pauli master equation). ^7Li post-v2.3 $\approx 1.64 \times 10^{-10}$ at $+0.13\sigma$ pull (TESTABLE). v2.2 D/H + Y_p ERO closure PRESERVED unchanged (B₇Be channel decoupled). 124/124 PASS.

— End of The Book of Z-Spin Cosmology v5.0 — Light Edition —

Compiled by: Claude (cross-paper integrator), under Kenny Kang's editorial direction. v4.0 — May 2026. Lighthouse function: navigation aid for AI agents engaging with the Z-Spin Cosmology research programme. No claim of physical realisation is made by this Light edition; canonical statements live in the source papers ZS-F0 through ZS-Q9 v1.2. v4.0 strictly preserves all v3.5 content (no-deletion rule); the v3.5 body was 1518 lines, v4.0 is ~1730 lines after PART XII absorption. The minimal triple $(A, Q, \dim(Z)) = (35/437, 11, 2)$ remains LOCKED across all 121 source papers.

Version History (v5.0 closing): v5.0 (May 2026): 121-paper coverage. PART XIII added between PART XII and Master Bypass Index. Consolidates from internal Z-Spin Collaboration research notes up to v4.0 + post-v4.0 papers ZS-F19 v2.2, ZS-F20 v1.1, ZS-M40 v1.1, ZS-Q10

v4.0, ZS-A11 v1.1, ZS-A12 v1.5, ZS-T9 v2.3. Previous version (v4.0) supersession chain preserved.